

LUND UNIVERSITY

Adaptive Control of Positive Systems

Bencherki, Fethi

2025

Document Version:
Publisher's PDF, also known as Version of record

Link to publication

Citation for published version (APA):
Bencherki, F. (2025). *Adaptive Control of Positive Systems*. [Licentiate Thesis, Department of Automatic Control]. Department of Automatic Control, Lund University.

Total number of authors:
1

Creative Commons License:
CC BY-NC-ND

General rights
Unless other specific re-use rights are stated the following general rights apply:
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.
• Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
• You may not further distribute the material or use it for any profit-making activity or commercial gain
• You may freely distribute the URL identifying the publication in the public portal

Read more about Creative commons licenses: https://creativecommons.org/licenses/

Take down policy
If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

LUND UNIVERSITY

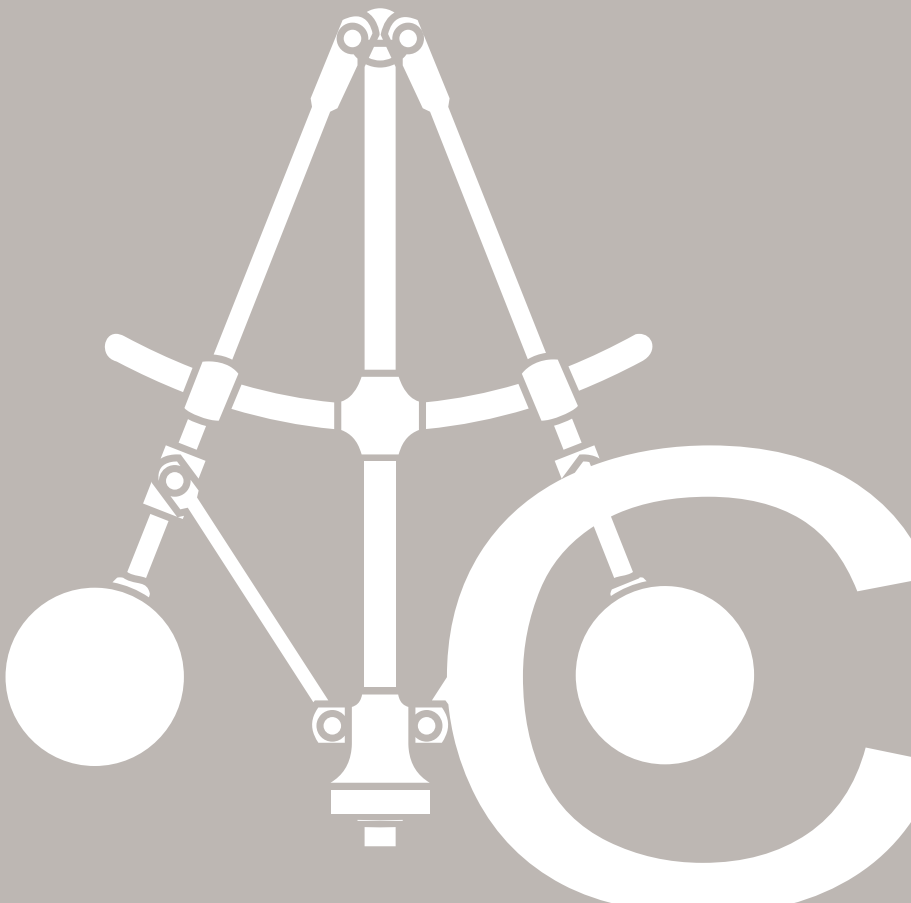
PO Box 117
221 00 Lund
+46 46-222 00 00

Download date: 07. Nov. 2026

Adaptive Control of Positive Systems

FETHI BENCHERKI
DEPARTMENT OF AUTOMATIC CONTROL | LUND UNIVERSITY





Adaptive Control of Positive Systems

Fethi Bencherki



LUND
UNIVERSITY

Department of Automatic Control

Licentiate Thesis TFRT-3284
ISSN 0280–5316

Department of Automatic Control
Lund University
Box 118
SE-221 00 LUND
Sweden

© 2025 by Fethi Bencherki. All rights reserved.
Printed in Sweden by Media-Tryck.
Lund 2025

Abstract

The analysis, control, and design of complex networks and systems represent an essential and ever-expanding field of study. At its foundation lie mathematical models that capture complex physical phenomena while remaining amenable to both analysis and design. These models prove instrumental in representing diverse systems such as power grids, cellular telecommunication networks, large-scale manufacturing processes, and computer systems. However, as technological and societal needs grow, these systems continue to increase in both complexity and scale, making their controlling tasks increasingly challenging. Traditional control approaches often fail to capture the uncertainties inherent in the modeling process, resulting in significant performance deterioration. Furthermore, these methods frequently suffer from poor scalability, rendering them unsuitable for large scale systems.

In response to these challenges, robustness and adaptation have emerged as critical design principles. Robustness ensures resilience against unmodeled dynamics, while adaptation addresses unexpected events and changes in the dynamics. Scalability, on the other hand, can be achieved by deriving localized control laws, thereby alleviating the limitations of centralized control approaches. The increasing affordability and accessibility of data measurements further provide a foundation for advancing these objectives. This has paved the way for data-driven approaches, which are now gaining traction as a preferred strategy.

The first paper of this thesis, *Robust Simultaneous Stabilization via Minimax Adaptive Control*, certifies the existence of adaptive minimax controllers. These controllers are a superior class of robust adaptive nonlinear controllers that can stabilize the plant in the presence of large model uncertainties. The paper focuses on the robust simultaneous stabilization of many linear time-invariant (LTI) plants and analyzes conditions under which these controllers are guaranteed to exist.

The second paper, *Robust adaptive data-driven control of positive systems with application to learning in SSP problems*, presents another type of robust adaptive controllers. This controller is designed for a positive systems class, where the property of positivity is exploited to enable the scalability of these control solutions. The controller uses the certainty-equivalence principle, relying on estimates of the Q-factor. Leveraging the equivalence between this optimal control problem and

stochastic shortest path (SSP) problems, this paper also contributes a model-free solution to the growing body of research on parameter-free learning in SSP.

The third and final paper, *Data-Driven Adaptive Dispatching Policies for Processing Networks*, applies the framework from the second paper to processing networks. In this case, the processing rates of the different units are initially unknown. The paper analyzes the online performance of the learning-based adaptive policy. It also establishes properties that enable scalability, such as diagonal dominance, to improve numerical efficiency of the proposed scheme.

Acknowledgements

First and foremost, I would like to express my deepest gratitude to my supervisor, Prof. Anders Rantzer, for his invaluable guidance and unwavering support throughout this journey. Your expertise and insights have been instrumental in shaping this thesis, and it would not have been possible without your mentorship. Thank you for the engaging research discussions and thought-provoking remarks that significantly enhanced the quality of this work.

Secondly, I want to thank the Department of Automatic Control for the refreshing atmosphere, very helpful administrative staff, and amazing colleagues. You've all made this so much more fun and engaging. I can't wait for many more fikas with you!

Lastly, a special thanks to my family, friends and to Isabella, for your support and encouragement. Thank you for being there during stressful times.

Large-Language Models

The author acknowledges the use of Large Language Models (LLMs) for soft editing. These tools were utilized to improve the clarity, coherence, and grammatical accuracy of the text.

Financial Support

The author is with the ELLIIT Strategic Research Area at Lund University, Lund, Sweden. This work is partially supported by the Wallenberg AI, Autonomous Systems and Software Program (WASP) funded by the Knut and Alice Wallenberg Foundation, and the European Research Council (ERC) under the European Union's Horizon 2020 Research and Innovation Programme under grant agreement No 834142 (ScalableControl).

Contents

1. Introduction **11**
1.1 Outline 12
2. Background **13**
2.1 Adaptive Control 13
2.1.1 Minimax Adaptive Control 15
2.2 Positive Systems 17
2.3 Data-Driven Control 18
2.3.1 Statistical Learning Theory for Control 19
2.3.2 Online Convex Optimization for Control 19
2.3.3 Data-Driven Identification and Control of Positive Systems 20
2.3.4 Learning Based on Q-factor 21
3. Contributions **23**
4. Discussion **26**
4.1 Minimax Adaptive Stabilization 26
4.2 Adaptive Positive Stabilization 26
4.3 Robustness 27
4.4 Scalability 27
4.5 Outlook for Future Works 28
Bibliography **29**
Paper I. Robust Simultaneous Stabilization via Minimax Adaptive Control **39**
1 Introduction 40
1.1 Notation 41
2 Problem Formulation 41
3 Main Results 43
3.1 Guarantees on Stabilization of two P-Stabilizable MIMO LTI Plants 43
3.2 Explicit Value of the Achievable $\mathcal{L}_2$-Gain Bound 45
3.3 A Specialized $\bar{\gamma}$ Bound for the Single-Input Single-Output (SISO) Case 47

4 Numerical Examples . . . 47
4.1 Systems with Unknown Input Direction . . . 47
4.2 Inverted Pendulum with Uncertain Actuator Gain . . . 48
4.3 Performance Comparison to Periodic Deadbeat Controllers 50
5 Conclusions and Future Works . . . 51
A Preliminary Lemmata . . . 52
References . . . 52

Paper II. Robust Adaptive Data-Driven Control of Positive Systems with Application to Learning in SSP Problems 55

1 Introduction . . . 56
1.1 Contributions and Outline of the Paper . . . 56
1.2 Notation . . . 57
2 Problem Setup . . . 58
2.1 Solution to Problem 1 . . . 59
2.2 Model-Free Optimal Control of Positive Systems via Q-Factor . . . 59
2.3 Problem Formulation . . . 61
2.4 Solution to Data-driven Algebraic Equation in the $q(t)$-Parameter . . . 62
3 Main Results . . . 63
3.1 Performance Analysis . . . 64
4 Numerical Experiment . . . 66
5 Conclusion . . . 68
A Proofs . . . 69
A.1 Proof of Lemma 1 . . . 69
A.2 Proof of Lemma 2 . . . 69
A.3 Proof of Theorem 1 . . . 70
A.4 Proof of Theorem 2 . . . 72
A.5 Proof of Corollary 1 . . . 72
References . . . 73

Paper III. Data-Driven Adaptive Dispatching Policies for Processing Networks 77

1 Introduction . . . 78
1.1 Outline and Contribution of the Letter . . . 78
1.2 Notation . . . 79
2 Problem Setup . . . 79
2.1 A Multi-Unit Processing Network Model . . . 79
2.2 An Optimal Control Problem . . . 79
2.3 Optimal Policy for Known Processing Rates . . . 80
2.4 Model-Free Optimal Control via Q-factor . . . 81
2.5 Problem Formulation . . . 82
3 Main Results . . . 83

3.1 Positive Least-Squares Estimation 83
3.2 Scalability Properties by Virtue of Diagonal Dominance . 84
3.3 Performance Analysis 85
3.4 The Choice of Explorations ϵ_t 86
4 Numerical Examples 86
4.1 Performance on a Random Processing Network 86
4.2 Adaptation in the Face of Unmodelled Dynamics 87
5 Conclusion 88
A Preliminaries 89
A.1 Constrained Optimal Control 89
B Proofs 89
B.1 Bounds on (q^x, q^0) Parameters 89
B.2 Proof of Lemma 1 89
B.3 Proof of Lemma 2 90
B.4 Proof of Theorem 1 90
B.5 Proof of Theorem 2 91
References 92

1

Introduction

As systems and network infrastructures grow in size and complexity, the methods used to control them must also become more powerful to meet these increasing demands [Meyn, 2008]. For example, the challenge of modeling every physical aspect of a system's dynamics led to the development of robust control. This approach helps tolerate inaccuracies in system models. Additionally, adaptive control techniques were introduced to handle time-variations and unexpected behaviors in the system by incorporating real-time learning mechanisms.

The big data era has sparked renewed interest in online robust adaptive control and the related field of reinforcement learning [Annaswamy, 2023]. Both aim to provide real-time optimal decision-making for complex tasks in large-scale systems, where centralized controllers may struggle with the curse of dimensionality. This has led to research into distributed control architectures that take advantage of the inherent structure of certain systems, such as in positive systems.

Positivity is a natural physical constraint and appears in many applications. For example, queue lengths, levels, heights, concentrations, fluid pressure, population sizes, and state variables in compartmental systems are all inherently positive quantities. This highlights the importance of the class of positive systems and motivates the development of a specialized theory for scalable robust adaptive control for this class of systems.

Figure 1.1 illustrates the feedback connection of a large-scale linear positive system with uncertain parameters, controlled by an adaptive controller. The system is affected by disturbances w, which could be external disturbances or unmodeled dynamics in the original plant, or both. The goal of the adaptive policy is to stabilize the system while maintaining its positivity. This leads to the *adaptive positive stabilization problem*, which is the central focus of this thesis.

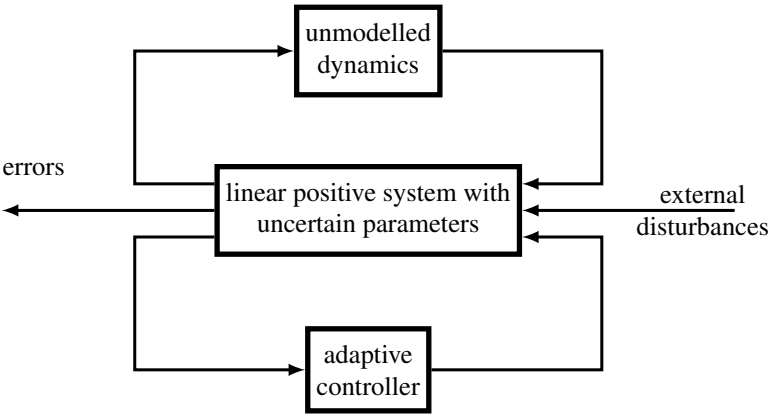


Figure 1.1 Feedback interconnection of a large-scale linear positive system with an adaptive controller in the presence of disturbances w.

1.1 Outline

This thesis addresses the positive stabilization problem from the perspective of robust adaptive control. Chapter 2 provides an overview of the relevant theory and literature on adaptive control, data-driven control, and positive systems. Specifically, it introduces the broad field of adaptive control, highlighting its major historical developments and discussing two popular types of adaptive controllers. Next, we introduce the minimax adaptive control synthesis framework, which leads to robust adaptive controllers. While the setup in Papers II and III is not minimax, their analysis is a worst-case analysis, aligning with the assumptions made in Paper I, thus linking all three papers with a common goal of robustness and adaptation against potential adversarial disturbances.

Chapter 2 also covers recent developments in data-driven control literature, particularly focusing on data-driven control and identification methods for positive systems, showing that the field is still emerging and offers many opportunities for future research. Additionally, we present a method of conducting adaptive control using the Q-factor, a concept rooted in reinforcement learning. This approach is applied in Papers II and III to design robust adaptive controllers.

Chapter 3 summarizes the key contributions of the included papers and credits them to their respective authors. Chapter 4 presents a discussion of the results and explores potential directions for future research. The main body of the thesis is made up of three papers.

2

Background

Since the core contributions of this thesis are focused on adaptive control, it is important to introduce this well-established field, albeit far from being an exhaustive one. Doing so also helps highlight the connections between the contributions of this thesis and the existing body of work.

2.1 Adaptive Control

According to [Åström and Wittenmark, 1995], *an adaptive controller is a controller with adjustable parameters and a mechanism for adjusting the parameters*. This definition is adopted in this thesis.

Adaptive control has its origins in the 1950s, when the focus was on designing autopilot systems for aircraft with high performance. The goal was to create controllers that could adapt to changing environmental conditions. For example, aircraft operate across a wide range of speeds and altitudes, with nonlinear and time-varying dynamics, making the performance of conventional controllers with constant gain and linear feedback poor. Later, gain scheduling [Leith and Leithead, 2000] proved to be effective for flight control systems.

At that time, the conceptual understanding of adaptive control systems was still in its early stages. However, many breakthroughs in control theory emerged, leading to a better understanding of the field. Modern control theory, which adopted state-space models, was introduced. Bellman also developed the principle of dynamic programming [Bellman, 1957], which helped in understanding adaptive systems. Alongside this, the dual control framework was coined by Feldbaum in [Feldbaum, 1960]. This framework suggested that control should both probe and direct; in other words, the controller should inject test signals to gather better information about the underlying dynamics. This idea gave rise to the exploration-exploitation tradeoff, which is popular in reinforcement learning [Sutton and Barto, 2018].

Further progress was made with the work in [Tsypkin and Nikolic, 1971], which showed that several learning and adaptive control schemes could be described using a common framework. Meanwhile, in the parallel field of operations research,

the problem of decision making under uncertainty was addressed. As a solution, the certainty equivalence principle was proposed [Simon, 1956], which suggested treating estimates as the true values when performing control synthesis. Many adaptive control schemes rely on this principle.

The 1970s marked a significant increase in research on adaptive control, highlighted by the landmark paper [Åström and Wittenmark, 1973], which addressed controlling systems with constant but unknown parameters. This work introduced a simple control law based on least-squares parameter estimation, achieving asymptotically optimal performance. Around the same time, model reference adaptive control (MRAC) emerged, driven by advancements in system identification. These developments enabled the integration of estimation techniques with design methods, resulting in successful applications, although theoretical analysis was initially limited. By the late 1970s and into the 1990s, research efforts shifted toward understanding stability, convergence, and robustness in adaptive control schemes. Studies such as [Ljung, 1977] examined the convergence of adaptive algorithms, while [Goodwin et al., 1981] introduced an adaptive control algorithm with proven mean square stability. Work on convergence of least-squares-based self-tuning regulators appeared in [Guo, Chen, et al., 1991; Guo, 1995], where as issues of instability were explored in [Ioannou and Kokotovic, 1984; Egardt, 1979].

While a detailed review of the field is beyond the scope of this thesis, readers can refer to foundational textbooks [Åström and Wittenmark, 1995; Goodwin and Sin, 2014; Narendra and Annaswamy, 2012; Ioannou and Sun, 2012] and the recent survey [Annaswamy and Fradkov, 2021]. Two important instances of adaptive controllers are worth noting. The self-tuning regulator, introduced in [Åström and Wittenmark, 1973], relies on the certainty equivalence principle and is closely related to the adaptive controllers described in papers II and III. They differ primarily in their underlying assumptions about disturbances, as well as in the parameters on which the adaptation is based. Another example is the dual adaptive controllers proposed in [Wittenmark, 1995], which offers an alternative approach to adaptive control under a different design strategy.

Self-tuning Regulators [Åström and Wittenmark, 1973] Self-tuning regulators (STR) are a type of adaptive regulators built on the principle of online system identification. They were originally inspired by needs in the process control industry [Åström, 2021], where steady-state regulation was a key problem. The mathematical framework for STRs typically involves discrete-time input-output stochastic systems, as exemplified by the following single-input, single-output case

$$\begin{aligned} y(t) + a_1 y(t-1) + \cdots + a_n y(t-1) = b_1 u(t-1) + \cdots + b_n u(t-n) \\ + c_1 w(t-1) + \cdots + c_n w(t-n) + e(t), \end{aligned}$$

where u is the control signal, y is the process output, w is a measured disturbance, e is a stochastic disturbance, and $\{a_k, b_k, c_k\}_{k=1}^{n}$ are unknown parameters. While

STRs are often implemented in this input-output framework, they can also be applied to state-space models by aggregating past inputs and outputs into a state vector. However, this approach may result in non-minimal realizations.

STRs employ least-squares parameter estimation alongside a control design that minimizes output variance. It was later established that if the parameter estimates converge, the controller will asymptotically approach the minimum variance controller. STRs have proven particularly useful when deployed as feedforward controllers, which depend on the availability of an accurate process model to achieve desired control objectives.

Adaptive dual controllers The dual control framework, introduced in [Feldbaum, 1960; Feldbâum, 1963], considers the minimization of a cumulative performance index function under incomplete prior knowledge about the plant's dynamics. The resulting optimal controller is characterized by a dual objective: (a) performing probing actions to improve learning of the system's true dynamics, which are long-term oriented, and (b) executing greedy actions aimed at achieving short-term optimal performance. This balance between exploration and exploitation gives rise to the term "dual control." The optimal controller in this framework solves a functional equation, commonly referred to as the Bellman equation.

While deriving the Bellman equation is straightforward, solving it for large systems is computationally infeasible due to the curse of dimensionality. To address this, many works have developed tractable numerical solutions by focusing on special cases, as seen in [Bernhardsson, 1989; Åström and Helmersson, 1986]. For a detailed exploration of the topic, readers are encouraged to consult surveys like [Wittenmark, 1995; Filatov and Unbehauen, 2000] and the comprehensive reference book [Filatov and Unbehauen, 2004]. Notably, minimax adaptive control, discussed next, can be viewed as a dual control framework, as it also integrates learning and control in a unified approach.

2.1.1 Minimax Adaptive Control

The term *minimax adaptive control*, coined in [Didinsky and Basar, 1994], refers to a control synthesis approach that addresses parameter uncertainties and disturbances by considering worst-case scenarios. This method ensures robustness [Zhou and Doyle, 1998] and is particularly useful in handling unmodeled dynamics and time-varying plant characteristics, where disturbances may be non-stochastic. The framework relies on a game-theoretical formulation [Başar and Bernhard, 2008; Başar and Olsder, 1998], where the controller seeks to minimize the cost function while adversaries, representing disturbances and uncertain parameters, aim to maximize it.

By framing the problem as a zero-sum, two-player game, the controller accounts for adversarial disturbances and the worst realizations of uncertain parameters. This results in an adaptive controller that guarantees an $\mathcal{L}_2$-gain bound from disturbances to errors across all possible parameter realizations. The game-theoretical nature of

minimax adaptive control can thus be viewed as an attempt to find a saddle point in the associated zero sum two-player game.

The original work in [Didinsky and Basar, 1994] introduced the general framework for minimax adaptive control, using nonlinear, time-varying differential equations to represent the unknown dynamics. It provided sufficient conditions for cases where the parameters are constrained to a convex set, ensuring the existence of a stabilizing policy and a finite value for the dynamic game. This framework was later extended in [Pan and Basar, 1998] to address single-input, single-output (SISO) nonlinear systems described in a parametric strict-feedback form. Further developments explored the achievable $\mathcal{L}_2$-gain in adaptive control under specific uncertainties. For example, [Vinnicombe, 2004] examined first-order SISO systems with sign uncertainties in state and input coefficients. The case of parameters constrained to an interval was addressed in [Megretski and Rantzer, 2003], which, along with [Megretski, 2004], derived fundamental limitations in minimax adaptive control.

Minimax adaptive control has been revisited recently in [Rantzer, 2021], focusing on discrete-time minimax optimization problems. The problem is defined as follows

$$\begin{aligned} \inf_{\mu} \sup_{w,A,B,T} & \sum_{t=0}^{T} (|x_t|_Q^2 + |u_t|_R^2 - \gamma^2 |w_t|^2) \\ \text{s.t. } & x_{t+1} = Ax_t + Bu_t + w_t, \quad t \geq 0 \\ & u_t = \mu_t(x_0, \ldots, x_t, u_0 \ldots u_{t-1}), \\ & (A, B) \in \mathcal{M},\ Q, R \succ 0, \end{aligned} \tag{2.1}$$

where $\mathcal{M}$ is a compact set defining the uncertain dynamics (A, B), and $|x|_A^2 = x^\top A x$. The work addressed the special case of a finite set of linear time-invariant systems, where $\mathcal{M} \coloneqq \{(A_1, B_1), \ldots, (A_N, B_N)\}$. The solution reformulates the problem in terms of historical data matrices Z_t, recursively defined as

$$Z_{t+1} = Z_t + \begin{bmatrix} -v_t \\ x_t \\ u_t \end{bmatrix} \begin{bmatrix} -v_t \\ x_t \\ u_t \end{bmatrix}^\top, \qquad Z_0 = 0.$$

This reformulation confines the dependence of the cost and dynamics on the unknown parameters (A, B) to the penalty of the final state, enabling the use of dynamic programming. Specialized cases, such as unknown input directions $(A, \pm B)$ and unknown state matrix signs $(\pm A, B)$, were also addressed in [Rantzer, 2021; Rantzer, 2020], respectively. However, the success of dynamic programming relies on solving nonlinear matrix inequalities. To address this, [Cederberg et al., 2022] proposed a linearization technique to solve the inequalities. The first contribution of this thesis, Paper I, analyzes the feasibility of these matrix inequalities for specific choices of the uncertainty set $\mathcal{M}$.

The minimax adaptive control framework was extended to network settings in [Renganathan et al., 2024] and to the output feedback case in [Kjellqvist and

Rantzer, 2022a; Kjellqvist and Rantzer, 2024; Kjellqvist, 2024]. Additionally, it was implemented in a distributed manner in [Renganathan et al., 2024], with further analysis from an online learning perspective in [Renganathan et al., 2023]. In a similar direction, minimax adaptive estimation was explored in [Kjellqvist and Rantzer, 2022b], extending the concepts to estimation problems under uncertain dynamics.

The thesis focuses on the adaptive control of linear positive systems. A brief introduction to this important class of systems is provided next.

2.2 Positive Systems

Positive systems are systems where the state evolution remains nonnegative, provided the initial state is nonnegative [Farina and Rinaldi, 2011; Kaczorek, 2012]. This class of systems is particularly important due to its wide range of applications, such as in social sciences, epidemiology, biology, pharmacology, biochemical engineering, econometrics and stochastic processes. Many real-world variables, like population density, prices, probabilities and masses inherently have non-negativity constraints [Rantzer and Valcher, 2018]. A classic reference on this subject is [Berman and Plemmons, 1994]. The theory for monotone systems [Hirsch and Smith, 2006] extends these concepts to nonlinear systems, where the partial ordering of initial states is preserved during the evolution of the system.

The study of positive systems is particularly relevant for the analysis and design of large-scale systems, as these systems are numerically more tractable due to their inherent structure. For example, in the stability analysis of positive LTI systems, one can use linear Lyapunov functions and storage functions instead of quadratic ones [Rantzer and Valcher, 2018], which significantly simplifies the complexity of stability analysis. In the context of controller synthesis, linear algebraic equations [Rantzer, 2022; Li and Rantzer, 2024] or linear programming techniques can be applied, avoiding the need for solving quadratic algebraic Riccati equations and semidefinite programs (SDPs). Additionally, scalable identification methods for positive systems exists, as demonstrated in [Umenberger and Manchester, 2016; Revay et al., 2021]. Moreover, by exploiting positivity, distributed control theory for positive systems has been developed, enabling more efficient decentralized control strategies, as seen in [Rantzer, 2011; Zhang et al., 2020; Zhang et al., 2023].

A discrete-time LTI state-space model of the form

$$x_+ = Ax,$$

is positive if and only if the matrix A is nonnegative [Berman and Plemmons, 1994]. In continuous systems, the counterpart is that the matrix A is Metzler [Kaczorek, 2012]. When considering the effect of inputs, the state-space model becomes

$$x_+ = Ax + Bu,$$

where the objective is to find stabilizing inputs that preserve the positivity of the system. This leads to the so-called positive stabilization problem [Shafai et al., 2022], which is one of the key focuses of this thesis.

While recent literature on data-driven control predominantly emphasizes offline methods over adaptive approaches, we offer a brief summary of the latest developments in this area. This discussion is especially relevant given that the adaptive schemes presented in the contributed papers similarly focus on finite-time guarantees, which is a key common interest.

2.3 Data-Driven Control

The concept of learning controllers from data is not new and dates back to the origins of adaptive control. System identification techniques [Ljung, 1999] have long been used to develop models from data, which can then be integrated with robust controller design methods [Zhou et al., 1996] to handle uncertainties and ensure reliable system performance.

Recent developments in data-driven control have shifted away from traditional system identification, often bypassing it or making it implicit, instead using data directly for controller synthesis. Examples of such approaches include iterative feedback tuning [Hjalmarsson et al., 1998], intelligent PID control [Fliess, 2009], and direct adaptive control [Kaufman et al., 2012]. Moreover, methods built upon Willems' fundamental lemma [Willems et al., 2005], which states that future trajectories can be predicted linearly from past ones, provided the data is persistently exciting, have started to emerge. This foundational idea has led to the works in [De Persis and Tesi, 2019] and [Berberich et al., 2020] for linear systems, and [De Persis et al., 2023] for nonlinear stabilization. These approaches are often classified under the behavioral approach to data-driven control [Markovsky and Dörfler, 2021]. Although these methods rely on persistently exciting data, which could also be used for system identification then controller design, the recent data-informativity framework [Van Waarde et al., 2020; Van Waarde et al., 2023] provides necessary and sufficient conditions for data to be informative enough for control tasks, eliminating the need for PE data in some cases. This further strengthens the case for direct data-driven control, positioning it as superior to traditional controller design strategies.

Recently, there has been a surge in research within the statistical machine learning literature focusing on finite-time analysis. This area of study is concerned with understanding the performance and behavior of algorithms over a finite time horizon, as opposed to asymptotic results. In the next subsection, we provide a brief overview of this literature, highlighting key advancements and findings in the field.

2.3.1 Statistical Learning Theory for Control

Traditional results on identification and control focus on the asymptotic scenario [Ljung and Wahlberg, 1992; Chiuso and Picci, 2004]. In contrast, decision-making in RL applications relies on finitely collected data. This has driven a shift towards finite-sample guarantees for learning and control algorithms. These guarantees are derived using tools from high-dimensional statistics and learning theory. The primary objective is to quantify sample efficiency and establish the fundamental limits of learning algorithms. This work spans a variety of control and identification setups. For example, [Jedra and Proutiere, 2019; Jedra and Proutiere, 2020] examine the sample complexity of least-squares estimators, the work in [He et al., 2024] does so for subspace identification, while studies on the linear quadratic regulator (LQR) are extensive and can be found in [Abbasi-Yadkori and Szepesvári, 2011; Dean et al., 2020; Simchowitz and Foster, 2020; Jedra and Proutiere, 2022]. Comprehensive surveys on these topics are available in [Matni et al., 2019; Tsiamis et al., 2023].

Despite the previously mentioned significant advances in data-driven control and finite-time analysis, the methods presented are primarily offline and often rely on stochastic assumptions. In contrast, the work in this thesis moves away from these approaches, focusing on an adaptive setting with potentially non-stochastic disturbances affecting the plant.

2.3.2 Online Convex Optimization for Control

Online nonstochastic control [Hazan and Singh, 2022] applies techniques from online convex optimization [Hazan et al., 2016] to develop control schemes with provable guarantees in classical optimal and robust control settings. It deals with linear systems, which may be unknown, and subject to bounded, potentially adversarial disturbances, with cost functions revealed sequentially after each control action. The state-feedback problem without model uncertainty was addressed in [Agarwal et al., 2019], while the case with uncertain parameters and an initial stabilizing controller is considered in [Hazan et al., 2020]. The output feedback case was studied in [Simchowitz et al., 2020], and the black-box control setting with a stabilizing controller was examined in [Chen and Hazan, 2021].

A key distinction between minimax adaptive control, the approach considered in Paper I, and the online control approach lies in how disturbances are treated. In online control, the adversary selects disturbance sequences a priori, meaning they are predetermined and independent of the realization of the control signals. Minimax adaptive control, on the other hand, introduces a significant extension by allowing disturbances to dynamically depend on both the control signals and the estimated system parameters. This added flexibility more accurately captures real-world scenarios where disturbances are reactive rather than static, resulting in a more challenging and realistic control problem. Moreover, it eliminates the need for an initial stabilizing policy, providing another important advantage.

Although the literature on adaptive control for positive systems is still in its in-

fancy, foundational work on identification and data-driven control for these systems has begun to emerge. The following section provides a detailed overview of this growing body of research.

2.3.3 Data-Driven Identification and Control of Positive Systems

Only a few works on adaptive control for positive systems have been reported in the literature. Among these, the study in [Sun et al., 2011] introduces an $\mathcal{L}_1$ adaptive controller for positive LTI state-space models. Its primary objective is to ensure that the system output tracks a given nonnegative reference signal with an unknown but Hurwitz state matrix, under the so-called matching assumption. A defining feature of $\mathcal{L}_1$ adaptive control is its ability to decouple the adaptation loop from the control loop, enabling arbitrarily fast adaptation without compromising robustness. This approach provides uniform performance bounds for both transient and steady-state operation [Cao and Hovakimyan, 2010].

On the other hand, the work in [Guiver et al., 2016] develops a high-gain adaptive tuning mechanism for pest management, modeled as a discrete LTI positive state-space model. This method ensures stability and guarantees nonnegative states and inputs under system uncertainties. The approach was further extended to nonlinear systems in [Edholm et al., 2022], where a class of forced nonlinear positive difference equations was analyzed. This extension was applied to pest population dynamics, effectively addressing the challenge of maintaining stability in systems constrained to nonnegative values. Lastly, high-gain adaptive controllers for component-wise nonnegative continuous-time linear systems were explored in [Haddad et al., 2010, Chapter 15], particularly in the context of models for general anesthesia.

With the rise of data-driven control approaches and finite-time analysis, the linear quadratic regulator (LQR) problem has received extensive attention and thorough treatment. In contrast, the class of positive systems has been less explored. While recent works have started to address this problem, the research is still far from comprehensive, underscoring the need for further investigation and development.

Following the trend in mainstream data-driven control literature, recent works on finite-sample identification and control of positive systems have started to emerge. The positive stabilization of linear systems through state-feedback using input-output data was explored in [Shafai et al., 2022], which applied data-driven control formulas from [De Persis and Tesi, 2019] and incorporated the structural constraint of positivity via input-output data-based linear matrix inequalities (LMIs). However, this approach does not account for noise in state measurements and is offline. In contrast, [Iwata et al., 2024] introduces a data-informativity approach for distributed positive stabilization, providing a necessary and sufficient condition for the problem in the form of LMIs.

Another notable contribution is [Makdah and Pasqualetti, 2023], which adopts a behavioral approach to synthesize dynamic output-feedback controllers that ensure the closed-loop system remains externally positive. Externally positive systems are defined as systems that produce positive outputs when positive inputs are applied [Grussler et al., 2017]. This work leverages the fact that in the space of input-output behaviors, a dynamic controller can be represented as a static behavioral feedback gain. Additionally, [Padoan et al., 2023] presents a data-driven representation of conical constraints within the behavioral systems theory framework, with positive systems typically characterized by behaviors that preserve a cone invariance.

In [Miller et al., 2023], a data-consistency approach is used to solve the problem of data-driven stabilization of linear positive systems, employing the extended Farkas Lemma and achieving a solution through linear programming. Additionally, [Bianchi et al., 2025] introduces data-driven Lyapunov and Metzler inequalities to perform switched positive linear systems control by solving a set of bilinear matrix inequalities. Subspace identification of positive systems is discussed in [Wang and Shafai, 2024a; Wang and Shafai, 2023], and subspace identification of externally positive systems appears in [Grussler et al., 2017]. Kernel-based identification methods for positive systems are explored in [Khosravi and Smith, 2019; Khosravi and Smith, 2025]. Finally, the work in [Wang and Shafai, 2024b] provides both indirect positive stabilization via subspace identification, as well as direct stabilization by solving data-dependent LMIs derived from formulas for data-driven control.

2.3.4 Learning Based on Q-factor

Dynamic programming, introduced in [Bellman, 1957], is a powerful optimization framework used to solve general optimal control problems. It is based on the principle of optimality, which asserts that if the sequence of inputs $\{u_t\}_{t=0}^{\infty}$ is optimal for the original problem starting from an initial state x_0, then any subsequence $\{u_t\}_{t=t_1}^{\infty}$ must also be optimal for the subproblem starting from a state xt_1 for $t_1 > 0$. This decomposition of the original problem into smaller subproblems is a key feature of dynamic programming.

A key result is the Bellman optimality equation, which characterizes the optimal cost function $J^*(x)$ as

$$J^*(x) = \min_{u \in \mathcal{U}(x)} [g(x,u) + J^*(f(x,u))] \quad \forall x \in \mathcal{X}. \tag{2.2}$$

Here, $g(x,u)$ represents the stage cost function, $x_+ = f(x,u)$ describes the system dynamics, $\mathcal{U}(x)$ is the set of input constraints and $\mathcal{X}$ is the state space. The solution to this equation provides the optimal cost incurred starting from any initial state x. Techniques for solving the Bellman equation include value iteration (VI) and policy iteration (PI) [Bertsekas, 2015; Bertsekas, 2012], both of which are guaranteed to converge under mild assumptions, thanks to the monotonicity of the Bellman operator. This approach leads to the optimal cost function $J^*(x)$ and the optimal

stationary policy, which is determined by

$$\mu^*(x) \coloneqq \underset{u\in\mathcal{U}(x)}{\arg\min}[g(x,u)+J^*(f(x,u))].$$

Solving the Bellman equation in (2.2) requires knowledge of the system dynamics $f(x,u)$. In cases where this knowledge is unavailable but state measurements are accessible, the goal is to still obtain the optimal policy. To address this, we define the Q-factor [Sutton and Barto, 2018; Bradtke et al., 1994] as

$$Q(x,u) \coloneqq g(x,u)+J^*(f(x,u)),$$

which leads to

$$J^*(x) = \min_{u\in\mathcal{U}(x)} Q(x,u).$$

Furthermore, the Bellman equation can be rewritten in its model-free form as

$$Q(x,u) = g(x,u) + \min_{u_+\in\mathcal{U}(x)} Q(x_+,u_+). \tag{2.3}$$

The term Q-learning and its associated algorithmic concepts were first introduced in Watkins' thesis [Watkins, 1989]. According to this seminal work, the original and most widely known Q-learning algorithm is a stochastic adaptation of the value iteration (VI) algorithm.

The equation in 2.3 eliminates the direct dependence on the system dynamics $f(x,u)$, replacing it with a dependence on measurements (u,x,x_+). This model-free formulation of the Bellman equation serves as the foundation for numerous reinforcement learning-based control algorithms. Many such approaches, including those presented in [Lee and Hu, 2018; Lopez et al., 2023; Fan and Xiong, 2024; Farjadnasab and Babazadeh, 2022; Rizvi and Lin, 2018a; Kiumarsi et al., 2014; Yaghmaie et al., 2022; Al-Tamimi et al., 2007; Rizvi and Lin, 2018b; Yánez and Souza, 2022; Borghesi et al., 2024; Yaghmaie and Gustafsson, 2019], are offline in nature and generally address deterministic setups or noisy case under stochastic assumptions.

A more recent contribution in [Rantzer, 2024] introduces an online adaptive certainty equivalence policy for addressing the online LQR problem in the presence of non-stochastic disturbances. This approach simultaneously estimates the Q-factor and synthesizes a policy using a data-driven Riccati equation. Similarly, the control schemes proposed in Papers II and III are also built upon adaptation in the Q-factor..

3

Contributions

We briefly introduce each of the three papers included in the thesis and outline the contribution of each.

Paper I

F. Bencherki and A. Rantzer (2023). "Robust simultaneous stabilization via minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 2503–2508

Summary

The paper investigates the existence of stabilizing minimax adaptive controllers for model sets consisting of a finite number of LTI plants, subject to potentially adversarial disturbances. This is accomplished by analyzing the feasibility of nonlinear matrix inequalities to ensure a solution to the minimax Bellman inequality. Using a constructive approach, the paper demonstrates that a robustly stabilizing minimax adaptive controller can be synthesized for the special case of two multiple-input multiple-output (MIMO) plants, provided that their closed-loop stability can be certified with a common quadratic Lyapunov function. This guarantees the simultaneous stabilization of both plants while maintaining a finite gain bound from disturbances to states. Such an approach is especially valuable in controller design when large plant uncertainties are present, where linear controllers often fail to meet the desired performance objectives.

Author Contributions

The main result in Theorem 1 was derived through a joint effort between the two authors. The remaining theoretical results, literature review, and numerical results were produced by F. Bencherki. The manuscript was written in its entirety by F. Bencherki and revised by A. Rantzer.

Paper II

F. Bencherki and A. Rantzer (2024b). *Robust adaptive data-driven control of positive systems with application to learning in ssp problems*. arXiv: 2412.17012 [math.OC]

Summary

Acknowledging its relevance in modeling and addressing a variety of network routing problems, this paper proposes an adaptive solution to the optimal control problem of a class of positive linear systems introduced in [Ohlin et al., 2024]. The proposed solution handles process disturbances that may not necessarily be stochastic, providing a more robust approach. The paper provides a comprehensive treatment of the problem, beginning with a from-scratch formulation of the data-driven adaptive policy and progressing to a detailed online analysis of its performance. While briefly, the paper also highlights similarities and differences with existing literature on data-driven control. However, unlike the widespread focus on offline data-driven control and stochastic setups, this paper takes an online non-stochastic approach. Finally, drawing on the similarity between the considered problem and the stochastic shortest path problem, a numerical case study is presented. It compares the performance of the proposed adaptive policy with existing parameter-free algorithms for learning the Stochastic Shortest Path (SSP).

Author Contributions

The underlying idea of the paper was developed through joint discussions between the authors. The literature survey, main results, and numerical examples were produced by F. Bencherki. The manuscript was written in its entirety by F. Bencherki and revised by A. Rantzer.

Paper III

F. Bencherki and A. Rantzer (2024a). "Data-driven adaptive dispatching policies for processing networks". *IEEE Control Systems Letters* **8**, pp. 2841–2846. DOI: 10.1109/LCSYS.2024.3516637

Summary

As a special case of the framework presented in Paper II, this paper addresses the adaptive optimal dispatching problem for processing networks. Unlike the saturation-based models commonly found in queueing network literature, we propose a linear processing network model that allows for an explicit solution to the

Bellman optimality equation. This linear model is further perturbed by additive non-stochastic process disturbances, which enhances its ability to capture scenarios that the basic linear model cannot, thus enriching its applicability. The proposed data-driven adaptive dispatching policy is built upon the explicit solution of the model-free Bellman equation in the Q-factor. The paper analyzes its online performance by deriving cost suboptimality bounds. These bounds account for the impact of model misspecification due to estimation errors and disturbances, providing insights into their effects on performance.

Author Contributions

The underlying idea of the paper, including the proposed linear processing network model, was formulated through joint discussions between the authors. The literature survey, main results, and numerical examples were produced by F. Bencherki. The manuscript was written in its entirety by F. Bencherki and revised by A. Rantzer.

4

Discussion

4.1 Minimax Adaptive Stabilization

The results presented in Paper I highlight the need for more sophisticated controller architectures when dealing with plant uncertainties that cannot be addressed by fixed-gain controllers. As demonstrated, a minimax adaptive controller—one such example—achieves simultaneous stabilizability of multiple LTI plants by employing a simple certainty-equivalence principle. Another key motivation of this work is to show that complex controllers are not always necessary to achieve challenging design objectives. In fact, simple architectures, like the one considered in Paper I, have shown significant potential, with their simplicity enabling a detailed formal analysis. This work also emphasizes how the pursuit of robustness against adversaries, in the face of uncertain plant parameters, leads to controllers that balance both exploration and exploitation.

4.2 Adaptive Positive Stabilization

The results presented in Paper II and III demonstrate a method for conducting adaptive control for a class of positive systems under model uncertainty. This approach integrates learning mechanisms into the control synthesis process while preserving the structural constraint of positivity. Notably, the adaptation here occurs point-wise in time, unlike some procedures in the literature, which apply adaptation episodically [Bradtke et al., 1994]. The Q-based adaptive controller construction here aligns with the broader adaptive control literature, providing a framework that is not only intuitive but also amenable to thorough analysis and performance quantification. Furthermore, the results emphasize the importance of cases where the Bellman equation, in its model-based form, admits an explicit solution—such as the one considered here or in [Blanchini et al., 2023]. This is valuable because such instances enable the construction of efficient, simple policies with online performance guarantees in a model-free setting.

Paper III builds on the results of Paper II by showing how Q-factor-based adaptation can be applied to linear processing networks. It focuses on solving the problem of optimal routing, where tasks are directed to the best processing unit even in the presence of adversaries. This demonstrates that the methods from Paper II can be used in real-world routing problems, proving the broader usefulness of the presented adaptive optimal control framework for positive systems.

4.3 Robustness

A common theme across the three papers is that the adaptive control schemes considered enhance robustness against potentially non-stochastic disturbances, setting them apart from mainstream literature, which typically assumes disturbances are stochastic—except for a few notable works. These approaches offer an advantage in that they can operate effectively under a wider range of external environmental conditions that influence the process but aren't accounted for in the model. This capability allows them to capture plant nonlinearities and other unmodeled dynamics, making the design both safe and reliable. However, this increased robustness comes with a trade-off in conservatism, which becomes evident when the disturbances follow a specific distribution. In such cases, tailored, instance-specific solutions are recommended, as emphasized in the online convex optimization literature [Agarwal et al., 2019].

4.4 Scalability

One of the primary goals of this thesis is to propose scalable control solutions, which is pursued here through the structured class of positive systems. The results in Paper II and III demonstrate significant computational advantages. In Paper II, an algebraic equation is solved with a complexity that scales linearly with the number of states, in contrast to the quadratic scaling seen in the traditional linear quadratic regulator (LQR) approach. This solution can also be efficiently obtained using linear programming (LP), further enhancing its scalability.

In Paper III, the analysis reveals structure within the data-correlation matrices, leading to diagonal dominance. This property proves to be computationally efficient when performing least-squares estimation of the q-vector over time. It also ensures numerical stability when using numerical methods to solve large systems of equations. This is essential when scaling the number of processing units in practical applications.

Additionally, applying the framework from Paper II to other routing problems could reveal further properties that enhance scalability. This insight motivates further exploration of these extensions, opening up promising avenues for future research.

4.5 Outlook for Future Works

There are several promising directions for future work, some of which are outlined below:

- In Paper I, the guarantees are currently limited to two LTI plants. A key goal is to generalize these guarantees to the case of N LTI pairs (A_i, B_i), as well as to other set structures. Additionally, we aim to provide tighter bounds on the achievable $\mathcal{L}_2$-gains for these more general cases.
- The data-driven adaptive control framework introduced in Paper II could be extended to address the minimax settings discussed in [Gurpegui et al., 2023; Gurpegui et al., 2024], thus broadening the applicability of the proposed method. This extension is relatively straightforward to implement.
- For the adaptive policies implemented in Paper II and III, we have so far used ϵ-greedy explorations, which has been shown to work through numerical inspection. However, a more efficient exploration strategy should be developed, and guarantees for these strategies should be derived.
- The class of positive systems considered in Paper II is equivalent to the stochastic shortest path (SSP) problem in the expected sense. This provides an opportunity to compare the performance of our approach to existing model-free SSP literature and draw inspiration from those methods, particularly regarding exploration strategies. This also opens the door for exchanging ideas and tools between the two domains.
- The positive systems framework from Paper II has the potential to model routing problems. Future work will focus on exploring additional instances of such problems and evaluating the performance of the adaptive policy in these contexts, leveraging the problem structure to improve scalability guarantees.

Bibliography

Abbasi-Yadkori, Y. and C. Szepesvári (2011). "Regret bounds for the adaptive control of linear quadratic systems". In: *Proceedings of the 24th Annual Conference on Learning Theory*. JMLR Workshop and Conference Proceedings, pp. 1–26.

Agarwal, N., B. Bullins, E. Hazan, S. Kakade, and K. Singh (2019). "Online control with adversarial disturbances". In: *International Conference on Machine Learning*. PMLR, pp. 111–119.

Annaswamy, A. M. (2023). "Adaptive control and intersections with reinforcement learning". *Annual Review of Control, Robotics, and Autonomous Systems* **6**:1, pp. 65–93.

Annaswamy, A. M. and A. L. Fradkov (2021). "A historical perspective of adaptive control and learning". *Annual Reviews in Control* **52**, pp. 18–41.

Åström, K. (2021). "History of adaptive control". In: *Encyclopedia of Systems and Control*. Springer, pp. 902–909.

Åström, K. J. and A. Helmersson (1986). "Dual control of an integrator with unknown gain". *Computers & Mathematics with Applications* **12**:6, pp. 653–662.

Åström, K. J. and B. Wittenmark (1973). "On self tuning regulators". *Automatica* **9**:2, pp. 185–199.

Åström, K. and B. Wittenmark (1995). *Adaptive Control (2 ed.)* English. Addison-Wesley. ISBN: 0-201-55866-1. DOI: 10.2307/1269433.

Başar, T. and P. Bernhard (2008). *H-infinity optimal control and related minimax design problems: a dynamic game approach*. Springer Science & Business Media.

Başar, T. and G. J. Olsder (1998). *Dynamic noncooperative game theory*. SIAM.

Bellman, R. (1957). "Dynamic programming, princeton univ". *Press Princeton, New Jersey* **39**.

Bencherki, F. and A. Rantzer (2023). "Robust simultaneous stabilization via minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 2503–2508.

Bencherki, F. and A. Rantzer (2024a). "Data-driven adaptive dispatching policies for processing networks". *IEEE Control Systems Letters* **8**, pp. 2841–2846. DOI: 10.1109/LCSYS.2024.3516637.

Bencherki, F. and A. Rantzer (2024b). *Robust adaptive data-driven control of positive systems with application to learning in ssp problems*. arXiv: 2412.17012 [math.OC].

Berberich, J., A. Koch, C. W. Scherer, and F. Allgöwer (2020). "Robust data-driven state-feedback design". In: *2020 American Control Conference (ACC)*. IEEE, pp. 1532–1538.

Berman, A. and R. J. Plemmons (1994). *Nonnegative matrices in the mathematical sciences*. SIAM.

Bernhardsson, B. (1989). "Dual control of a first-order system with two possible gains". *International Journal of Adaptive Control and Signal Processing* **3**:1, pp. 15–22.

Bertsekas, D. (2012). *Dynamic programming and optimal control: Volume I*. Vol. 4. Athena scientific.

Bertsekas, D. P. (2015). "Value and policy iterations in optimal control and adaptive dynamic programming". *IEEE transactions on neural networks and learning systems* **28**:3, pp. 500–509.

Bianchi, M., S. Grammatico, and J. Cortés (2025). "Data-driven stabilization of switched and constrained linear systems". *Automatica* **171**, p. 111974.

Blanchini, F., P. Bolzern, P. Colaneri, G. De Nicolao, and G. Giordano (2023). "Optimal control of compartmental models: the exact solution". *Automatica* **147**, p. 110680.

Borghesi, M., A. Bosso, and G. Notarstefano (2024). "Mr-arl: model reference adaptive reinforcement learning for robustly stable on-policy data-driven lqr". *arXiv preprint arXiv:2402.14483*.

Bradtke, S. J., B. E. Ydstie, and A. G. Barto (1994). "Adaptive linear quadratic control using policy iteration". In: *Proc. Am. Control Conf.* Vol. 3. IEEE, pp. 3475–3479.

Cao, C. and N. Hovakimyan (2010). "Stability margins of $\mathcal{L}_1$ adaptive control architecture". *IEEE Transactions on Automatic Control* **55**:2, pp. 480–487.

Cederberg, D., A. Hansson, and A. Rantzer (2022). "Synthesis of minimax adaptive controller for a finite set of linear systems". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. IEEE, pp. 1380–1384.

Chen, X. and E. Hazan (2021). "Black-box control for linear dynamical systems". In: *Conference on Learning Theory*. PMLR, pp. 1114–1143.

Chiuso, A. and G. Picci (2004). "The asymptotic variance of subspace estimates". *Journal of Econometrics* **118**:1-2, pp. 257–291.

De Persis, C., M. Rotulo, and P. Tesi (2023). "Learning controllers from data via approximate nonlinearity cancellation". *IEEE Transactions on Automatic Control* **68**:10, pp. 6082–6097.

De Persis, C. and P. Tesi (2019). "Formulas for data-driven control: stabilization, optimality, and robustness". *IEEE Transactions on Automatic Control* **65**:3, pp. 909–924.

Dean, S., H. Mania, N. Matni, B. Recht, and S. Tu (2020). "On the sample complexity of the linear quadratic regulator". *Foundations of Computational Mathematics* **20**:4, pp. 633–679.

Didinsky, G. and T. Basar (1994). "Minimax adaptive control of uncertain plants". In: *Proceedings of 1994 33rd IEEE Conference on Decision and Control*. Vol. 3. IEEE, pp. 2839–2844.

Edholm, C., C. Guiver, R. Rebarber, B. Tenhumberg, and S. Townley (2022). "Stabilization by adaptive feedback control for positive difference equations with applications in pest management". *SIAM Journal on Control and Optimization* **60**:4, pp. 2214–2245.

Egardt, B. (1979). *Stability of adaptive controllers*. Springer.

Fan, W. and J. Xiong (2024). "Q-learning methods for lqr control of completely unknown discrete-time linear systems". *IEEE Transactions on Automation Science and Engineering*.

Farina, L. and S. Rinaldi (2011). *Positive linear systems: theory and applications*. John Wiley & Sons.

Farjadnasab, M. and M. Babazadeh (2022). "Model-free lqr design by q-function learning". *Automatica* **137**, p. 110060.

Feldbaum, A. A. (1960). "Dual control theory. i". *Avtomatika i Telemekhanika* **21**:9, pp. 1240–1249.

Feldbâum,
bibinitperiod (1963). "Dual control theory problems". *IFAC Proceedings Volumes* **1**:2, pp. 541–550.

Filatov, N. M. and H. Unbehauen (2000). "Survey of adaptive dual control methods". *IEE Proceedings-Control Theory and Applications* **147**:1, pp. 118–128.

Filatov, N. M. and H. Unbehauen (2004). *Adaptive dual control: Theory and applications*. Vol. 302. Springer Science & Business Media.

Fliess, M. (2009). "Model-free control and intelligent pid controllers: towards a possible trivialization of nonlinear control?" *IFAC Proceedings Volumes* **42**:10, pp. 1531–1550.

Goodwin, G. C., P. J. Ramadge, and P. E. Caines (1981). "Discrete time stochastic adaptive control". *SIAM Journal on Control and Optimization* **19**:6, pp. 829–853.

Goodwin, G. C. and K. S. Sin (2014). *Adaptive filtering prediction and control*. Courier Corporation.

Grussler, C., J. Umenberger, and I. R. Manchester (2017). "Identification of externally positive systems". In: *2017 IEEE 56th annual conference on decision and control (CDC)*. IEEE, pp. 6549–6554.

Guiver, C., C. Edholm, Y. Jin, M. Mueller, J. Powell, R. Rebarber, B. Tenhumberg, and S. Townley (2016). "Simple adaptive control for positive linear systems with applications to pest management". *SIAM Journal on Applied Mathematics* **76**:1, pp. 238–275.

Guo, L. (1995). "Convergence and logarithm laws of self-tuning regulators". *Automatica* **31**:3, pp. 435–450.

Guo, L., H.-F. Chen, et al. (1991). "The astrom-wittenmark self-tuning regulator revisited and els-based adaptive trackers". *IEEE Transactions on Automatic Control* **36**:7, pp. 802–812.

Gurpegui, A., M. Jeeninga, E. Tegling, and A. Rantzer (2024). "Minimax linear regulator problems for positive systems". *arXiv preprint arXiv:2411.04809*.

Gurpegui, A., E. Tegling, and A. Rantzer (2023). "Minimax linear optimal control of positive systems". *IEEE Control Systems Letters*.

Haddad, W. M., V. Chellaboina, and Q. Hui (2010). *Nonnegative and compartmental dynamical systems*. Princeton University Press.

Hazan, E. et al. (2016). "Introduction to online convex optimization". *Foundations and Trends® in Optimization* **2**:3-4, pp. 157–325.

Hazan, E., S. Kakade, and K. Singh (2020). "The nonstochastic control problem". In: *Algorithmic Learning Theory*. PMLR, pp. 408–421.

Hazan, E. and K. Singh (2022). "Introduction to online nonstochastic control". *arXiv preprint arXiv:2211.09619*.

He, J., I. Ziemann, C. R. Rojas, and H. Hjalmarsson (2024). "Finite sample analysis for a class of subspace identification methods". *arXiv preprint arXiv:2404.17331*.

Hirsch, M. W. and H. Smith (2006). "Monotone dynamical systems". *Handbook of differential equations: ordinary differential equations* **2**, pp. 239–357.

Hjalmarsson, H., M. Gevers, S. Gunnarsson, and O. Lequin (1998). "Iterative feedback tuning: theory and applications". *IEEE control systems magazine* **18**:4, pp. 26–41.

Ioannou, P. A. and P. V. Kokotovic (1984). "Instability analysis and improvement of robustness of adaptive control". *Automatica* **20**:5, pp. 583–594.

Ioannou, P. A. and J. Sun (2012). *Robust adaptive control*. Courier Corporation.

Iwata, T., S.-i. Azuma, R. Ariizumi, and T. Asai (2024). "Data informativity for distributed positive stabilization". *IEEE Control Systems Letters*.

Jedra, Y. and A. Proutiere (2019). "Sample complexity lower bounds for linear system identification". In: *2019 IEEE 58th Conference on Decision and Control (CDC)*. IEEE, pp. 2676–2681.

Jedra, Y. and A. Proutiere (2020). "Finite-time identification of stable linear systems optimality of the least-squares estimator". In: *2020 59th IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 996–1001.

Jedra, Y. and A. Proutiere (2022). "Minimal expected regret in linear quadratic control". In: *International Conference on Artificial Intelligence and Statistics*. PMLR, pp. 10234–10321.

Kaczorek, T. (2012). *Positive 1D and 2D systems*. Springer Science & Business Media.

Kaufman, H., I. Barkana, and K. Sobel (2012). *Direct adaptive control algorithms: theory and applications*. Springer Science & Business Media.

Khosravi, M. and R. S. Smith (2019). "Kernel-based identification of positive systems". In: *2019 IEEE 58th Conference on Decision and Control (CDC)*. IEEE, pp. 1740–1745.

Khosravi, M. and R. S. Smith (2025). "Regularized identification with internal positivity side-information". *SIAM Journal on Control and Optimization* **63**:1, pp. 26–56.

Kiumarsi, B., F. L. Lewis, H. Modares, A. Karimpour, and M.-B. Naghibi-Sistani (2014). "Reinforcement q-learning for optimal tracking control of linear discrete-time systems with unknown dynamics". *Automatica* **50**:4, pp. 1167–1175.

Kjellqvist, O. (2024). "Minimax dual control with finite-dimensional information state". In: *6th Annual Learning for Dynamics & Control Conference*. PMLR, pp. 299–311.

Kjellqvist, O. and A. Rantzer (2022a). "Learning-enabled robust control with noisy measurements". In: *Learning for Dynamics and Control Conference*. PMLR, pp. 86–96.

Kjellqvist, O. and A. Rantzer (2022b). "Minimax adaptive estimation for finite sets of linear systems". In: *2022 American Control Conference (ACC)*. IEEE, pp. 260–265.

Kjellqvist, O. and A. Rantzer (2024). "Output feedback minimax adaptive control". *arXiv preprint arXiv:2409.04115*.

Lee, D. and J. Hu (2018). "Primal-dual q-learning framework for lqr design". *IEEE Transactions on Automatic Control* **64**:9, pp. 3756–3763.

Leith, D. J. and W. E. Leithead (2000). "Survey of gain-scheduling analysis and design". *International journal of control* **73**:11, pp. 1001–1025.

Li, Y. and A. Rantzer (2024). "Exact dynamic programming for positive systems with linear optimal cost". *IEEE Trans. Autom. Control*.

Ljung, L. (1999). "System identification: theory for the user". *PTR Prentice Hall, Upper Saddle River, NJ* **28**, p. 540.

Ljung, L. (1977). "Analysis of recursive stochastic algorithms". *IEEE transactions on automatic control* **22**:4, pp. 551–575.

Ljung, L. and B. Wahlberg (1992). "Asymptotic properties of the least-squares method for estimating transfer functions and disturbance spectra". *Advances in Applied Probability* **24**:2, pp. 412–440.

Lopez, V. G., M. Alsalti, and M. A. Müller (2023). "Efficient off-policy q-learning for data-based discrete-time lqr problems". *IEEE Transactions on Automatic Control* **68**:5, pp. 2922–2933.

Makdah, A. A. A. and F. Pasqualetti (2023). "Model-based and data-based dynamic output feedback for externally positive systems". *arXiv preprint arXiv:2305.02472*.

Markovsky, I. and F. Dörfler (2021). "Behavioral systems theory in data-driven analysis, signal processing, and control". *Annual Reviews in Control* **52**, pp. 42–64.

Matni, N., A. Proutiere, A. Rantzer, and S. Tu (2019). "From self-tuning regulators to reinforcement learning and back again". In: *2019 IEEE 58th Conference on Decision and Control (CDC)*. IEEE, pp. 3724–3740.

Megretski, A. (2004). "A nonlinear dynamical game interpretation of adaptive l2 control: performance limitations and suboptimal controllers". In: *Proceedings of 16th International Symposium on Mathematical Theory of Networks and Systems (MTNS2004), Leuven*.

Megretski, A. and A. Rantzer (2003). *Bounds on the optimal l2 gain in adaptive control of a first order linear system*. Tech. rep. Technical Report ISRN IML.

Meyn, S. (2008). *Control techniques for complex networks*. Cambridge University Press.

Miller, J., T. Dai, M. Sznaier, and B. Shafai (2023). "Data-driven control of positive linear systems using linear programming". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 1588–1594.

Narendra, K. S. and A. M. Annaswamy (2012). *Stable adaptive systems*. Courier Corporation.

Ohlin, D., E. Tegling, and A. Rantzer (2024). "Optimal control of linear cost networks". *Eur. J. Control*, p. 101068.

Padoan, A., F. Dörfler, and J. Lygeros (2023). "Data-driven representations of conical, convex, and affine behaviors". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 596–601.

Pan, Z. and T. Basar (1998). "Adaptive controller design for tracking and disturbance attenuation in parametric strict-feedback nonlinear systems". *IEEE Transactions on Automatic Control* **43**:8, pp. 1066–1083.

Rantzer, A. (2011). "Distributed control of positive systems". In: *2011 50th IEEE conference on decision and control and European control conference*. IEEE, pp. 6608–6611.

Rantzer, A. (2020). "Minimax adaptive control for state matrix with unknown sign". *IFAC-PapersOnLine* **53**:2, pp. 58–62.

Rantzer, A. (2021). "Minimax adaptive control for a finite set of linear systems". In: *Learning for Dynamics and Control*. PMLR, pp. 893–904.

Rantzer, A. (2022). "Explicit solution to bellman equation for positive systems with linear cost". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. IEEE, pp. 6154–6155.

Rantzer, A. (2024). "A data-driven riccati equation". In: *6th Annual Learning for Dynamics & Control Conference*. PMLR, pp. 504–513.

Rantzer, A. and M. E. Valcher (2018). "A tutorial on positive systems and large scale control". In: *2018 IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 3686–3697.

Renganathan, V., A. Iannelli, and A. Rantzer (2023). "An online learning analysis of minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 1034–1039.

Renganathan, V., A. Rantzer, and O. Kjellqvist (2024). "Distributed adaptive control for uncertain networks". In: *2024 European Control Conference (ECC)*. IEEE, pp. 1789–1794.

Revay, M., J. Umenberger, and I. R. Manchester (2021). "Distributed identification of contracting and/or monotone network dynamics". *IEEE Transactions on Automatic Control* **67**:7, pp. 3410–3425.

Rizvi, S. A. A. and Z. Lin (2018a). "Output feedback q-learning control for the discrete-time linear quadratic regulator problem". *IEEE transactions on neural networks and learning systems* **30**:5, pp. 1523–1536.

Rizvi, S. A. A. and Z. Lin (2018b). "Output feedback q-learning for discrete-time linear zero-sum games with application to the h-infinity control". *Automatica* **95**, pp. 213–221.

Shafai, B., A. Moradmand, and M. Siami (2022). "Data-driven positive stabilization of linear systems". In: *2022 8th International Conference on Control, Decision and Information Technologies (CoDIT)*. Vol. 1. IEEE, pp. 1031–1036.

Simchowitz, M. and D. Foster (2020). "Naive exploration is optimal for online lqr". In: *International Conference on Machine Learning*. PMLR, pp. 8937–8948.

Simchowitz, M., K. Singh, and E. Hazan (2020). "Improper learning for non-stochastic control". In: *Conference on Learning Theory*. PMLR, pp. 3320–3436.

Simon, H. A. (1956). "Dynamic programming under uncertainty with a quadratic criterion function". *Econometrica, Journal of the Econometric Society*, pp. 74–81.

Sun, H., Q. Zhu, N. Hovakimyan, and T. Başar (2011). "$\mathcal{L}_1$ adaptive control for positive lti systems". In: *Proceedings of the 2011 American Control Conference*. IEEE, pp. 13–18.

Sutton, R. S. and A. G. Barto (2018). *Reinforcement learning: An introduction*. MIT press.

Al-Tamimi, A., F. L. Lewis, and M. Abu-Khalaf (2007). "Model-free q-learning designs for linear discrete-time zero-sum games with application to h-infinity control". *Automatica* **43**:3, pp. 473–481.

Tsiamis, A., I. Ziemann, N. Matni, and G. J. Pappas (2023). "Statistical learning theory for control: a finite-sample perspective". *IEEE Control Systems Magazine* **43**:6, pp. 67–97.

Tsypkin, Y. Z. and Z. J. Nikolic (1971). *Adaptation and learning in automatic systems*. Vol. 73. Academic Press New York.

Umenberger, J. and I. R. Manchester (2016). "Scalable identification of stable positive systems". In: *2016 IEEE 55th Conference on Decision and Control (CDC)*. IEEE, pp. 4630–4635.

Van Waarde, H. J., J. Eising, M. K. Camlibel, and H. L. Trentelman (2023). "The informativity approach: to data-driven analysis and control". *IEEE Control Systems Magazine* **43**:6, pp. 32–66.

Van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). "Data informativity: a new perspective on data-driven analysis and control". *IEEE Transactions on Automatic Control* **65**:11, pp. 4753–4768.

Vinnicombe, G. (2004). "Examples and counterexamples in finite $\mathcal{L}_2$-gain adaptive control". In: *Leuven: Sixteenth International Symposium on Mathematical Theory of Networks and Systems (MTNS2004)*.

Wang, Y. and B. Shafai (2023). "A subspace system identification method for positive systems". In: *2023 9th International Conference on Control, Decision and Information Technologies (CoDIT)*. IEEE, pp. 608–613.

Wang, Y. and B. Shafai (2024a). "Data driven positive subspace system identification". *IFAC-PapersOnLine* **58**:15, pp. 372–377.

Wang, Y. and B. Shafai (2024b). "Data-driven identification and control of positive systems". In: *Integrated Systems: Data Driven Engineering*. Springer, pp. 289–307.

Watkins, C. J. C. H. (1989). "Learning from delayed rewards".

Willems, J. C., P. Rapisarda, I. Markovsky, and B. L. De Moor (2005). "A note on persistency of excitation". *Systems & Control Letters* **54**:4, pp. 325–329.

Wittenmark, B. (1995). "Adaptive dual control methods: an overview". *Adaptive Systems in Control and Signal Processing 1995*, pp. 67–72.

Yaghmaie, F. A. and F. Gustafsson (2019). "Using reinforcement learning for model-free linear quadratic control with process and measurement noises". In: *2019 IEEE 58th Conference on Decision and Control (CDC)*. IEEE, pp. 6510–6517.

Yaghmaie, F. A., F. Gustafsson, and L. Ljung (2022). "Linear quadratic control using model-free reinforcement learning". *IEEE Transactions on Automatic Control* **68**:2, pp. 737–752.

Yánez, W. J. L. and F. d. C. de Souza (2022). "On the effect of probing noise in optimal control lqr via q-learning using adaptive filtering algorithms". *European Journal of Control* **65**, p. 100633.

Zhang, J., L. Zhang, and T. Raissi (2020). "A linear framework on the distributed model predictive control of positive systems". *Systems & Control Letters* **138**, p. 104665.

Zhang, J., S. Zhang, X. Deng, and Z. Huang (2023). "Adaptive event-triggered dynamic distributed control of switched positive systems with switching faults". *Nonlinear Analysis: Hybrid Systems* **48**, p. 101328.

Zhou, K., J. Doyle, and K. Glover (1996). *Robust and Optimal Control*. Feher/Prentice Hall Digital and. Prentice Hall. ISBN: 9780134565675. URL: `https://books.google.se/books?id=RPSOQgAACAAJ`.

Zhou, K. and J. C. Doyle (1998). *Essentials of robust control*. Vol. 104. Prentice hall Upper Saddle River, NJ.

Paper I

Robust Simultaneous Stabilization via Minimax Adaptive Control

Fethi Bencherki Anders Rantzer

Abstract

The paper explores the usage of minimax adaptive controllers to guarantee finite $\mathcal{L}_2$-gain simultaneous stabilization of linear time-invariant (LTI) plants. It is shown that a minimax adaptive controller simultaneously stabilizes any two multiple-input multiple-output (MIMO) P-stabilizable LTI plants when no LTI controller can achieve that, and the worst attained $\mathcal{L}_2$-gain bound for the transient dynamics is readily computable.

©2023 IEEE. Reprinted, with permission, from F. Bencherki and A. Rantzer (2023). "Robust simultaneous stabilization via minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 2503–2508

1. Introduction

Recently, there has been a surge of research interest at the intersection of machine learning, system identification and adaptive control; see, for example, [Matni et al., 2019] for a review. Most work concerns the stochastic setting, but recently works relating to worst-case disturbances started to appear, see [Simchowitz, 2020; Agarwal et al., 2019]. Based on a game theoretical formulation of the $\mathcal{H}_\infty$ optimal control introduced in [Başar and Bernhard, 2008], the minimax adaptive control approach was presented in [Didinsky and Basar, 1994; Rantzer, 2021]. It focuses on worst-case models of disturbances and uncertain parameters and assumes no prior knowledge of a stabilizing controller. The exploration-exploitation strategy exhibited by these adaptive controllers ensure stability when faced with large uncertainties that can not be tackled using robust LTI controllers.

The relevance of the problem of simultaneous stabilization of a finite set of plants could be justified in the sense that besides stabilizing a nominal plant of the underlying system, it is also desirable to stabilize discrete perturbations of this nominal plant induced by structural changes or component failure [Stoustrup and Blondel, 2004]. It could also be motivated within the scope of nonlinear systems where the concern is to design controllers around several operating points of interest [Jia and Ackermann, 2001].

The problem of simultaneous stabilization of linear time-invariant plants has been studied extensively in the literature. It was considered in [Vidyasagar and Viswanadham, 1982; Saeks and Murray, 1982; Ghosh and Byrnes, 1983; Jia and Ackermann, 2001] with the restriction that the controller is linear time-invariant. However, as discussed in [Vinnicombe, 2004], there are counterexamples where an LTI controller fails to stabilize even a pair of single-input single-output systems simultaneously. This then calls for the need to employ a more complex controller architecture in attempting a solution to this problem.

The works in [Khargonekar et al., 1985; Khargonekar et al., 1988b; Khargonekar et al., 1988a] show that using periodic time-varying compensators, it is possible to $\mathcal{L}_2$ stabilize any finite collection of discrete-time LTI plants. The same conclusion was reached in [OLBROT, 1987]. The use of periodic time-varying controllers for the simultaneous placement of the closed-loop poles of N SISO LTI plants was also reported in [Das and Kar, 2003]. On the other hand, the work in [French and Trenn, 2005] explores using switching deadbeat controllers to stabilize a finite set of scalar systems. This was generalized to the MIMO setting in [Buchstaller and French, 2008] under the condition that the systems are controllable. However, one difficulty of these works is that extracting $\mathcal{L}_2$-gain bounds, which reflects on the performance of these controllers, is cumbersome and complicated, especially for MIMO models.

The current work concerns using minimax adaptive controllers to provide guarantees on finite $\mathcal{L}_2$-gain stabilization of LTI plants, possibly MIMO, in the presence of adversarial disturbances. In doing that, We relax the controllability assumption

needed in [Buchstaller and French, 2008] to only stabilizability. We also provide an explicit easily computable $\mathcal{L}_2$-gain bound achieved by the adaptive controller.

The outline of the paper is as follows: Section 2 presents the framework of minimax adaptive control synthesis and formulates the problem; section 3, on the other hand, establishes the main results of the paper. Section 4 is dedicated for numerical examples and section 5 concludes the paper.

1.1 Notation

The set of $n \times m$ matrices with real coefficients is denoted by $\mathbb{R}^{n\times m}$. For a symmetric matrix $A \in \mathbb{R}^{n\times n}$, we indicate that A is positive definite by writing $A \succ 0$ and positive semi-definite by writing $A \succeq 0$. I_n denotes the identity matrix of dimension $n \times n$. Given $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$, the notation $|x|_A^2$ means $x^T Ax$. Given a positive definite matrix $P \prec \gamma^2 I_n$ and a scalar $\gamma > 0$, we define the positive definite matrix $S_{P,\gamma^2} \coloneqq (P^{-1} - \gamma^{-2} I_n)^{-1}$. We let $\|A\|$ denote the spectral radius of a symmetric matrix $A \in \mathbb{R}^{n\times n}$, i.e., $\|A\| \coloneqq \max_j |\lambda_j(A)|$, where $\lambda_j(A)$ $j = 1, \ldots, n$ are the eigenvalues of A.

2. Problem Formulation

Minimax adaptive control synthesis was presented in [Rantzer, 2021]. Given a compact set $\mathcal{M} \subset \mathbb{R}^{n\times n} \times \mathbb{R}^{n\times m}$, an initial state $x_0 \in \mathbb{R}^n$, and a scalar $\gamma > 0$, we seek a solution to the following optimization problem

$$\begin{aligned} \inf_{\mu} \sup_{w,A,B,T} & \sum_{t=0}^{T} (|x_t|_Q^2 + |u_t|_R^2 - \gamma^2 |w_t|^2) \\ \text{s.t. } & x_{t+1} = Ax_t + Bu_t + w_t, \quad t \geq 0 \\ & u_t = \mu_t(x_0, \ldots, x_t, u_0 \ldots u_{t-1}), \\ & (A, B) \in \mathcal{M}. \end{aligned} \tag{1}$$

The problem is a zero sum dynamic game where the μ-player minimizes the cost and the (w, A, B)-player maximizes it. Note that the adversary not only is allowed to choose the disturbance w but also the model pair (A, B) [Başar and Bernhard, 2008].

The following result from [Rantzer, 2021] presents an explicit solution to (1) in terms of an adaptive controller that guarantees a pre-specified $\mathcal{L}_2$-gain bound from disturbances to errors for model sets $\mathcal{M}$ of the type $\mathcal{M} \coloneqq \{(A_1, B_1), \ldots, (A_N, B_N)\}$.

PROPOSITION 1
Given $A_1, \ldots, A_s \in \mathbb{R}^{n\times n}$, $B_1, \ldots, B_N \in \mathbb{R}^{n\times m}$ and positive definite matrices $Q \in \mathbb{R}^{n\times n}$, $R \in \mathbb{R}^{m\times m}$, suppose there exist $K_1, \ldots, K_N \in \mathbb{R}^{m\times n}$ and $P_{ij} \in \mathbb{R}^{n\times n}$ with $0 \prec$

$P_{ij} = P_{ji} \prec \gamma^2 I$ *and*

$$|x|^2_{P_{jk}} \geq |x|^2_Q + |K_k x|^2_R + \left|(A_i - B_i K_k + A_j - B_j K_k)x/2\right|^2_{S_{P_{ij},\gamma^2}} - \gamma^2 \left|(A_i - B_i K_k - A_j + B_j K_k)x/2\right|^2 \tag{2}$$

for $x \in \mathbb{R}^n$ *and* $i, j, k \in \{1, \ldots, N\}$*, excluding the case* $i \neq j = k$*. Then, the input-output gain is bounded by* γ *for the adaptive control law given by*

$$\begin{aligned} u_t &= -K_{k_t} x_t, \\ k_t &= \arg\min_i \sum\nolimits_{\tau=1}^{t-1} |A_i x_\tau + B_i u_\tau - x_{\tau+1}|^2 . \end{aligned} \tag{3}$$

The gain bound γ provides robustness guarantees to unmodelled (possibly nonlinear and infinite-dimensional) dynamics following the small gain theorem [Zhou and Doyle, 1998]. Therefore, in solving (2) for γ, K_i, and P_{ij} for $i, j \in \{1, \cdots, N\}$, one wishes to acquire as small as possible values of γ. We specialize Proposition 1 to the case $\mathcal{M} \coloneqq \{(A_1, B_1), (A_2, B_2)\}$, i.e., the simultaneous stabilization of two LTI plants, possibly MIMO.

COROLLARY 1
Consider the model set $\mathcal{M} \coloneqq \{(A_1, B_1), (A_2, B_2)\}$*. It suffices for the existence of the adaptive minimax controller that the following set of inequalities hold*

$$\begin{aligned} P_i &\succeq Q + K_i^T R K_i + (A_i - B_i K_i)^T S_{P_i,\gamma^2}(A_i - B_i K_i), \\ T &\succeq Q + K_k^T R K_k + (A_i - B_i K_k)^T S_{P_i,\gamma^2}(A_i - B_i K_k), \\ T &\succeq Q + K_k^T R K_k + \frac{1}{4}|(A_k - B_k K_k + A_i - B_i K_k)|^2_{S_{T,\gamma^2}} \\ &\quad - \frac{\gamma^2}{4}|(A_k - B_k K_k - A_i + B_i K_k)|^2, \end{aligned}$$

where $i, k \in \{1, 2\}$ *and* $i \neq k$*, and the quadruplet* $\left(P_1, P_2, T, \gamma^2\right)$ *adheres to the construction* $0 \prec P_i \prec T$ *and* $0 \prec T \prec \gamma^2 I_n$*.*

Proof. The first inequality is obtained by selecting $i = j = k$ in (2). The second one, on the other hand, is obtained by selecting $i = j \neq k$ and defining $T \coloneqq P_{12} = P_{21}$. Lastly, to obtain the third inequality, we select $i = k \neq j$. By consequence, we have demonstrated that the set of matrix inequalities in (2) collapses to the matrix inequalities mentioned in the statement of the Corollary when $\mathcal{M} \coloneqq \{(A_1, B_1), (A_2, B_2)\}$. ∎

The natural question that arises then is if the set of inequalities in (2) are feasible for any $\mathcal{M}$ of the type $\mathcal{M} \coloneqq \{(A_1, B_1), (A_2, B_2)\}$ with a finite $\mathcal{L}_2$-gain γ under the condition that each pair is stabilizable. This question is the center of this paper. We capture this question in the following problem.

PROBLEM 1
Suppose the two pairs $(A_1, B_1), (A_2, B_2)$ *are stabilizable. Show that a minimax adaptive controller stabilizes the two pairs in the set simultaneously.*

The main results of the paper are presented in the sequel.

3. Main Results

3.1 Guarantees on Stabilization of two P-Stabilizable MIMO LTI Plants

We first introduce the notion of P-stabilization.

DEFINITION 1
Call the model set $\mathcal{M} \coloneqq \{(A_1, B_1), (A_2, B_2)\}$ *P-stabilizable if* $\exists P \succ 0$ *and* $\gamma > 0$ *such that the following holds*

$$P \succeq Q + K_i^T R K_i + (A_i - B_i K_i)^T S_{P,\gamma^2}(A_i - B_i K_i),$$

for $i \in \{1, 2\}$*, for some stabilizing controllers* K_1 *and* K_2*.*

REMARK 3.1
Two systems are P-stabilizable if they accept the same solution $P \succ 0$ *to the* $\mathcal{H}_\infty$ *discrete algebraic Riccati inequality (DARI). Another way to look at it is that they are stabilizable with the same closed loop quadratic Lyapunov function* $V(x) = x^T P x$*.*

EXAMPLE 1
Consider two plants $(A_1, B_1), (A_2, B_2)$ *in the controllability canonical form. It is then possible to place the closed loop poles of the two plants inside the unit circle such that it holds that* $A_1 - B_1 K_1 = A_2 - B_2 K_2$ *for some* K_1 *and* K_2*; hence,* $\exists P \succ 0$ *common among the two plants for some large enough* $\gamma > 0$ *and therefore they are P-stabilizable.*

The main result of the paper is stated next.

THEOREM 1
Suppose (A_1, B_1) *and* (A_2, B_2) *are P-stabilizable. Then, there exists a minimax adaptive controller that stabilizes both plants simultaneously in the* $\mathcal{L}_2$*-gain sense.*

Proof. The proof is constructive and constitutes of showing the feasibility of the set of inequalities given in Corollary 1 for all P-stabilizable pairs $(A_1, B_1), (A_2, B_2)$ for the choices $P_1 = P_2 = P$ and $T \coloneqq \alpha P$ for some $\alpha > 1$, i.e., showing the feasibility

of

$$P \succeq Q + K_i^T R K_i + (A_i - B_i K_i)^T S_{P,\gamma^2}(A_i - B_i K_i) \tag{4}$$

$$\alpha P \succeq Q + K_k^T R K_k + (A_i - B_i K_k)^T S_{P,\gamma^2}(A_i - B_i K_k) \tag{5}$$

$$\begin{aligned}\alpha P \succeq Q + K_k^T R K_k + \frac{1}{4}|(A_k - B_k K_k + A_i - B_i K_k)|^2_{S_{\alpha P,\gamma^2}} \\ - \frac{\gamma^2}{4}|(A_k - B_k K_k - A_i + B_i K_k)|^2\end{aligned} \tag{6}$$

Where $i, k \in \{1, 2\}$, $i \neq k$, and the triplet (P, T, γ^2) is selected such that $0 \prec P \prec T$, and $0 \prec T \prec \gamma^2 I_n$. The two pairs being P-stabilizable means that (4) accepts a feasible solution pair (P, γ) for $i \in \{1, 2\}$. Fix γ and select α large enough such that (5) holds, i.e., the following holds

$$\alpha P \succeq Q + K_k^T R K_k + (A_i - B_i K_k)^T S_{P,\gamma^2}(A_i - B_i K_k), \tag{7}$$

where $i \neq k$. It remains to show the feasibility of (6) for some $\bar{\gamma} > \gamma$. Define the matrix variables

$$\begin{aligned} M_k &\coloneqq A_k - B_k K_k, \\ \Delta_{ik} &\coloneqq \frac{(A_i - B_i K_k) - (A_k - B_k K_k)}{2}, \end{aligned}$$

then (6) could be rewritten succinctly as

$$\alpha P \succeq Q + K_k^T R K_k + (M_k + \Delta_{ik})^T S_{\alpha P,\bar{\gamma}^2}(M_k + \Delta_{ik}) - \bar{\gamma}^2 \Delta_{ik}^T \Delta_{ik}. \tag{8}$$

Assuming that (4) holds, multiplying by α the following also holds

$$T = \alpha P \succeq Q + K_k^T R K_k + M_k^T S_{\alpha P,\alpha\gamma^2} M_k.$$

Then, (8) holds if the following holds

$$(M_k + \Delta_{ik})^T S_{\alpha P,\bar{\gamma}^2}(M_k + \Delta_{ik}) \preceq \bar{\gamma}^2 \Delta_{ik}^T \Delta_{ik} + M_k^T S_{\alpha P,\alpha\gamma^2} M_k. \tag{9}$$

Expanding the left hand side of (9) yields

$$\begin{aligned}(M_k + \Delta_{ik})^T S_{\alpha P,\bar{\gamma}^2}(M_k + \Delta_{ik}) = M_k^T S_{\alpha P,\bar{\gamma}^2} M_k \\ + \Delta_{ik}^T S_{\alpha P,\bar{\gamma}^2} \Delta_{ik} + M_k^T S_{\alpha P,\bar{\gamma}^2} \Delta_{ik} + \Delta_{ik}^T S_{\alpha P,\bar{\gamma}^2} M_k.\end{aligned} \tag{10}$$

Using Lemma 1 at the Appendix-A by taking $E = M_k$, $F = \Delta_{ik}$, $M = S_{\alpha P,\bar{\gamma}^2}$, and a non-zero scalar β yields the inequality

$$M_k^T S_{\alpha P,\bar{\gamma}^2} \Delta_{ik} + \Delta_{ik}^T S_{\alpha P,\bar{\gamma}^2} M_k \preceq \beta^2 M_k^T S_{\alpha P,\bar{\gamma}^2} M_k + \beta^{-2} \Delta_{ik}^T S_{\alpha P,\bar{\gamma}^2} \Delta_{ik}. \tag{11}$$

Combining (9-11) we obtain the sufficient condition

$$M_k^T S_{\alpha P,\bar{\gamma}^2} M_k + \Delta_{ik}^T S_{\alpha P,\bar{\gamma}^2} \Delta_{ik} + \beta^2 M_k^T S_{\alpha P,\bar{\gamma}^2} M_k + \beta^{-2} \Delta_{ik}^T S_{\alpha P,\bar{\gamma}^2} \Delta_{ik} \preceq \bar{\gamma}^2 \Delta_{ik}^T \Delta_{ik} + M_k^T S_{\alpha P,\alpha\gamma^2} M_k.$$

Re-arranging gives

$$M_k^T \left(S_{\alpha P,\alpha\gamma^2} - \left(1+\beta^2\right) S_{\alpha P,\bar{\gamma}^2} \right) M_k + \Delta_{ik}^T \left(\bar{\gamma}^2 I_n - \left(1+\beta^{-2}\right) S_{\alpha P,\bar{\gamma}^2} \right) \Delta_{ik} \succeq 0.$$

Hence, it is sufficient to show the existence of a $\bar{\gamma}^2 > \alpha\gamma^2$ such that the next two inequalities hold

$$S_{\alpha P,\alpha\gamma^2} - \left(1+\beta^2\right) S_{\alpha P,\bar{\gamma}^2} \succeq 0, \tag{12}$$

$$\bar{\gamma}^2 I_n - \left(1+\beta^{-2}\right) S_{\alpha P,\bar{\gamma}^2} \succeq 0. \tag{13}$$

(12-13) could be satisfied by selecting a small enough β and large enough $\bar{\gamma}$ as will be demonstrated in the next section.

To conclude, we have shown that (4-6) could be made feasible for any two P-stabilizable pairs (A_1, B_1), (A_2, B_2) and therefore the theorem is proven. ■

3.2 Explicit Value of the Achievable $\mathcal{L}_2$-Gain Bound

We provide an explicit value for $\bar{\gamma}$ that the adaptive controller achieves. This is done by solving the inequalities in (12-13). This bound is provided in the next Theorem.

THEOREM 2
Let (A_1, B_1), (A_2, B_2) be stabilizable with a common tuple (P, γ). Then, a minimax adaptive controller stabilizes both plants with an $\mathcal{L}_2$-gain bound given by

$$\bar{\gamma}^2 = \alpha(2+\beta^{-2})\|P\|,$$

where

$$\beta^2 := -\frac{\|P\|^{-1} + \|P^{-1}\| - 3\gamma^{-2}}{4\left(\|P^{-1}\| - \gamma^{-2}\right)} + \frac{1}{4}\sqrt{\Lambda},$$

$$\Lambda := \left(\frac{\|P\|^{-1} + \|P^{-1}\| - 3\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}} \right)^2 + \frac{8\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}},$$

and $\alpha > 1$ is selected such that

$$\alpha P \succeq Q + K_k^T R K_k + (A_i - B_i K_k)^T S_{P,\gamma^2} (A_i - B_i K_k),$$

where $i, k \in \{1, 2\}$, $i \neq k$.

Proof. Consider the condition in (13). Written explicitly

$$\bar{\gamma}^2 I_n - \left(1+\beta^{-2}\right)\left((\alpha P)^{-1} - \bar{\gamma}^{-2} I_n\right)^{-1} \succeq 0.$$

Solving for $\bar{\gamma}$ yields

$$\bar{\gamma}^2 I_n \succeq \alpha(2+\beta^{-2})P.$$

Therefore, the lowest achievable $\bar{\gamma}^2$ is

$$\bar{\gamma}^2 = \alpha(2+\beta^{-2})\|P\|. \tag{14}$$

To get a feasible value for β we plug (14) in (12). This leads to

$$\left((\alpha P)^{-1} - \left(\alpha\gamma^2\right)^{-1} I_n\right)^{-1} - \left(1+\beta^2\right)\left((\alpha P)^{-1} - \left(\alpha\left(1+\beta^{-2}\right)\|P\|\right)^{-1} I_n\right)^{-1} \succeq 0,$$

canceling out the α term and simplifying further yields

$$\left(\left(3\beta^2+2\beta^4+1\right)\gamma^{-2} - \beta^2\|P\|^{-1}\right) I_n \succeq \left(\beta^2+2\beta^4\right) P^{-1}.$$

Therefore, β has to satisfy

$$\left(3\beta^2+2\beta^4+1\right)\gamma^{-2} - \beta^2\|P\|^{-1} \succeq \left(\beta^2+2\beta^4\right)\|P^{-1}\|.$$

Rearranging yields the scalar inequality in β

$$2\beta^4 + \frac{\|P\|^{-1} + \|P^{-1}\| - 3\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}}\beta^2 - \frac{\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}} \geq 0.$$

Since we are interested in the smallest value of β, we solve for the equality instead

$$2\beta^4 + \frac{\|P\|^{-1} + \|P^{-1}\| - 3\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}}\beta^2 - \frac{\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}} = 0.$$

First, we must make sure the existence of a $\beta^2 \in \mathbb{R}$ such that the equation holds. Secondly, such β^2 has to be positive. The first condition is met since the discriminant, which we denote by Λ, is positive, that is

$$\Lambda = \left(\frac{\|P\|^{-1} + \|P^{-1}\| - 3\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}}\right)^2 + \frac{8\gamma^{-2}}{\|P^{-1}\| - \gamma^{-2}} > 0.$$

This is the case since $\|P^{-1}\| - \gamma^{-2} > 0$. This can be concluded since by construction $\lambda_{\min}(P) \leq \lambda_{\max}(P) < \gamma^2 \iff \|P^{-1}\| = \frac{1}{\lambda_{\min}(P)} > \gamma^{-2} \iff \|P^{-1}\| - \gamma^{-2} > 0$. The next step is to show that one of the solutions

$$\beta_{1,2}^2 = -\frac{\|P\|^{-1} + \|P^{-1}\| - 3\gamma^{-2}}{4\left(\|P^{-1}\| - \gamma^{-2}\right)} \pm \frac{1}{4}\sqrt{\Lambda}$$

is positive. β_2^2 is obviously rejected since it yields a negative solution. However, β_1^2 is clearly admissible since $\frac{1}{4}\sqrt{\Lambda} > \frac{\|P\|^{-1}+\|P^{-1}\|-3\gamma^{-2}}{4\left(\|P^{-1}\|-\gamma^{-2}\right)}$. Therefore we take

$$\beta^2 = -\frac{\|P\|^{-1}+\|P^{-1}\|-3\gamma^{-2}}{4\left(\|P^{-1}\|-\gamma^{-2}\right)} + \frac{1}{4}\sqrt{\Lambda}.$$

To get the lowest achievable $\bar{\gamma}$, we plug this value of β in (14). This concludes the proof. ∎

3.3 A Specialized $\bar{\gamma}$ Bound for the Single-Input Single-Output (SISO) Case

in the SISO case we have $\|P\|^{-1} = \|P^{-1}\| = P^{-1}$, and therefore the discriminant from before simplifies to

$$\Lambda = \left(\frac{2P^{-1}-3\gamma^{-2}}{P^{-1}-\gamma^{-2}}\right)^2 + \frac{8\gamma^{-2}}{P^{-1}-\gamma^{-2}} = \left(\frac{2P^{-1}-\gamma^{-2}}{P^{-1}-\gamma^{-2}}\right)^2,$$

and β^2 simplifies to

$$\beta^2 = \frac{\gamma^{-2}}{2\left(P^{-1}-\gamma^{-2}\right)} = \frac{P}{2\left(\gamma^2-P\right)} > 0,$$

and the lowest achievable $\bar{\gamma}^2$ is given by

$$\bar{\gamma}^2 = \alpha(2+\beta^{-2})P = \alpha P\left(2+\frac{2\left(\gamma^2-P\right)}{P}\right) \iff \bar{\gamma} = 2\alpha\gamma^2.$$

4. Numerical Examples

4.1 Systems with Unknown Input Direction

Consider the unstable SISO system with unknown input direction

$$x_{t+1} = ax_t \pm u_k + w_t, \quad a > 1.$$

We attempt to stabilize the system and extract an $\mathcal{L}_2$-gain bound we call it $\bar{\gamma}$ and compare it to the optimal bound given in [Vinnicombe, 2004]. Stated in our context, we are trying to simultaneously stabilize the two pairs $(a, 1)$ and $(a, -1)$. To establish ground for comparison, we follow [Vinnicombe, 2004] by taking $Q = 1$ and $R = 0$, which corresponds to bounding the $\mathcal{L}_2$-gain from w to x.

We start by finding a pair (P, γ) that commonly solves the $\mathcal{H}_\infty$ DARI for both systems. We pick $K_1 = -K_2 = K$ where K is the $\mathcal{H}_\infty$ controller given by

$$K = \frac{BS_{P,\gamma^2}}{R + B^2 S_{P,\gamma^2}} A = a,$$

which corresponds to a deadbeat controller. Plugging $K = a$ in the $\mathcal{H}_\infty$ DARI yields the solution $P = Q = 1$ and the condition $\gamma^2 > P = 1 \iff \gamma > 1$. Let's consider the pair $(1, \gamma)$ to be a feasible pair, where $\gamma > 1$. Next, we select α such that it holds

$$T \coloneqq \alpha P = \alpha = 1 + \frac{4a^2}{1-\gamma^{-2}} = 1 + \frac{4a^2\gamma^2}{\gamma^2 - 1}.$$

Last step would be to solve the inequalities (12-13) for a feasible tuple $(\bar{\gamma}, \beta)$. From section (3.3) and having $P = 1$, the smallest achievable $\bar{\gamma}$ is given by

$$\bar{\gamma} = \sqrt{2\gamma^2 + \frac{8\gamma^4 a^2}{\gamma^2 - 1}}, \quad \gamma > 1.$$

In [Vinnicombe, 2004], it is conjectured that no controller can achieve a better $\mathcal{L}_2$-gain bound $a + \sqrt{a^2 + 1}$. We compare our obtained bound to this optimal bound for different values of a and for a fixed value of $\gamma = 1.4$; this comparison is given in figure 1. We justify the non-tightness by the fact that unlike [Vinnicombe, 2004], our bound account for a much larger uncertainty set as compared to that of only unknown input sign, and also works in the case $R \neq 0$, and therefore when applied to special cases such as this it tends to be conservative. However, we believe that the techniques used in the constructive proof could be refined to produce much tighter bounds when the model set $\mathcal{M}$ is more accurately defined a priori.

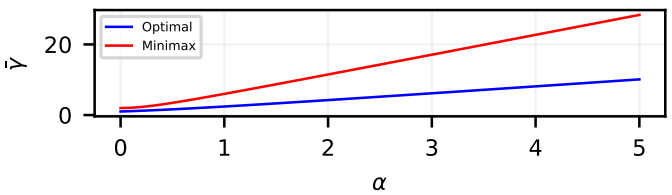


Figure 1. A comparison between The optimal $\mathcal{L}_2$-gain bound and the one achieved by the minimax controller for the system given in example 4.1.

4.2 Inverted Pendulum with Uncertain Actuator Gain

Consider the unstable state-space model corresponding to an inverted pendulum with uncertain input matrix

$$x_{t+1} = \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix} x_t + \begin{bmatrix} b \\ 0 \end{bmatrix} u_t + w_t,$$

where b takes two values, $b_1 = 1$ or $b_2 = 3$. By taking $K = [k_1 \; k_2]$, it could be checked by an application of Jury's test for stability [Ogata, 1995] that such system

can not be stabilized by a static linear state feedback law. To see this, we plug in K and obtain the following characteristic polynomial in the variable z

$$D(z) = z^2 + z(bk_1 - 3) + bk_2 + 1.$$

We attempt to ensure that the polynomial $D(z)$ has roots inside the unit circle through the design variables k_1 and k_2. Following Jury's Stability test, the following necessary conditions must first hold

(i) $D(1) > 0$;

(ii) $D(-1) > 0$;

(iii) $|a_0| < a_2$;

where $D(z)$ is considered in the form $D(z) = a_2 z^2 + a_1 z + a_0$. These conditions translate to the following

(i) $b(k_1 + k_2) > 1$;

(ii) $b(k_2 - k_1) > -5$;

(iii) $|bk_2 + 1| < 1$;

solving the inequalities for $b = 1$ and $b = 3$ yields the solutions given in table 1.

Table 1

$b = 1$	$b = 3$
$3 < k_1 < 5$	$1 < k_1 < \frac{5}{3}$
$-2 < k_2 < 0$	$-\frac{2}{3} < k_2 < 0$

Notice that there exist no value of k_1 for which we could place the poles of the characteristic polynomial inside the unit circle for both values of b. This asserts that our design objectives are not met using a static state feedback control law.

The goal now is to find an adaptive feedback law that stabilizes the system for both values of b. We select $Q = I$ and $R = 1$. We first find a feasible pair (P, γ) that solves the DARI in (4) for both pairs. We get

$$P = \begin{bmatrix} 9.62 & -2.68 \\ -2.68 & 2 \end{bmatrix} \quad \text{and} \quad \gamma = 10.$$

Next, we fix $T \coloneqq \alpha P$ where α is found as in (7); we get $\alpha = 35$ and therefore

$$T = \begin{bmatrix} 336.75 & -93.91 \\ -93.91 & 70 \end{bmatrix};$$

this choice of T renders (5) feasible. Last step is to find a $\bar{\gamma} > \gamma$ that will make (6) feasible for the same choice of T. This problem breaks down to finding a pair $(\bar{\gamma}, \beta)$ that renders (12-13) feasible. One way is to follow the explicit solutions given in theorem 2. Another alternative is to bisect over $\bar{\gamma}$ until (6) is made feasible. Following the second approach yields $\bar{\gamma} = 27$.

We run the adaptive control law given in (3) on this system in the presence of a stochastic disturbance w under the assumption that the true system corresponds to an input matrix with $b = b_2 = 3$. We compare the performance of the minimax adaptive controller to the optimal $\mathcal{H}_\infty$ controller which knows the actuator gain, and a $2n$-periodic switching deadbeat controller. The 4-periodic switching deadbeat controller is given by

$$\bar{u}_t \coloneqq \begin{cases} -\bar{K}_1 x_t, & t \bmod 4 = 0 \\ -\bar{K}_1 x_t, & t \bmod 4 = 1 \\ -\bar{K}_2 x_t, & t \bmod 4 = 2 \\ -\bar{K}_2 x_t, & t \bmod 4 = 3 \end{cases}$$

where $\bar{K}_1$ and $\bar{K}_2$ are the deadbeat controller gains corresponding to the two systems respectively and the mod operator is the remainder of division. The results are presented in figure 2. As can be seen, the minimax adaptive controller exhibits a nearly optimal performance and outperforms the switching deadbeat controller. The minimax controller increases the control activity in the beginning for the sake of exploration. Once the true dyanmics is learned, it behaves similar to the $\mathcal{H}_\infty$ controller.

4.3 Performance Comparison to Periodic Deadbeat Controllers

We compare the $\mathcal{L}_2$-gain achieved by the adaptive controller to that of the switching deadbeat controller and the optimal $\mathcal{H}_\infty$ controller for the system in 4.2. To obtain a tighter bound on the $\mathcal{L}_2$ of that of the minimax controller we resort to the work in [Cederberg et al., 2022] which treats the synthesis of minimax adaptive controllers. The approach presented in [Cederberg et al., 2022] proved to deliver tighter lower bounds on the $\mathcal{L}_2$-gain. To obtain the $\mathcal{L}_2$-gain of the system driven by the 4-periodic switching deadbeat controller, note that in such case the closed loop system is periodic with period $2n = 4$. This suggest representing it via an LTI system according to [Khargonekar et al., 1985]. To this end, we define the new state and disturbance sequence

$$X_t \coloneqq \begin{bmatrix} x_{4t} \\ \vdots \\ x_{4t+3} \end{bmatrix}, \eta_t \coloneqq \begin{bmatrix} w_{4t} \\ \vdots \\ w_{4t+6} \end{bmatrix},$$

this yields the LTI representation of our periodic systems given by

$$X_{t+1} = A_p X_t + B_p \eta_t,$$

where

$$A_p \coloneqq \begin{bmatrix} \bar{A}_{22}^2 \bar{A}_{21}^2 & 0 & 0 & 0 \\ 0 & \bar{A}_{21} \bar{A}_{22}^2 \bar{A}_{21} & 0 & 0 \\ 0 & 0 & \bar{A}_{21}^2 \bar{A}_{22}^2 & 0 \\ 0 & 0 & 0 & \bar{A}_{22} \bar{A}_{21}^2 \bar{A}_{22} \end{bmatrix},$$

$$B_p \coloneqq \begin{bmatrix} \bar{A}_{22}^2 \bar{A}_{21} & A_{22}^2 & A_{22} & I_n & 0 & 0 & 0 \\ 0 & \bar{A}_{21} A_{22}^2 & \bar{A}_{21} \bar{A}_{22} & \bar{A}_{21} & I_n & 0 & 0 \\ 0 & 0 & \bar{A}_{21}^2 \bar{A}_{21} & \bar{A}_{21}^2 & \bar{A}_{21} & I_n & 0 \\ 0 & 0 & 0 & \bar{A}_{22} \bar{A}_{21}^2 & \bar{A}_{22} \bar{A}_{21} & \bar{A}_{22} & I_n \end{bmatrix},$$

and

$$\begin{aligned} \bar{A}_{22} &\coloneqq A - B_2 \bar{K}_2, \\ \bar{A}_{21} &\coloneqq A - B_2 \bar{K}_1, \\ B_2 &\coloneqq \begin{bmatrix} b_2 \\ 0 \end{bmatrix}. \end{aligned}$$

The $\mathcal{L}_2$-gain of this system is the $\mathcal{H}_\infty$ norm of the transfer function from η to X. Let's denote such transfer function by $G(z)$; then, the $\mathcal{L}_2$-gain is

$$\| G(z) \|_\infty = \| \left(z I_{4n} - A_p \right) B_p \|_\infty.$$

The obtained numerical results are given in table 2.

Table 2. $\mathcal{L}_2$-gain from disturbance to state for system (4.2) for different controllers

Optimal $\mathcal{H}_\infty$	Minimax	Periodic deadbeat
1.88	8.8	59.1

5. Conclusions and Future Works

The paper discussed the simultaneous stabilization of LTI plants via minimax adaptive control. We have established that any two P-stabilizable LTI plants could be stabilized via a minimax adaptive controller when no static state-feedback controller could accomplish that. We as well provided an explicit $\mathcal{L}_2$-gain bound on the achievable worst-case performance. Future work concerns scaling our guarantees to cover stabilization of more than two systems and providing tighter lower bound on the achievable $\mathcal{L}_2$-gain performance.

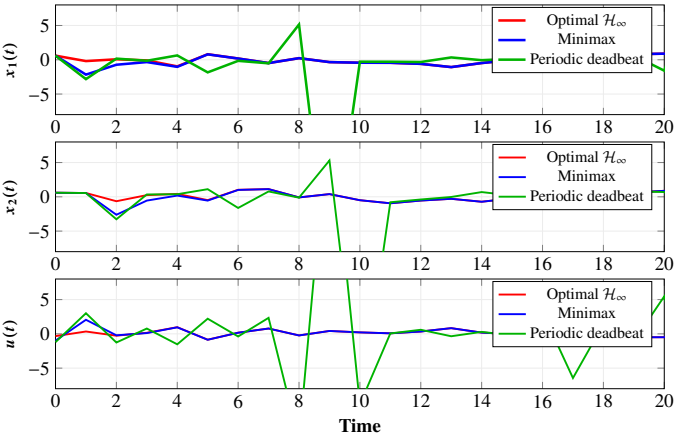


Figure 2. The states and the control input trajectories when using the minimax adaptive controller, the optimal $\mathcal{H}_\infty$ controller, and a 4-periodic switching deadbeat controller.

A. Preliminary Lemmata

LEMMA 1
Let $E, F \in \mathbb{R}^{n\times n}$ be two matrices and $M \in \mathbb{R}^{n\times n}$ be a positive definite matrix. Then, for any scalar $\beta \neq 0$, it holds that

$$E^T MF + F^T ME \preceq \beta^2 E^T ME + \beta^{-2} F^T MF.$$

Proof. Start by noting that for any two matrices $E, F \in \mathbb{R}^{n\times n}$ and any nonzero constant β it holds that

$$(\beta E - \beta^{-1} F)^T M(\beta E - \beta^{-1} F) \succeq 0,$$

expanding the left hand side of the inequality gives

$$\beta^2 E^T ME + \beta^{-2} F^T MF - E^T MF - F^T ME \succeq 0.$$

Then, it follows immediately that

$$E^T MF + F^T ME \preceq \beta^2 E^T ME + \beta^{-2} F^T MF.$$

Note that in the event $E = F$ along with the selection $\beta = 1$, we get equality. This shows that the upper bound provided in the lemma is tight. ∎

References

Agarwal, N., B. Bullins, E. Hazan, S. Kakade, and K. Singh (2019). "Online control with adversarial disturbances". In: *International Conference on Machine Learning*. PMLR, pp. 111–119.

Başar, T. and P. Bernhard (2008). *H-infinity optimal control and related minimax design problems: a dynamic game approach*. Springer Science & Business Media.

Bencherki, F. and A. Rantzer (2023). "Robust simultaneous stabilization via minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 2503–2508.

Buchstaller, D. and M. French (2008). "Gain bounds for multiple model switched adaptive control of general mimo lti systems". In: *2008 47th IEEE Conference on Decision and Control*. IEEE, pp. 5330–5335.

Cederberg, D., A. Hansson, and A. Rantzer (2022). "Synthesis of minimax adaptive controller for a finite set of linear systems". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. IEEE, pp. 1380–1384.

Das, S. K. and P. Kar (2003). "Simultaneous pole placement of m discrete-time plants using a m-periodic controller". *IEEE transactions on automatic control* **48**:11, pp. 2045–2050.

Didinsky, G. and T. Basar (1994). "Minimax adaptive control of uncertain plants". In: *Proceedings of 1994 33rd IEEE Conference on Decision and Control*. Vol. 3. IEEE, pp. 2839–2844.

French, M. and S. Trenn (2005). "$\mathcal{L}_p$ gain bounds for switched adaptive controllers". In: *Proceedings of the 44th IEEE Conference on Decision and Control*. IEEE, pp. 2865–2870.

Ghosh, B. and C. Byrnes (1983). "Simultaneous stabilization and simultaneous pole-placement by nonswitching dynamic compensation". *IEEE Transactions on Circuits and Systems* **30**:6, pp. 422–428.

Jia, Y. and J. Ackermann (2001). "Condition and algorithm for simultaneous stabilization of linear plants". *Automatica* **37**:9, pp. 1425–1434.

Khargonekar, P., K. Poolla, and A. Tannenbaum (1985). "Robust control of linear time-invariant plants using periodic compensation". *IEEE Transactions on Automatic Control* **30**:11, pp. 1088–1096.

Khargonekar, P. P., A. Pascoal, and R. Ravi (1988a). "Stabilization of linear time-varying systems strong, simultaneous and reliable stabilization". In: *1988 American Control Conference*. IEEE, pp. 2477–2482.

Khargonekar, P. P., A. M. Pascoal, and R. Ravi (1988b). "Strong, simultaneous, and reliable stabilization of finite-dimensional linear time-varying plants". *IEEE transactions on automatic control* **33**:12, pp. 1158–1161.

Matni, N., A. Proutiere, A. Rantzer, and S. Tu (2019). "From self-tuning regulators to reinforcement learning and back again". In: *2019 IEEE 58th Conference on Decision and Control (CDC)*. IEEE, pp. 3724–3740.

Ogata, K. (1995). *Discrete-time control systems*. Prentice-Hall, Inc.

OLBROT, A. W. (1987). "Robust stabilization of uncertain systems by periodic feedback". *International Journal of Control* **45**:3, pp. 747–758.

Rantzer, A. (2021). "Minimax adaptive control for a finite set of linear systems". In: *Learning for Dynamics and Control*. PMLR, pp. 893–904.

Saeks, R. and J. Murray (1982). "Fractional representation, algebraic geometry, and the simultaneous stabilization problem". *IEEE Transactions on Automatic Control* **27**:4, pp. 895–903.

Simchowitz, M. (2020). "Making non-stochastic control (almost) as easy as stochastic". *Advances in Neural Information Processing Systems* **33**, pp. 18318–18329.

Stoustrup, J. and V. D. Blondel (2004). "Fault tolerant control: a simultaneous stabilization result". *IEEE transactions on automatic control* **49**:2, pp. 305–310.

Vidyasagar, M. and N. Viswanadham (1982). "Algebraic design techniques for reliable stabilization". *IEEE Transactions on Automatic Control* **27**:5, pp. 1085–1095.

Vinnicombe, G. (2004). "Examples and counterexamples in finite $\mathcal{L}_2$-gain adaptive control". In: *Leuven: Sixteenth International Symposium on Mathematical Theory of Networks and Systems (MTNS2004)*.

Zhou, K. and J. C. Doyle (1998). *Essentials of robust control*. Vol. 104. Prentice hall Upper Saddle River, NJ.

Paper II

Robust Adaptive Data-Driven Control of Positive Systems with Application to Learning in SSP Problems

Fethi Bencherki Anders Rantzer

Abstract

An adaptive data-driven controller is proposed and analysed for the class of infinite-horizon optimal control of positive linear system problems presented in [Ohlin et al., 2024b]. This controller is synthesized from the solution of a "data-driven algebraic equation" constructed from the model-free Bellman equation used in Q-learning. This algebraic equation is driven by data correlation matrices that does not scale in the number of data points, making for an efficient online implementation. As a result, a sufficient condition that ensures stability and robustness against unmodeled dynamics in the plant is extracted. The derived results also allows for a quantitative characterization of the interplay between explorations and robustness to unmodeled dynamics. The class of optimal control problems considered here bears equivalence to Stochastic Shortest Path (SSP) problems, allowing for a performance comparison between the proposed adaptive policy and model-free algorithms for learning the stochastic shortest path, as presented in the numerical experiment.

Submitted to 7th Annual Learning for Dynamics & Control Conference (L4DC). Preprint available: F. Bencherki and A. Rantzer (2024b). *Robust adaptive data-driven control of positive systems with application to learning in ssp problems*. arXiv: `2412.17012 [math.OC]`

1. Introduction

Positive systems characterise a class of systems for which the state never goes negative when starting from a positive initial state. A lot of physical quantities such as concentrations, buffer sizes, queue lengths, charge levels, prices, etc are intrinsically positive, giving rise to positive mathematical models incorporating this sign constraint. Such models have seen much success in many application areas such us biology and pharmacology [Carson and Cobelli, 2013; Blanchini and Giordano, 2014], thermodynamics [Haddad et al., 2010], traffic and congestion modelling [Shorten et al., 2006] and epidemiology [Hernandez-Vargas and Middleton, 2013]. This therefore motivated and led to a plethora of works in regard to this class of systems focusing both on analysis and controller design, see the works [De Leenheer and Aeyels, 2001; Rami and Tadeo, 2007; Blanchini et al., 2015; Rantzer, 2015; Ebihara et al., 2016; Gurpegui et al., 2023; Ohlin et al., 2024b] and the comprehensive tutorial [Rantzer and Valcher, 2018].

Despite the rich literature on the optimal control of positive systems, most of it is still offline model-based. The current work is motivated by the recent advances/interest in statistical machine learning [Tsiamis et al., 2023] and finite-time analysis, working towards an online data-driven (adaptive) approach to the problem. This direction has received increasing interest lately, see the works [De Persis and Tesi, 2019; Dörfler et al., 2023; Zhao et al., 2024] that treats the linear quadratic regulator (LQR) case and the works [Shafai et al., 2022; Miller et al., 2023; Padoan et al., 2023; Iwata et al., 2024; Makdah and Pasqualetti, 2023; Wang and Shafai, 2024] targeted towards the positive systems class.

In the current manuscript, we adopt a worst-case approach in terms of the type of disturbances and uncertain plant parameters, in line with the works presented in [Rantzer, 2021; Kjellqvist and Rantzer, 2022b; Kjellqvist and Rantzer, 2022a; Bencherki and Rantzer, 2023; Renganathan et al., 2023]. However, the disturbances in this paper are subject to a bound constraint in terms of past states and inputs in a similar fashion to the work presented in [Rantzer, 2024], where the linear quadratic optimal control problem in the presence of non-stochastic process disturbances is addressed. This constraint brings forward a different flavor as compared to the above mentioned works.

1.1 Contributions and Outline of the Paper

Contributions The paper proposes and analyzes a robust data-driven adaptive control scheme for the class of positive systems presented in [Ohlin et al., 2024b]. A data-driven algebraic equation is constructed from the model-free counterpart of the Bellman optimality equation. This equation allows for a direct extraction of the adaptive policy bypassing an explicit system identification step. The data-driven algebraic equation is constructed using data correlation matrices that assumes the availability of sate measurements contaminated with additive process noise. To tolerate uncertainties in regard to unmodelled dynamics and time-variations in the

plant, we avoid putting stochastic assumptions on the noise. This equips the proposed approach with robustness guarantees. The proposed approach could also be readily extended to cover the problems classes reported in [Gurpegui et al., 2023; Gurpegui et al., 2024], which are natural extensions to the problem class tackled here.

Applications The considered problem class could capture various network routing problem settings. A special instance of this appeared in [Bencherki and Rantzer, 2024a] (paper III), where the authors considered the problem of learning the optimal processing rate in a multi-unit processing network in the presence of disturbances. Another interesting instance is the Stochastic Shortest Path (SSP) problem class, by virtue of the work in [Ohlin et al., 2024a] where the authors show the equivalence between the two problems classes. This equivalence motivates considering a numerical study comparing the performance of the model-free policy presented here to the existing parameter-free algorithms for SSP.

Outline The outline of this paper is as follows: In section 2, we give a detailed problem setup description. In particular, we touch on several aspects of the problem in an attempt to provide a comprehensive treatment. This is achieved by following a natural build up starting from presenting the model-based optimal control problem as laid out originally in [Ohlin et al., 2024b], its solution, how to solve it and ending up in formulating its model-free counterpart. This gives rise to the so called model-free algebraic equation, based on which we construct the data-driven adaptive policy. Section 3 is opened by first presenting a set of supporting lemmas that plays an essential role in paving the way to a comprehensive formal performance analysis of the proposed online adaptive policy that appears later in the section. Section 4 compares the numerical performance of the adaptive policy to that of the Q-learning algorithm presented in [Yu and Bertsekas, 2013] in learning the stochastic shortest path. Lastly, Section 5 concludes the paper and provides potential future directions. For the sake of neatness of presentation, all the proofs are kept in the appendix.

1.2 Notation

Inequalities are applied element-wise for both matrices and vectors. Furthermore, the notation $\mathbb{R}^n_+$ is used to denote the closed positive orthant of dimension n. $|X|$ means the element-wise absolute value of the entries of the matrix (or vector) X. The operator $\min\{A, 0\}$ extracts the minimum element of A, yielding zero if A contains no negative elements. The expressions $\mathbf{1}_{n\times m}$ and $\mathbf{0}_{n\times m}$ signify a matrix of ones or zeros, respectively, of the indicated dimension, with subscript omitted when the size is clear from the considered context. The operation $\otimes$ is the Kronecker product. $U(a, b)$ is the random uniform distribution between a and b.

2. Problem Setup

We consider the infinite-horizon optimal control problems class presented in [Ohlin et al., 2024b]

$$\begin{aligned}
\text{Minimize} \quad & \sum_{t=0}^{\infty}[s^\top x(t)+r^\top u(t)] \text{ over } \{u(t)\}_{t=0}^{\infty} \\
\text{subject to} \quad & x(t+1) \quad = Ax(t)+Bu(t) \\
& u(t) \quad \geq 0, \quad x(0)\in\mathbb{R}_+^n \\
& \mathbf{1}^\top u_1(t) \leq E_1^\top x(t) \\
& \qquad\vdots \\
& \mathbf{1}^\top u_M(t) \leq E_M^\top x(t)
\end{aligned} \tag{1}$$

where $A\in\mathbb{R}^{n\times n}$ and $B=\begin{bmatrix}B_1 & \cdots & B_M\end{bmatrix}\in\mathbb{R}^{n\times m}$ with $B_i\in\mathbb{R}^{n\times m_i}$ defining the linear dynamics. The input signal $u\in\mathbb{R}^m$ is partitioned into M subvectors u_i, each containing m_i elements, so that $m=\sum_{i=1}^M m_i$. The cost vectors connected to the states and actions are $s\in\mathbb{R}_+^n$ and $r\in\mathbb{R}_+^m$ with $r_i\in\mathbb{R}^{m_i}$ following the partition of u. The constraints on the input signal u are given by $E=\begin{bmatrix}E_1 & \cdots & E_M\end{bmatrix}^\top\in\mathbb{R}_+^{M\times n}$. Furthermore, we define the extended constraint matrix $\bar{E}\coloneqq\begin{bmatrix}\mathbf{1}_{m_1}^\top\otimes E_1 & \cdots & \mathbf{1}_{m_M}^\top\otimes E_M\end{bmatrix}^\top\in\mathbb{R}_+^{m\times n}$ and the set of indices $\mathcal{V}=\{1,\dots,n\}$. Let $K=\begin{bmatrix}K_1^\top\cdots K_M^\top\end{bmatrix}^\top$ be a feedback matrix with $K_i\in\mathbb{R}_+^{m_i\times n}$ and define the set of feasible gains

$$\mathbf{K}(E)\coloneqq\{K:(\forall i\in\mathbf{V})\ \mathbf{1}_{m_i}^\top K_i=E_i^\top \text{ or } \mathbf{1}_{m_i}^\top K_i=\mathbf{0}_n\} \tag{2}$$

and correspondingly the state feedback law of the i-th control subvector $u_i=K_ix$. The set $\mathcal{K}(E)$ characterizes all feedback gain matrices that either grants full or no actuation of the control inputs u_i, $i\in\mathcal{V}$. We impose the following two assumptions on the sextuplets $(A,B,E,\bar{E},s,r)$.

ASSUMPTION 2.1—[OHLIN ET AL., 2024A]
The matrices A,B and the set $\mathcal{K}(E)$ satisfy

$$(A+BK)x\geq 0$$

for all $K\in\mathcal{K}(E)$ and all reachable states $x\in\mathbb{R}_+^n$.

ASSUMPTION 2.2
The triplet $(s,\bar{E},r)$ satisfies $s>\bar{E}^\top r$.

REMARK 2.1
Assumption 2.1 warrants the positivity the closed loop dynamics. Assumption 2.2, on the other hand, will prove useful in establishing the main results as shall be seen in later.

2.1 Solution to Problem 1

Under Assumption 2.1 and via dynamic programming [Bellman, 1966], it was shown in [Ohlin et al., 2024b, Theorem 1] that if problem (1) has a finite value for every $x(0) \in \mathbb{R}^n_+$, then the optimal cost would be $p^\top x(0)$, where $p \in \mathbb{R}^n_+$ is the solution to the following model-based algebraic equation

$$p = s + A^\top p + \sum_{i=1}^{M} \min\{r_i + B_i^\top p, 0\} E_i. \tag{3}$$

Furthermore, the optimal policy is a linear state feedback law $u(t) = Kx(t)$ where $K = \begin{bmatrix} K_1^\top & \dots & K_M^\top \end{bmatrix}^\top$ and

$$K_i \coloneqq \begin{bmatrix} \mathbf{0}_{j-1 \times n} \\ E_i^\top \\ \mathbf{0}_{m_i - j \times n} \end{bmatrix} \in \mathbb{R}^{m_i \times n}, \tag{4}$$

where the vector $E_i^\top$ enters at the j-th row with j being the index of the minimal element of $r_i + B_i^\top p$, provided it is negative. If all elements are nonnegative, then $K_i = \mathbf{0}_{m_i \times n}$. The solution to (3) could for instance be found via value iteration (VI), i.e. performing the following fixed point iteration in the p-parameter until convergence

$$p^{k+1} = s + A^\top p^k + \sum_{i=1}^{M} \min\{r_i + B_i^\top p^k, 0\} E_i, \quad p^0 = 0. \tag{5}$$

2.2 Model-Free Optimal Control of Positive Systems via Q-Factor

The cost-to-go function from time t for the optimization problem in (1) for a control policy u beginning at time t from an initial state $x(t)$ is

$$J(x(t)) \coloneqq \min_u \sum_{k=t}^{\infty} s^\top x(k) + r^\top u(k),$$

where the stage cost given by $c(x(k), u(k)) \coloneqq s^\top x(k) + r^\top u(k)$ for $k \geq t$. This optimization problem gives the objective value $J(x(t)) = p^\top x(t)$ where p is solved for from (3), which requires the knowledge of the system model (A, B). The optimal Q-function was defined in [Bradtke et al., 1994] to be

$$Q(x(t), u(t)) \coloneqq c(x(t), u(t)) + J(x(t+1)), \tag{6}$$

which represents the cost of taking action an action $u(t)$ starting at some state $x(t)$ and following by the optimal policy u^* after that for all time. The optimal Q-function then takes the form

$$Q^*(x(t), u(t)) = \begin{bmatrix} s^\top + p^\top A & r^\top + p^\top B \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} = \begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} = q^\top \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}, \tag{7}$$

where $q \coloneqq \begin{bmatrix} q^x \\ q^u \end{bmatrix} \coloneqq \begin{bmatrix} s + A^\top p \\ r + B^\top p \end{bmatrix}$. Note that the following also holds

$$J(x(t)) = \min_{u(t) \in \mathcal{U}(x(t))} \left[c(x(t), u(t)) + J(x(t+1)) \right] = \min_{u(t) \in \mathcal{U}(x(t))} Q(x(t), u(t)),$$

where we denote the set of inputs satisfying the constraints in (1) at time t as $\mathcal{U}(x(t))$. Plugging this in the right hand side of (6) yields

$$Q(x(t), u(t)) = c(x(t), u(t)) + \min_{u(t+1) \in \mathcal{U}(x(t+1))} Q\left(x(t+1), u\left(t+1\right)\right), \tag{8}$$

and the optimal policy could be found via $u^*(t) = \arg\min_{u(t) \in \mathcal{U}(x(t))} Q\left(x(t), u(t)\right)$. One can think of (8) as the Bellman equation in the Q-factor [Sutton and Barto, 2018]. Replacing (7) in (8) yields

$$\begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} = s^\top x(t) + r^\top u(t) + \min_{u(t+1) \in \mathcal{U}(x(t+1))} \begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} x(t+1) \\ u(t+1) \end{bmatrix},$$

which by the virtue of the definition of $\mathcal{K}(E)$ in (2) is equivalent to

$$\begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} x(t) \\ u(t) \end{bmatrix} = s^\top x(t) + r^\top u(t) + \min_{K \in \mathcal{K}(E)} \begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} I \\ K \end{bmatrix} x(t+1). \tag{9}$$

In the absence of the knowledge of the dynamics (A, B), (9) makes it possible to get information about the true q-parameter by collecting triplets $(x(t), u(t), x(t+1))$. In fact, collecting t consecutive data points leads to the equation

$$\left(q - \begin{bmatrix} s \\ r \end{bmatrix} \right)^\top \begin{bmatrix} x(0) & \cdots & x(t-1) \\ u(0) & \cdots & u(t-1) \end{bmatrix} = \min_{K \in \mathcal{K}(E)} q^\top \begin{bmatrix} I \\ K \end{bmatrix} \begin{bmatrix} x(1) & \cdots & x(t) \end{bmatrix}. \tag{10}$$

Let's define $Z(t, \lambda) \coloneqq \begin{bmatrix} \lambda^{t-1} x(0) & \lambda^{t-2} x(1) & \cdots & x(t-1) \\ \lambda^{t-1} u(0) & \lambda^{t-2} u(1) & \cdots & u(t-1) \end{bmatrix} \in \mathbb{R}^{(n+m) \times t}$ for a forgetting factor $\lambda \in (0, 1]$. Multiply (10) by $Z^\top(t, \lambda)$ from the right and define the data correlation matrices

$$\begin{aligned} \Sigma(t) &\coloneqq \sum_{k=0}^{t-1} \lambda^{t-1-k} \begin{bmatrix} x(k) \\ u(k) \end{bmatrix} \begin{bmatrix} x(k) \\ u(k) \end{bmatrix}^\top + \lambda^t \Sigma(0) \\ \bar{\Sigma}(t) &\coloneqq \sum_{k=0}^{t-1} \lambda^{t-1-k} x(k+1) \begin{bmatrix} x(k) \\ u(k) \end{bmatrix}^\top, \end{aligned} \tag{11}$$

gives *the data-driven algebraic equation* in the $q(t)$-parameter

$$\left(q(t) - \begin{bmatrix} s \\ r \end{bmatrix} \right)^\top \Sigma(t) = \min_{K(t) \in \mathcal{K}(E)} q^\top(t) \begin{bmatrix} I \\ K(t) \end{bmatrix} \bar{\Sigma}(t), \tag{12}$$

where we denote the data-based solution as $(q\left(t\right), K\left(t\right))$ in contrast to the model-based (or ground truth) solution (q, K). Equation (12) is at the center of the proposed controller construction as shall be seen next.

2.3 Problem Formulation

Inspired by the constructed data-driven algebraic equation in (12), for the linear system

$$x(t+1) = Ax(t) + Bu(t) + w(t), \tag{13}$$

we consider data-driven policies of the type

$$\begin{cases} \Sigma(t) = \lambda\Sigma(t-1) + \begin{bmatrix} x^\top(t-1) & u^\top(t-1) \end{bmatrix}^\top \begin{bmatrix} x^\top(t-1) & u^\top(t-1) \end{bmatrix}, \quad \Sigma(0) \succ 0, \\ \bar{\Sigma}(t) = \lambda\bar{\Sigma}(t-1) + x(t) \begin{bmatrix} x^\top(t-1) & u^\top(t-1) \end{bmatrix}, \quad \bar{\Sigma}(t) = 0, \\ u(t) = K(t)x(t) + \epsilon(t). \end{cases} \tag{14}$$

The controller states $\Sigma(t)$ and $\bar{\Sigma}(t)$ collect correlation data with a forgetting factor λ. Given $\Sigma(t)$ and $\bar{\Sigma}(t)$, the controller $K(t)$ is obtained as the minimizing argument in (12) and $\epsilon(t)$ is added to provide explorations towards learning the true dynamics (A, B).

REMARK 2.2
The introduction of the term $\Sigma(0) \succ 0$ guarantees the invertibility of $\Sigma(t)$ for all t, which makes it possible to deploy the policy (14) immediately. However, it introduces an initial bias in the model estimate $\begin{bmatrix} \hat{A}(t) & \hat{B}(t) \end{bmatrix}$, which is diminishing as more data is collected. This explains why the derived results, such as in Theorems 1, 2 and Corollary 1 present guarantees which are only valid after some initial time t_0 has elapsed.

The following definition constrain the considered model set.

DEFINITION 1
Let the model parameter set $\mathcal{M}_\beta$ be the set of plants (A, B, E) pertaining to assumptions 2.1 and 2.2 such that the algebraic equation in (3) has a solution p satisfying $s \leq p \leq \left(\beta \min_{i,\, s_i \neq 0} s_i\right) \mathbf{1} \leq \beta s$.

β reflects on the degree of stabilizability of the system. Larger values of β corresponds to systems that are harder to stabilize or less stabilizable for that matter. Different than the type of assumption in Definition 1, the work in [Mania et al., 2019] extracted regret bounds in terms of the degree of controllability, a stronger requirement than stabilizability. On the other hand, the work in [Simchowitz and Foster, 2020], presents regret bounds that scales in the positive definite matrix solution P to the discrete algebraic Riccati equation (DARE) of the system, making it more aligned with the work presented here, however, and differently from that work we consider a non-stochastic setting.

REMARK 2.3
Controllers taking the form (14) are denoted by adaptive certainty equivalence controllers in the adaptive control literature [Åström and Wittenmark, 1995], since $u(t)$ is extracted using the model estimate $\begin{bmatrix}\hat{A}(t) & \hat{B}(t)\end{bmatrix} \coloneqq \bar{\Sigma}(t)\Sigma^{-1}(t)$ in (12) with a complete disregard for estimation errors.

Definition 1 imposes prior constraints on the model. Under such constraints, we shall establish guarantees for stability and robustness of the closed loop system under the policy in (14) given assumptions on the disturbance w, but we first provide ways to obtain the solution of the algebraic data-driven equation in (12), since its solution lies at the heart of the controller construction and implementation.

2.4 Solution to Data-driven Algebraic Equation in the $q(t)$-Parameter

Similarly to the p-parameter based algebraic equation in (3), (12) could be solved via either value iteration, policy iteration or linear programming.

(i) **Value iteration.** This achieved by performing the following fixed point iteration in the $q(t)$-parameter starting at $q^0(t) = 0$ until convergence

$$\begin{aligned} q^{k+1}(t) &= \Sigma^{-1}(t)\bar{\Sigma}^\top(t)\begin{bmatrix} I & \left(K^k(t)\right)^\top \end{bmatrix} q^k(t) + \begin{bmatrix} s \\ r \end{bmatrix}, \quad q^0(t) = 0, \\ K_i^k(t) &= \begin{cases} \begin{bmatrix} \mathbf{0}_{j-1\times n} \\ E_i^\top \\ \mathbf{0}_{m_k-j\times n} \end{bmatrix}, \min\left\{\left(q_i^u\right)^k(t), 0\right\} < 0 \\ 0_{m_i\times n}, \quad \text{otherwise} \end{cases}, i = 1,\dots,M, \end{aligned} \tag{15}$$

and $K^k(t) = \begin{bmatrix}\left(K_1^k(t)\right)^\top & \dots & \left(K_1^M(t)\right)^\top\end{bmatrix}^\top$, j being the index of the minimal negative entry of $\left(q_i^u\right)^k(t)$.

(ii) **Policy iteration.** This assumes the existence of an initial stabilizing policy $K^0(t) \in \mathcal{K}(E)$ and performs the following updates until convergence

$$\begin{aligned} q^{k+1}(t) &= \left(I - \Sigma^{-1}(t)\bar{\Sigma}^\top(t)\begin{bmatrix} I \\ K^k(t) \end{bmatrix}^\top\right)^{-1}\begin{bmatrix} s \\ r \end{bmatrix}, \\ K_i^{k+1}(t) &= \begin{cases} \begin{bmatrix} \mathbf{0}_{j-1\times n} \\ E_i^\top \\ \mathbf{0}_{m_k-j\times n} \end{bmatrix}, \min\left(\left(q_i^u\right)^{k+1}(t)\right) < 0 \\ 0_{m_i\times n}, \quad \text{otherwise} \end{cases}, i = 1,\dots,M. \end{aligned} \tag{16}$$

where $K^{k+1}(t) = \begin{bmatrix}\left(K_1^{k+1}(t)\right)^\top & \dots & \left(K_M^{k+1}(t)\right)^\top\end{bmatrix}^\top$ and j is the index of the minimal element of $\left(q_i^u\right)^{k+1}(t)$.

(iii) **Linear programming.** Instead of a fixed point iteration, the solution could be obtained via linear programming. To this end, define $z \coloneqq \left[(z^x)^\top \quad (z^u)^\top\right]^\top$ where $z^x \in \mathbb{R}^n_+$ and $z^u \in \mathbb{R}^m_+$. We propose the following optimization problem to solve for $q(t)$.

$$\begin{aligned} \text{Maximize} \quad & \mathbf{1}^\top \begin{bmatrix} I & -E^\top \end{bmatrix} z \text{ over } z \in \mathbb{R}^{n+m}_+, q(t) \in \mathbb{R}^{n+m} \\ \text{subject to} \quad & \Sigma^{-1}(t)\bar{\Sigma}^\top(t) \begin{bmatrix} I & -E^\top \end{bmatrix} z = q(t) - \begin{bmatrix} s \\ r \end{bmatrix} \\ & z^x = q^x(t),\ z^u \geq q^u(t),\ z^u \geq 0. \end{aligned}$$

When the plant (A, B) is known, $\Sigma^{-1}(t)\bar{\Sigma}^\top(t)$ is replaced by $\begin{bmatrix} A & B \end{bmatrix}^\top$ and a model-based counterpart to (15) is

$$\begin{aligned} q^{k+1} &= \begin{bmatrix} A & B \end{bmatrix}^\top \begin{bmatrix} I & \left(K^k\right)^\top \end{bmatrix} q^k + \begin{bmatrix} s \\ r \end{bmatrix}, \quad q^0 = 0, \\ K_i^k &= \begin{cases} \begin{bmatrix} \mathbf{0}_{j-1\times n} \\ E_i^\top \\ \mathbf{0}_{m_k - j\times n} \end{bmatrix}, \min\left\{\left(q_i^u\right)^k, 0\right\} < 0 \\ 0_{m_i \times n}, \quad \text{otherwise} \end{cases}, i = 1, \ldots, M, \end{aligned} \tag{17}$$

where $K^k = \left[\left(K_1^k\right)^\top \ \ldots \ \left(K_M^k\right)^\top\right]^\top$ and j is the index of the minimal negative element of $\left(q_i^u\right)^k$, and $q(t)$ was replaced by the optimal q-parameter since the two coincide in this scenario.

3. Main Results

Before stating the main results of the paper, we first establish important supporting lemmata.

LEMMA 1
Iterating on q in (17) is algebraically equivalent to iterating on p in (5).

Proof. See Appendix-A.1. ∎

A similar result to that of Lemma 1 was also exploited in [Al-Tamimi et al., 2007] to establish the convergence of a model-free Q-learning based algorithm for the $\mathcal{H}$-infinity optimal control problem formulated as a zero-sum game.

REMARK 3.1
We extract two key observations from Lemma 1:

(i) Lemma 1 asserts that value iteration in the q-parameter is algebraically equivalent to value iteration in the p-parameter.

(ii) According to Lemma 1, given $(q(t), K(t))$, the solution to the the data-driven algebraic equation in (12), if we define $p(t) \coloneqq \begin{bmatrix} I & (K(t))^\top \end{bmatrix} q(t)$, then $p(t)$ satisfies the following model-based algebraic equation in the estimated dynamics $\left(\hat{A}(t), \hat{B}(t)\right)$

$$p(t) - s = \hat{A}^\top(t)p(t) + \sum_{i=1}^{M} \min\left\{r_i + \hat{B}_i^\top p(t), 0\right\} E_i, \tag{18}$$

where $\begin{bmatrix} \hat{A}(t) & \hat{B}(t) \end{bmatrix} \coloneqq \bar{\Sigma}(t)\Sigma^{-1}(t)$.

This is key to establishing proofs to the main results of the paper as shall be seen in Section 3.1.

LEMMA 2
Let $p \in \mathbb{R}_+^n$ and $\hat{p} \in \mathbb{R}_+^n$ be such that they satisfy the algebraic equation and inequality respectively, i.e. they satisfy

$$p = s + A^\top p + \sum_{i=1}^{M} \min\left\{r_i + B_i^\top p, 0\right\} E_i,$$
$$\hat{p} \geq s + A^\top \hat{p} + \sum_{i=1}^{M} \min\left\{r_i + B_i^\top \hat{p}, 0\right\} E_i.$$

Then, it holds that $\hat{p} \geq p$.

Proof. See Appendix-A.2. ∎

3.1 Performance Analysis

Prior to presenting our first main result, we define

$$\tilde{\Sigma}(t) \coloneqq \begin{bmatrix} \Sigma^{wx}(t) & \Sigma^{wu}(t) \end{bmatrix} \coloneqq \sum_{k=0}^{t-1} w(k) \begin{bmatrix} x^\top(k) & u^\top(k) \end{bmatrix},$$

implying that $\bar{\Sigma}(t) = \begin{bmatrix} A & B \end{bmatrix} \Sigma(t) + \tilde{\Sigma}(t)$ and therefore

$$\begin{bmatrix} \hat{A} & \hat{B} \end{bmatrix} = \bar{\Sigma}(t)\Sigma^{-1}(t) = \begin{bmatrix} A & B \end{bmatrix} + \begin{bmatrix} \tilde{A}(t) & \tilde{B}(t) \end{bmatrix}, \tag{19}$$

where $\begin{bmatrix} \tilde{A}(t) & \tilde{B}(t) \end{bmatrix} \coloneqq \tilde{\Sigma}(t)\Sigma^{-1}(t) = \begin{bmatrix} \Sigma^{wx}(t)\Sigma^{-1}(t) & \Sigma^{wu}(t)\Sigma^{-1}(t) \end{bmatrix}$ is the model misspecification. The first main result of the paper is stated next.

THEOREM 1
Consider $\beta, \rho \in \mathbb{R}_+$ satisfying $\rho\beta < 1$, $\Sigma(t)$ and $\bar{\Sigma}(t)$ as in (11). Let $(A, B) \in \mathcal{M}_\beta$. Additionally, suppose that

$$\left\|E^\top\right\|_\infty \left\|A^\top - \begin{bmatrix} I & 0 \end{bmatrix} \Sigma^{-1}(t)\bar{\Sigma}^\top(t)\right\|_\infty + \left\|B^\top - \begin{bmatrix} 0 & I \end{bmatrix} \Sigma^{-1}(t)\bar{\Sigma}^\top(t)\right\|_\infty \leq \rho, \quad \forall t \geq t_0. \tag{20}$$

Let $p \in \mathbb{R}^n_+$ be the solution to the model-based algebraic equation in (3), and let $q(t)$ be the solution of the data-based algebraic equation in (12) with $K(t)$ being the minimizing argument. Define $p(t) \coloneqq \begin{bmatrix} I & K^\top(t) \end{bmatrix} q(t)$. Then, it holds that

$$\hat{\alpha} p \leq p(t) \leq \check{\alpha}^{-1} p \tag{21}$$

for positive constants satisfying $\check{\alpha} = 1 - \rho\beta$ and $\hat{\alpha} = 1 - \check{\alpha}^{-1}\rho\beta$.

Proof. The proof is given in Appendix-A.3. ∎

REMARK 3.2
The result in theorem 1 is obtained from a perturbation analysis to the solution of the algebraic equations in (3) and (18). Perturbation analysis of algebraic Riccati equations is a well studied problem in the literature, see the works [Sun, 1998; Konstantinov et al., 1993; Konstantinov et al., 2003].

Assumption (20) could be written as

$$\left\|E^\top\right\|_\infty \left\|\Sigma^{-1}(t)\left(\Sigma^{wx}(t)\right)^\top\right\|_\infty + \left\|\Sigma^{-1}(t)\left(\Sigma^{wu}(t)\right)^\top\right\|_\infty \leq \rho,$$

and will be valid as long as two conditions are met:

(i) The condition number of the matrix $\Sigma(t)$ must not be too large. This is conditioned on choosing proper explorations $\epsilon(t)$.

(ii) The disturbance sequence $w(t)$ must be sufficiently small compared to $x(t)$. If $w(t)$ is taken to represent unmodelled dynamics, this means that such dynamics are subject to a gain bound from state to disturbance.

Furthermore, If one impose stochastic assumptions on w instead, larger values of t_0 typically will make it possible to satisfy (20) with a smaller value of ρ. However, if faced with adversarial disturbances, the true dynamics may never be learned since the adversary w could play a policy that fool the controller forever.

THEOREM 2
Consider $\beta, \rho \in \mathbb{R}_+$ satisfying $\rho\beta < 1$, $\Sigma(t)$ and $\bar{\Sigma}(t)$ as in (11). Let $(A, B) \in \mathcal{M}_\beta$ with $p \in \mathbb{R}^n_+$ the solution to the model-based algebraic equation in (3). Suppose that (20) holds for all $t \geq t_0$ and let $K(t)$ be the minimizer in (12). Then, it holds that

$$\check{\alpha}^{-1}\left(1 + \rho\beta\left(1 + \beta\left\|A + |B|\bar{E}\right\|_1\right)\right)p - s \geq A^\top p + K^\top(t)\left(r + B^\top p\right). \tag{22}$$

where the constant $\check{\alpha}$ is as in Theorem 1.

Proof. See Appendix-A.4. ∎

REMARK 3.3
Theorem 1 aims at demonstrating how the storage function $p^\top x$ is influenced when deploying the data-driven gain $K(t)$ instead of the optimal gain K. Note that as $\rho \to 0$, we get that $\breve{\alpha} \to 1$ and the right hand side of (22) goes to 1 and with that closing the suboptimality gap. Larger values of β corresponds to systems that are harder to stabilize, mandating ρ to be smaller to satisfy the condition $\rho\beta < 1$, which characterizes the need for more accurate data.

COROLLARY 1
Consider $(A, B) \in \mathcal{M}_\beta$ with p being the positive solution to the algebraic equation (3), while β, ρ are positive numbers satisfying $\rho\beta < 1$. Let

$$x(t+1) = Ax(t) + Bu(t) + w(t), \qquad u(t) = K(t)x(t) + \epsilon(t),$$

where $\Sigma(t)$ and $\bar{\Sigma}(t)$ are defined as in (11) while (20) is assumed to hold for all $t \geq t_0$. let $q(t)$ be the solution of the data-based algebraic equation in (12) with $K(t)$ being its minimizing argument. Then, it holds that

$$\sum_{t=t_0}^{T-1} s^\top x(t) + r^\top K(t)x(t) \leq \gamma^{-1} \left(p^\top x(t_0) + \sum_{t=t_0}^{T-1} \beta s^\top |B\epsilon(t) + w(t)| \right)$$

where $0 < \gamma \leq 1$ satisfies

$$\gamma \left(s - \bar{E}^\top |r| \right) \leq s - \bar{E}^\top |r| - \left(\breve{\alpha}^{-1} - 1 + \rho\beta \left(1 + \beta \left\| A + |B| \, \bar{E} \right\|_1 \right) \right) \beta s$$

and the constant $\breve{\alpha}$ is as in Theorem 1.

Proof. See Appendix-A.5. ∎

REMARK 3.4
Corollary 1 gives a quantification of the suboptimality incurred in the total cost due to deploying the data-driven controller instead of the optimal controller. The constant γ captures the effect of model misspecification and $\gamma \to 1$ if $\rho \to 0$, and with it tightening the suboptimality gap. The last term on the right hand side captures the cost incurred due to disturbances $w(t)$ and explorations $\epsilon(t)$.

4. Numerical Experiment

We consider the problem of learning the following stochastic shortest path

Stochastic Shortest Path (SSP) problem

$$\mathcal{T}(1) = \begin{bmatrix} 0.4 & 0 \\ 0 & 0.4 \\ 0.4 & 0.4 \\ 0.2 & 0.2 \end{bmatrix}, \mathcal{T}(2) = \begin{bmatrix} 0 & 0.3 & 0 \\ 0.6 & 0 & 0.1 \\ 0.4 & 0.7 & 0.4 \\ 0 & 0 & 0.5 \end{bmatrix}, \mathcal{T}(3) = \begin{bmatrix} 0 & 0.2 \\ 0 & 0.2 \\ 0.4 & 0 \\ 0.6 & 0.6 \end{bmatrix}, \mathcal{T}(4) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$c(1) = \begin{bmatrix} 1.5 & 2 \end{bmatrix}, c(2) = \begin{bmatrix} 1.5 & 2 & 2 \end{bmatrix}, c(3) = \begin{bmatrix} 1.5 & 2 \end{bmatrix} c(4) = 0, i_{\text{init}} = 1$$

Conversion according to [Ohlin et al., 2024a]

Reformulation as problem 1

$n = 3$, $m = 4$, $x(0) = [1\ 0\ 0]^\top$, $E = I$, $m_1 = 1$, $m_2 = 2$, $m_3 = 1$ and dynamics

$$A = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 0.6 & 0 \\ 0.4 & 0.4 & 0.4 \end{bmatrix}, B = \begin{bmatrix} -0.4 & 0.3 & 0 & 0.2 \\ 0.4 & -0.6 & -0.5 & 0.2 \\ 0 & 0.3 & 0 & -0.4 \end{bmatrix}$$

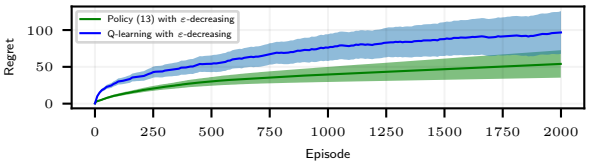
with associated cost vectors

$$s = \begin{bmatrix} 1.5 & 1.5 & 1.5 \end{bmatrix}^\top, r = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}^\top$$

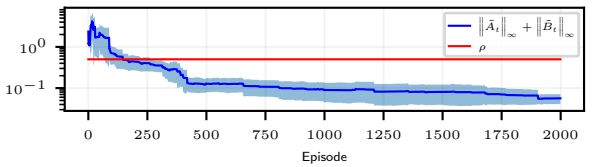
where $\mathcal{T}(i)$ for $i \in \mathcal{V} \cup \{v_g\}$, $v_g = 4$ being the fictitious goal state, are transition probabilities from state i to other states, with different columns corresponding to different actions. $c(i)$ is the expected stage cost vector of the transition from state i to other states for different actions. Note that both Assumptions 2.1 and 2.2 holds true for our system. We compare the performance of the adaptive policy to the that of the Q-learning algorithm in [Yu and Bertsekas, 2013] while employing ϵ-decreasing as an exploration strategy where $\epsilon = 0.05\alpha^h$ with $\alpha = 0.99$ and h corresponding to the episode number. In the context of problem 1, ϵ-greedy means picking a random gain $K \in \mathcal{K}(E)$ with probability ϵ and picking the estimated optimal gain $K(t)$ according to the solution of the data-driven algebraic equation (12) with probability $1 - \epsilon$. For the policy (14), we pick $\lambda = 1$ and initialize $\Sigma(0) = 10^{-6}I$. We contaminate the state measurements with additive positive disturbances drawn according to $w(t) \sim U(0, 0.01)$ for all t to render the learning task more difficult. The comparison between the two algorithms is conducted by considering the regret defined as

$$R(H) \coloneqq \left(\sum_{h=0}^{H-1} \sum_{t=0}^{T_h - 1} s^\top x(t) + r^\top u(t) \right) - HJ(x(0))$$

where H indicates the number of episodes and T_h is the time length of episode h. $J(x(0))$ is the optimal cumulative cost due to applying the optimal policy to the system subject to disturbances w. In the SSP domain, an episode terminates if the goal state ($i = 4$) is reached whereas in problem 1 the state vector becomes sufficiently small indicating that the measurements are purely noise. After an episode is concluded, we re-initialize the states in both problems domain and run the algorithms again while keeping the last estimates of the Q-factor in the Q-learning algorithm and the $q(t)$-parameter and $\Sigma(t)$ for policy (14). The achieved regrets are plotted in Figure 1a. Note that both algorithms exhibit sublinear regret and that our algorithm outperforms the Q-learning algorithm. Figure 1b, on the other hand, plots a



(a) Accumulated regret of each algorithm with ϵ-decreasing explorations where $\epsilon = 0.05\alpha^h$ for $\alpha = 0.99$ and h the episode number.



(b) The condition in (20) for $\rho = 0.3$ in the presence of disturbances. for this example, $\|E^\top\|_\infty = 1$.

Figure 1. Each plot is the average of 100 repeated runs, and the shaded area is 95% confidence interval.

test of the condition in (20) while selecting $\rho = 0.3$. Note that the condition holds true after sufficient number of episodes have elapsed indicating that better estimates of the system as attained as more data is collected.

5. Conclusion

The paper presented a robust adaptive data-driven control framework for the optimal control of positive system problems class presented in [Ohlin et al., 2024b]). This was achieved via the construction of a data-driven algebraic equation in the Q-factor, based on which the controller policy is updated in an online fashion. The designed policy proved to robustly stabilize the set $\mathcal{M}_\beta$ with robustness meant to be tolerance to a certain degree of unmodeled dynamics. The considered class witnesses applications in network routing problems among which are the Stochastic Shortest Path problems, allowing for performance comparison with the existing model-free methods of finding the stochastic shortest path. Future work concerns exploring better exploration strategies than ϵ-greedy, and the possibility of adapting efficient exploration methods from the SSP literature into our control setup.

A. Proofs

A.1 Proof of Lemma 1

Consider the iteration in (17) and define $p^k \coloneqq \begin{bmatrix} I & \left(K^k(t)\right)^\top \end{bmatrix} q^k$. This then gives

$$q^{k+1} = \begin{bmatrix} A^\top \\ B^\top \end{bmatrix} p^k + \begin{bmatrix} s \\ r \end{bmatrix}, \tag{23}$$

and also

$$p^{k+1} \coloneqq \begin{bmatrix} I & \left(K^{k+1}\right)^\top \end{bmatrix} q^{k+1}. \tag{24}$$

where

$$K_i^{k+1} = \begin{cases} \begin{bmatrix} \mathbf{0}_{j-1\times n} \\ E_i^\top \\ \mathbf{0}_{m_k-j\times n} \end{bmatrix}, \min\left\{\left(q_i^u\right)^{k+1}, 0\right\} \overset{(23)}{=} \min\left\{r_i + B_i^\top p^{k+1}, 0\right\} < 0 \\ 0_{m_i\times n}, \quad \text{otherwise} \end{cases} , i = 1, \dots, M. \tag{25}$$

Plugging (23) and (25) in (24) yields the desired result, which is

$$p^{k+1} = s + A^\top p^k + \sum\nolimits_{i=1}^{M} \min\{r_i + B_i^\top p^k, 0\} E_i.$$

A.2 Proof of Lemma 2

Start by defining the value functions $J(x) \coloneqq p^\top x$ and $\hat{J}(x) \coloneqq \hat{p}^\top x$ that solves the Bellman equation and inequality correspondingly, i.e., they satisfy

$$J(x) = \min_{u\in\mathcal{U}(x)} [s^\top x + r^\top u + J(Ax + Bu)],$$
$$\hat{J}(x) \geq \min_{\hat{u}\in\mathcal{U}(x)} [s^\top x + r^\top \hat{u} + \hat{J}(Ax + B\hat{u})],$$

for all $x \in \mathbb{R}^n_+$. Taking the minimizers $u = \begin{bmatrix} u_1^\top \dots u_M^\top \end{bmatrix}^\top$ with $u_i = -K_i x$ and $\hat{u} = \begin{bmatrix} \hat{u}_1^\top \dots \hat{u}_M^\top \end{bmatrix}^\top$ with $\hat{u}_i = -\hat{K}_i x$ for $i = 1, \dots, M$, where K_i is as in (4) and $\hat{K}_i$ is defined analogously leads to

$$\left(p - s - A^\top p - \sum\nolimits_{i=1}^{M} \min\left\{r_i + B_i^\top p, 0\right\} E_i\right)^\top x = 0,$$
$$\left(\hat{p} - s - A^\top \hat{p} + \sum\nolimits_{i=1}^{M} \min\left\{r_i + B_i^\top \hat{p}, 0\right\} E_i\right)^\top x \geq 0.$$

These last equation and inequality are guaranteed to hold if the equation and inequality in the statement of the Lemma hold and $x \in \mathbb{R}^n_+$. Then, it holds that $\hat{J} \geq J$ for all $x \in \mathbb{R}^n_+$, see Proposition 1.(a) in [Li and Rantzer, 2024]). In consequence, the element wise ordering $\hat{p} \geq p$ holds. ∎

A.3 Proof of Theorem 1

First, and according to Remark 4, we associate to the solution of the noisy (noiseless) data-driven algebraic equation in (12) the following model-based algebraic equations in the estimated (true) model

$$p(t) - s = \hat{A}^\top(t)p + \sum_{i=1}^{M} \min\{r_i + \hat{B}_i^\top(t)p, 0\}E_i \tag{26}$$

$$p - s = A^\top p + \sum_{i=1}^{M} \min\{r_i + B_i^\top p, 0\}E_i. \tag{27}$$

The two coincides when the uncertainty vanishes, i.e., when $\rho = 0$.

(i) Consider the algebraic equation in (27). Plug in

$$\begin{bmatrix} A & B \end{bmatrix} = \begin{bmatrix} \hat{A}(t) & \hat{B}(t) \end{bmatrix} - \begin{bmatrix} \tilde{A}(t) & \tilde{B}(t) \end{bmatrix},$$

yields

$$\begin{aligned} p - s &= \left(\hat{A}(t) - \tilde{A}(t)\right)^\top p + \sum_{i=1}^{M} \min\left\{r_i + \left(\hat{B}_i(t) - \tilde{B}_i(t)\right)^\top p, 0\right\} E_i \\ &= \left(\hat{A}(t) - \tilde{A}(t)\right)^\top p + \sum_{i=1}^{M} \min\left\{r_i + \hat{B}_i^\top(t)p - \tilde{B}_i^\top(t)p, 0\right\} E_i. \end{aligned}$$

since $\min\left\{r_i + \hat{B}_i^\top(t)p - \tilde{B}_i^\top(t)p, 0\right\} \geq \min\left\{r_i + \hat{B}_i^\top(t)p, 0\right\} - \left\|\tilde{B}_i^\top(t)p\right\|_\infty$, using the fact that the ∞ norm is sub-multiplicative and applying the triangle inequality yields

$$\begin{aligned} p - s &\geq \hat{A}^\top(t)p + \sum_{i=1}^{M} \min\{r_i + \hat{B}_i^\top(t)p, 0\}E_i - \sum_{i=1}^{M} \left\|\tilde{B}_i^\top(t)p\right\|_\infty E_i \\ &\qquad - \left|\tilde{A}^\top(t)p\right| \\ &\geq \hat{A}^\top(t)p + \sum_{i=1}^{M} \min\{r_i + \hat{B}_i^\top(t)p, 0\}E_i - \|p\|_\infty \sum_{i=1}^{M} \left\|\tilde{B}_i^\top(t)\right\|_\infty E_i \\ &\qquad - \left|\tilde{A}^\top(t)\right| |p|. \end{aligned}$$

Next, we have

$$\sum_{i=1}^{M} \left\|\tilde{B}_i^\top(t)\right\|_\infty E_i \leq E^\top \mathbf{1}_M \max_i \left\|\tilde{B}_i^\top(t)\right\|_\infty \leq \left\|E^\top\right\|_\infty \left\|\tilde{B}^\top(t)\right\|_\infty \mathbf{1}_n$$

and $\left\|\left|\tilde{A}^\top(t)\right| |p|\right\|_\infty \leq \left\|\tilde{A}^\top(t)\right\|_\infty \|p\|_\infty$, which implies

$$\begin{aligned} &\|p\|_\infty \sum_{i=1}^{M} \left\|\tilde{B}_i^\top(t)\right\|_\infty E_i + \left|\tilde{A}^\top(t)\right| |p| \\ &\qquad \leq \left(\left\|E^\top\right\|_\infty \left\|\tilde{B}^\top(t)\right\|_\infty + \left\|\tilde{A}^\top(t)\right\|_\infty\right) \|p\|_\infty \mathbf{1} \\ &\qquad \leq \left(\left\|E^\top\right\|_\infty \left\|\tilde{B}^\top(t)\right\|_\infty + \left\|\tilde{A}^\top(t)\right\|_\infty\right) \beta \min_{i,\, s_i \neq 0} s_i \mathbf{1} \\ &\qquad \leq \left(\left\|E^\top\right\|_\infty \left\|\tilde{B}^\top(t)\right\|_\infty + \left\|\tilde{A}^\top(t)\right\|_\infty\right) \beta s, \end{aligned}$$

and in consequence it holds from (20) and by defining $\breve{\alpha} \coloneqq 1 - \rho\beta$ that

$$p - \breve{\alpha} s \geq \hat{A}^\top(t)p + \sum_{i=1}^{M} \min\{r_i + \hat{B}_i^\top(t)p, 0\} E_i.$$

Divide both sides by $\breve{\alpha}$ and since $r_i \geq 0$ this yields

$$\begin{aligned}\breve{\alpha}^{-1}p - s &\geq \hat{A}^\top(t)\breve{\alpha}^{-1}p + \sum_{i=1}^{M} \min\left\{\breve{\alpha}^{-1}\left(r_i + \hat{B}_i^\top(t)p\right), 0\right\} E_i \\ &\geq \hat{A}^\top(t)\breve{\alpha}^{-1}p + \sum_{i=1}^{M} \min\left\{r_i + \hat{B}_i^\top(t)\breve{\alpha}^{-1}p, 0\right\} E_i.\end{aligned}$$

An application of lemma 3 by taking $\hat{p} = \breve{\alpha}^{-1}p$ reveals that

$$p(t) \leq \breve{\alpha}^{-1}p. \tag{28}$$

(ii) Now we consider the algebraic equation in (26). Plug in

$$\begin{bmatrix}\hat{A}(t) & \hat{B}(t)\end{bmatrix} = \begin{bmatrix}A & B\end{bmatrix} + \begin{bmatrix}\tilde{A}(t) & \tilde{B}(t)\end{bmatrix}$$

yields

$$\begin{aligned}p(t) - s &= \left(A + \tilde{A}(t)\right)^\top p + \sum_{i=1}^{M} \min\left\{r_i + \left(B_i + \tilde{B}_i(t)\right)^\top p(t), 0\right\} E_i \\ &= \left(A + \tilde{A}(t)\right)^\top p(t) + \sum_{i=1}^{M} \min\left\{r_i + B_i^\top p(t) + \tilde{B}_i^\top(t)p(t), 0\right\} E_i \\ &\geq A^\top p(t) + \sum_{i=1}^{M} \min\{r_i + B_i^\top p(t), 0\} E_i - \|p(t)\|_\infty \sum_{i=1}^{M} \left\|\tilde{B}_i^\top(t)\right\|_\infty E_i \\ &\qquad - \left|\tilde{A}^\top(t)\right| |p(t)| \\ &\geq A^\top p(t) + \sum_{i=1}^{M} \min\{r_i + B_i^\top p(t), 0\} E_i \\ &\qquad - \left(\left\|E^\top\right\|_\infty \left\|\tilde{B}^\top(t)\right\|_\infty + \left\|\tilde{A}^\top(t)\right\|_\infty\right) \|p(t)\|_\infty \mathbf{1}.\end{aligned}$$

Employing (28) and (20) we get that

$$p(t) - \left(1 - \rho\beta\breve{\alpha}^{-1}\right) s \geq A^\top p(t) + \sum_{i=1}^{M} \min\{r_i + B_i^\top p(t), 0\} E_i.$$

Define $\hat{\alpha} = 1 - \rho\beta\breve{\alpha}^{-1}$ and divide both sides by $\hat{\alpha}$ leads to

$$\hat{\alpha}^{-1}p(t) - s \geq A^\top \hat{\alpha}^{-1}p(t) + \sum_{i=1}^{M} \min\{r_i + B_i^\top \hat{\alpha}^{-1}p(t), 0\} E_i.$$

Applying lemma 3 by picking $\hat{p} = \hat{\alpha}^{-1}p(t)$ yields

$$p \leq \hat{\alpha}^{-1}p(t).$$

In consequence we get that

$$\hat{\alpha}p \leq p(t) \leq \breve{\alpha}^{-1}p,$$

for positive constants $\breve{\alpha} = 1 - \rho\beta$ and $\hat{\alpha} = 1 - \rho\beta\breve{\alpha}^{-1}$. ■

A.4 Proof of Theorem 2

The following holds by letting $\hat{B}(t) = B + \tilde{B}(t)$, $\hat{A}(t) = A + \tilde{A}(t)$ in (18) and $K^\top(t)$ as in (15)

$$\begin{aligned} p(t) - s &= \left(A + \tilde{A}(t)\right)^\top p(t) + K^\top(t)\left(r + \left(B + \tilde{B}(t)\right)^\top p(t)\right) \\ &= A^\top p(t) + K^\top(t)\left(r + B^\top p(t)\right) + \tilde{A}^\top(t)p(t) + K^\top(t)\tilde{B}^\top(t)p(t) \\ &= A^\top p + K^\top(t)\left(r + B^\top p\right) + \left(\tilde{A}^\top(t) + K^\top(t)\tilde{B}^\top(t)\right)p(t) + \left(A^\top + K^\top(t)B^\top\right)(p(t) - p) \\ &\geq A^\top p + K^\top(t)\left(r + B^\top p\right) + \left(\tilde{A}^\top(t) + K^\top(t)\tilde{B}^\top(t)\right)p(t) - \left(\left|A^\top\right| + \bar{E}^\top\left|B^\top\right|\right)|p(t) - p|, \end{aligned}$$

since $|K| \leq \bar{E}$ and $A^\top + K^\top(t)B^\top \leq A^\top + \bar{E}^\top|B^\top|$. (21) gives $|p(t) - p| \leq (1 - \hat{\alpha})\, p$ since $\breve{\alpha}^{-1} - 1 = 1 - \hat{\alpha}$. Then, we get

$$\begin{aligned} \left(A^\top + \bar{E}^\top\left|B^\top\right|\right)|p(t) - p| &\leq (1 - \hat{\alpha})\left\|A^\top + \bar{E}^\top\left|B^\top\right|\right\|_\infty \|p\|_\infty \mathbf{1} \\ &\leq (1 - \hat{\alpha})\left\|A^\top + \bar{E}^\top\left|B^\top\right|\right\|_\infty \beta s \leq (1 - \hat{\alpha})\left\|A + |B|\,\bar{E}\right\|_1 \beta p \end{aligned}$$

and using (20) we have

$$\left(\tilde{A}^\top(t) + K^\top(t)\tilde{B}^\top(t)\right)p(t) \leq \rho\breve{\alpha}^{-1}\beta s \leq \rho\beta\breve{\alpha}^{-1}p,$$

yielding consequently

$$p(t) - s \geq A^\top p + K^\top(t)\left(r + B^\top p\right) - \rho\beta\breve{\alpha}^{-1}p - (1 - \hat{\alpha})\,\|A + |B|\,\bar{E}\|_1 \beta p.$$

Next, using $p(t) \leq \breve{\alpha}^{-1}p$ in the right hand side of the last inequality and letting $1 - \hat{\alpha} = \rho\beta\breve{\alpha}^{-1}$, we get the desired result

$$\breve{\alpha}^{-1}\left(1 + \rho\beta\left(1 + \beta\left\|A + |B|\,\bar{E}\right\|_1\right)\right)p - s \geq A^\top p + K^\top(t)\left(r + B^\top p\right).$$

■

A.5 Proof of Corollary 1

The following holds

$$\begin{aligned} x_{t+1}^\top p &= \left((A + BK(t))\,x(t) + B\epsilon(t) + w(t)\right)^\top p \\ &= \left((A + BK(t))x(t)\right)^\top p + \left(B\epsilon(t) + w(t)\right)^\top p \\ &\leq \left((A + BK(t))x(t)\right)^\top p + |B\epsilon(t) + w(t)|^\top p \\ &= x^\top(t)(A + BK(t))^\top p + |B\epsilon(t) + w(t)|^\top p. \end{aligned}$$

Define $\theta \coloneqq \breve{\alpha}^{-1}\left(1 + \rho\beta\left(1 + \beta\left\|A + |B|\,\bar{E}\right\|_1\right)\right)$. The application of theorem 2 then yields

$$\begin{aligned} (A + BK(t))^\top p \leq \theta p - K^\top(t)r - s &= p - K^\top(t)r - s + (\theta - 1)\,p \\ &= p - \left(s + K^\top(t)r - (\theta - 1)\,p\right). \end{aligned}$$

Next, we let γ be a constant such that

$$s + K^\top(t)r - (\theta - 1)\, p \geq \gamma \left(s + K^\top(t)r\right) \Leftrightarrow (1-\gamma)\left(s + K^\top(t)r\right) \geq (\theta - 1)\, p.$$

Since $s + K^\top(t)r \geq s - \bar{E}^\top r$ and $p \leq \beta s$ it suffices that γ satisfies

$$(1-\gamma)\left(s - \bar{E}^\top r\right) \geq (\theta - 1)\beta s \Leftrightarrow \gamma\left(s - \bar{E}^\top r\right) \leq s - \bar{E}^\top r - (\theta - 1)\beta s.$$

Such γ is guaranteed to exist by virtue of Assumption 2.2. Then, it holds that

$$(A + BK(t))^\top p \leq p - \gamma\left(s + K^\top(t)r\right),$$

and also that

$$\begin{aligned} x_{t+1}^\top p &\leq x^\top(t)p - \gamma x^\top(t)\left(s + K^\top(t)r\right) + |B\epsilon(t) + w(t)|^\top p \\ &\leq x^\top(t)p - \gamma x^\top(t)\left(s + K^\top(t)r\right) + |B\epsilon(t) + w(t)|^\top \beta s. \end{aligned}$$

Re-adjusting yields

$$\gamma\left(x^\top(t)s + x^\top(t)K^\top(t)r\right) \leq x^\top(t)p - x_{t+1}^\top p + \beta\,|B\epsilon(t) + w(t)|^\top s.$$

Taking the transpose and dividing both sides by γ yields

$$s^\top x(t) + r^\top K(t)x(t) \leq \gamma^{-1}\left(p^\top x(t) - p^\top x_{t+1} + \beta s^\top |B\epsilon(t) + w(t)|\right)$$

Summing over $t_0 \leq t \leq T-1$ yields the following desired result

$$\sum_{t=t_0}^{T-1} s^\top x(t) + r^\top K(t)x(t) \leq \gamma^{-1}\left(p^\top x(t_0) + \sum_{t=t_0}^{T-1} \beta s^\top |B\epsilon(t) + w(t)|\right).$$

■

References

Åström, K. and B. Wittenmark (1995). *Adaptive Control (2 ed.)* English. Addison-Wesley. ISBN: 0-201-55866-1. DOI: 10.2307/1269433.

Bellman, R. (1966). "Dynamic programming". *Science* **153**:3731, pp. 34–37.

Bencherki, F. and A. Rantzer (2023). "Robust simultaneous stabilization via minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 2503–2508.

Bencherki, F. and A. Rantzer (2024a). "Data-driven adaptive dispatching policies for processing networks". *IEEE Control Systems Letters* **8**, pp. 2841–2846. DOI: 10.1109/LCSYS.2024.3516637.

Bencherki, F. and A. Rantzer (2024b). *Robust adaptive data-driven control of positive systems with application to learning in ssp problems*. arXiv: 2412.17012 [math.OC].

Blanchini, F., P. Colaneri, M. E. Valcher, et al. (2015). "Switched positive linear systems". *Foundations and Trends® in Systems and Control* **2**:2, pp. 101–273.

Blanchini, F. and G. Giordano (2014). "Piecewise-linear lyapunov functions for structural stability of biochemical networks". *Automatica* **50**:10, pp. 2482–2493.

Bradtke, S. J., B. E. Ydstie, and A. G. Barto (1994). "Adaptive linear quadratic control using policy iteration". In: *Proc. Am. Control Conf.* Vol. 3. IEEE, pp. 3475–3479.

Carson, E. and C. Cobelli (2013). *Modelling methodology for physiology and medicine*. Newnes.

De Leenheer, P. and D. Aeyels (2001). "Stabilization of positive linear systems". *Systems & control letters* **44**:4, pp. 259–271.

De Persis, C. and P. Tesi (2019). "Formulas for data-driven control: stabilization, optimality, and robustness". *IEEE Transactions on Automatic Control* **65**:3, pp. 909–924.

Dörfler, F., P. Tesi, and C. De Persis (2023). "On the certainty-equivalence approach to direct data-driven lqr design". *IEEE Transactions on Automatic Control* **68**:12, pp. 7989–7996.

Ebihara, Y., D. Peaucelle, and D. Arzelier (2016). "Analysis and synthesis of interconnected positive systems". *IEEE Transactions on Automatic Control* **62**:2, pp. 652–667.

Gurpegui, A., M. Jeeninga, E. Tegling, and A. Rantzer (2024). "Minimax linear regulator problems for positive systems". *arXiv preprint arXiv:2411.04809*.

Gurpegui, A., E. Tegling, and A. Rantzer (2023). "Minimax linear optimal control of positive systems". *IEEE Control Systems Letters*.

Haddad, W. M., V. Chellaboina, and Q. Hui (2010). *Nonnegative and compartmental dynamical systems*. Princeton University Press.

Hernandez-Vargas, E. A. and R. H. Middleton (2013). "Modeling the three stages in hiv infection". *Journal of theoretical biology* **320**, pp. 33–40.

Iwata, T., S.-i. Azuma, R. Ariizumi, and T. Asai (2024). "Data informativity for distributed positive stabilization". *IEEE Control Systems Letters*.

Kjellqvist, O. and A. Rantzer (2022a). "Learning-enabled robust control with noisy measurements". In: *Learning for Dynamics and Control Conference*. PMLR, pp. 86–96.

Kjellqvist, O. and A. Rantzer (2022b). "Minimax adaptive estimation for finite sets of linear systems". In: *2022 American Control Conference (ACC)*. IEEE, pp. 260–265.

Konstantinov, M. M., P. H. Petkov, and N. D. Christov (1993). "Perturbation analysis of the discrete riccati equation". *Kybernetika* **29**:1, pp. 18–29.

Konstantinov, M., D. W. Gu, V. Mehrmann, and P. Petkov (2003). *Perturbation theory for matrix equations*. Gulf Professional Publishing.

Li, Y. and A. Rantzer (2024). "Exact dynamic programming for positive systems with linear optimal cost". *IEEE Trans. Autom. Control*.

Makdah, A. A. A. and F. Pasqualetti (2023). "Model-based and data-based dynamic output feedback for externally positive systems". *arXiv preprint arXiv:2305.02472*.

Mania, H., S. Tu, and B. Recht (2019). "Certainty equivalence is efficient for linear quadratic control". *Advances in Neural Information Processing Systems* **32**.

Miller, J., T. Dai, M. Sznaier, and B. Shafai (2023). "Data-driven control of positive linear systems using linear programming". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 1588–1594.

Ohlin, D., A. Rantzer, and E. Tegling (2024a). "Heuristic search for linear positive systems". *arXiv preprint arXiv:2410.17220*.

Ohlin, D., E. Tegling, and A. Rantzer (2024b). "Optimal control of linear cost networks". *Eur. J. Control*, p. 101068.

Padoan, A., F. Dörfler, and J. Lygeros (2023). "Data-driven representations of conical, convex, and affine behaviors". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 596–601.

Rami, M. A. and F. Tadeo (2007). "Controller synthesis for positive linear systems with bounded controls". *IEEE Transactions on Circuits and Systems II: Express Briefs* **54**:2, pp. 151–155.

Rantzer, A. (2015). "On the kalman-yakubovich-popov lemma for positive systems". *IEEE Transactions on Automatic Control* **61**:5, pp. 1346–1349.

Rantzer, A. (2021). "Minimax adaptive control for a finite set of linear systems". In: *Learning for Dynamics and Control*. PMLR, pp. 893–904.

Rantzer, A. (2024). "A data-driven riccati equation". In: *6th Annual Learning for Dynamics & Control Conference*. PMLR, pp. 504–513.

Rantzer, A. and M. E. Valcher (2018). "A tutorial on positive systems and large scale control". In: *2018 IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 3686–3697.

Renganathan, V., A. Iannelli, and A. Rantzer (2023). "An online learning analysis of minimax adaptive control". In: *2023 62nd IEEE Conference on Decision and Control (CDC)*. IEEE, pp. 1034–1039.

Shafai, B., A. Moradmand, and M. Siami (2022). "Data-driven positive stabilization of linear systems". In: *2022 8th International Conference on Control, Decision and Information Technologies (CoDIT)*. Vol. 1. IEEE, pp. 1031–1036.

Shorten, R., F. Wirth, and D. Leith (2006). "A positive systems model of tcp-like congestion control: asymptotic results". *IEEE/ACM transactions on networking* **14**:3, pp. 616–629.

Simchowitz, M. and D. Foster (2020). "Naive exploration is optimal for online lqr". In: *International Conference on Machine Learning*. PMLR, pp. 8937–8948.

Sun, J.-G. (1998). "Perturbation theory for algebraic riccati equations". *SIAM Journal on Matrix Analysis and Applications* **19**:1, pp. 39–65.

Sutton, R. S. and A. G. Barto (2018). *Reinforcement learning: An introduction*. MIT press.

Al-Tamimi, A., F. L. Lewis, and M. Abu-Khalaf (2007). "Model-free q-learning designs for linear discrete-time zero-sum games with application to h-infinity control". *Automatica* **43**:3, pp. 473–481.

Tsiamis, A., I. Ziemann, N. Matni, and G. J. Pappas (2023). "Statistical learning theory for control: a finite-sample perspective". *IEEE Control Systems Magazine* **43**:6, pp. 67–97.

Wang, Y. and B. Shafai (2024). "Data-driven identification and control of positive systems". In: *Integrated Systems: Data Driven Engineering*. Springer, pp. 289–307.

Yu, H. and D. P. Bertsekas (2013). "On boundedness of q-learning iterates for stochastic shortest path problems". *Mathematics of Operations Research* **38**:2, pp. 209–227.

Zhao, F., F. Dörfler, A. Chiuso, and K. You (2024). "Data-enabled policy optimization for direct adaptive learning of the lqr". *arXiv preprint arXiv:2401.14871*.

Paper III

Data-Driven Adaptive Dispatching Policies for Processing Networks

Fethi Bencherki Anders Rantzer

Abstract

This letter presents and analyzes an adaptive data-driven controller that learns the optimal processing rate in a multi-unit processing network in the presence of disturbances. We formulate an optimization problem of linear cost, linear dynamics for the processing network model and an affine constraint on the dispatcher policy. A data-driven linear equation is constructed, based on which the online dispatcher policy is updated. An upper bound on the gap between the optimal cost and the cost incurred by the data-driven controller is extracted.

©2024 IEEE. Reprinted, with permission, from F. Bencherki and A. Rantzer (2024).
"Data-driven adaptive dispatching policies for processing networks". *IEEE Control Systems Letters* **8**, pp. 2841–2846. DOI: 10.1109/LCSYS.2024.3516637

1. Introduction

Processing networks take different forms and include communication networks, supply chains, resource allocation, manufacturing and even healthcare [Cochran and Roche, 2009]. The optimal control of processing networks is of great importance, since improving system efficiency and reducing operational costs hinges on it [Bertsimas et al., 2014; Bertsimas and Kim, 2023]. However, the high complexity of these systems renders this task quite challenging.

The optimal control of processing networks has been attempted under various assumptions, for instance the availability of processing rates [Chen et al., 2004; Tassiulas and Ephremides, 1990]; see the reference book [Meyn, 2008] and the tutorial in [Walton and Xu, 2021] for a detailed treatment of the subject. However, there are many scenarios where these parameters are unknown and the dispatcher needs to learn them first in order to be able to assign jobs optimally [Adler et al., 2022]. An instance of this appears in the recently emerging applications such as cloud computing where the servers may encounter interruptions and service slowdown. Therefore, the processing rates can be unknown, highly variable across the different units, and potentially changing over time [Choudhury et al., 2021].

The challenging nature of the optimal control problem of processing networks in the case of unknown processing rates instigated a plethora of research efforts, mainly focusing on learning-based control techniques to tackle the problem, see the works in [Krishnasamy et al., 2021; Stahlbuhk et al., 2021; Yang et al., 2023; Cohen et al., 2024; Qu et al., 2018; Adler et al., 2022; Liu et al., 2022]. The current work concerns employing adaptive dual control [Wittenmark, 1995] as an approach that both attempts to learn the processing rates and take optimal online decision making for a linear multi-unit processing network.

1.1 Outline and Contribution of the Letter

A discrete time linear multi-unit processing network model is proposed and an online adaptive data-driven dispatching policy is analyzed in the presence of process disturbances. The data-driven policy assumes the availability of noisy waiting jobs measurements collected across the different processing units. To tolerate uncertainties in regard to unmodelled dynamics in the proposed model, we refrain from putting any statistical assumptions on the disturbances. This grants the proposed scheme inherent performance robustness guarantees.

The outline of this work is as follows: Section 2 presents the problem setup and defines the problem. Section 3 presents the main results of the letter. Section 4 is dedicated to numerical examples to showcase the effectiveness of the proposed approach and lastly Section 5 gives concluding remarks. The proofs are kept in the appendix.

1.2 Notation

Inequalities are applied element-wise for matrices and vectors throughout. Furthermore, the notation $\mathbb{R}^n_+$ is used to denote the closed positive orthant of dimension n. The boldfaced $\mathbf{1}$ indicates the vector of all ones. $[x_t]_i$ is used to access i-th entry of a vector $x_t \in \mathbb{R}^n$ at time t. e_i is a standard basis vector with 1 at the i-th entry and zero elsewhere. The operation $\mathrm{diag}(x)$ returns a diagonal matrix with x along its diagonal. $U(a, b)$ is the random uniform distribution between a and b.

2. Problem Setup

2.1 A Multi-Unit Processing Network Model

We consider a multi-unit processing network model consisting of n units, with jobs awaiting to be processed at each unit. The number of external jobs arriving at the network is a deterministic quantity denoted by λ, to be forwarded by the dispatcher to the different units for processing. We let $[u_t]_i$ denotes the portion of jobs routed to unit i. The units process the jobs at different processing rates denoted by $\eta_i > 1$ for unit $i \in \{1, \ldots, n\}$. This setup is illustrated in Figure 1. We let the vector $[x_t]_i$ represent the number of jobs awaiting at unit i at time t, and adhering to the following linear dynamics

$$[x_{t+1}]_i = \tfrac{1}{\eta_i}[x_t]_i + [u_t]_i, \quad \eta_i > 1, \tag{1}$$

and in aggregation across all units

$$x_{t+1} = Mx_t + u_t, \quad M \coloneqq \mathrm{diag}\left(\eta_1^{-1}, \ldots, \eta_n^{-1}\right). \tag{2}$$

Note that according to (1), unit i possessing a large processing rate η_i indicates that it enjoys a faster linear convergence with a rate of $0 < \eta_i^{-1} < 1$. The inputs, on the other hand, adhere to the constraints

$$\mathbf{1}^\top u_t = \lambda \quad \text{and} \quad u_t \geq 0 \quad \forall t. \tag{3}$$

The equality constraint on the inputs could be justified in the sense that we would like the incoming jobs λ at time t to be fully distributed for processing. The inequality constraint on the other hand to signify that a job is a non-negative quantity.

2.2 An Optimal Control Problem

A typical performance objective is to minimize the sum of awaiting jobs across all units. This gives rise to the following optimization problem with a discount factor

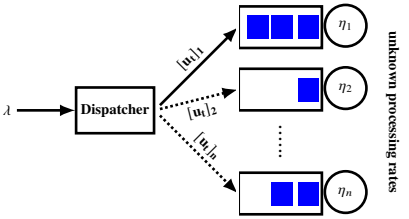


Figure 1. An illustration of the multi-unit processing network.

$\gamma \in (0,1)$

$$\begin{aligned} \text{Minimize} \quad & \sum_{t=0}^{\infty} \gamma^t \mathbf{1}^\top x_t \text{ over } \{u_t\}_{t=0}^{\infty} \\ \text{subject to} \quad & x_{t+1} = Mx_t + u_t, \\ & \mathbf{1}^\top u_t = \lambda \quad \text{and} \quad u_t \geq 0, \quad \forall t. \end{aligned} \tag{4}$$

This optimal control problem falls under the classes presented in [Li and Rantzer, 2024; Ohlin et al., 2024] with minor modifications.

2.3 Optimal Policy for Known Processing Rates

In the case the processing rates are known to the dispatcher, we could solve (4) via dynamic programming [Bellman, 1966]. The general constrained optimal control problem (22) presented in Appendix A.1 reduces to our problem setup (4) if we let

$$\begin{aligned} g(x,u) &\coloneqq \mathbf{1}^\top x, \quad f(x,u) \coloneqq Mx+u, \\ \mathcal{X} &\coloneqq \mathbb{R}_+^n, \quad \mathcal{U}(x_t,\lambda) \coloneqq \mathcal{U}(\lambda), \end{aligned}$$

where the set of inputs satisfying (3) are denoted as $\mathcal{U}(\lambda)$. This leads to the Bellman optimality equation

$$J(x) = \min_{u \in \mathbf{U}(\lambda)} \left[\mathbf{1}^\top x + \gamma J(Mx+u) \right]. \tag{5}$$

Since (4) is characterized by linear cost, linear dynamics and affine constraints on the inputs, we propose an affine ansatz for the value function J, i.e., we let $J(x) = p^\top x + p^0$ where $p \in \mathbb{R}_+^n$ and $p^0 \in \mathbb{R}_+$. Plugging in (5) and simplifying yields

$$p^\top x + p^0 = \mathbf{1}^\top x + \gamma p^\top Mx + \gamma p^0 + \gamma \min_{u \in \mathcal{U}(\lambda)} p^\top u.$$

Solving we get

$$p_i = \frac{\eta_i}{\eta_i - \gamma} \text{ for } i = 1, \ldots, n, \tag{6}$$

$$p^0 = \frac{\gamma \lambda \min_i (p_i)}{1-\gamma} \tag{7}$$

and the optimal policy is

$$u^* = \lambda e_i \quad \text{where} \quad i = \arg\min_j \frac{\eta_j}{\eta_j - \gamma} = \arg\max_j \eta_j.$$

The optimal policy expression promotes forwarding all incoming jobs to the unit with the highest processing rate.

2.4 Model-Free Optimal Control via Q-factor

The cost-to-go function from time t for the optimization problem in (4) for a control policy u beginning at time t from state x_t is $J(x_t) \coloneqq \min_u \sum_{k=t}^{\infty} \gamma^{k-t} \mathbf{1}^\top x_k$, with the stage cost given by $c(x_k, u_k) \coloneqq \gamma^{k-t} \mathbf{1}^\top x_k$ for $k \geq t$. As seen in Section 2.3, this optimization problem yields the objective value $J(x_t) = p^\top x_t + p^0$ where computing p and p^0 requires the knowledge of the processing rates vector η. The optimal Q-function as defined in [Bradtke et al., 1994] is given by

$$Q(x_t, u_t) \coloneqq c(x_t, u_t) + \gamma J(x_{t+1}), \tag{8}$$

which represents the cost of taking action u_t starting at state x_t and following by the optimal policy u^* afterwards. Then, it follows that $Q(x_t, u_t) = \mathbf{1}^\top x_t + \gamma p^\top (M x_t + u_t) + \gamma p^0$, where we used the fact that $J(x_{t+1}) = p^\top (M x_t + u_t) + p^0$. The optimal Q-factor is then given by

$$Q(x_t, u_t) = \begin{bmatrix} \mathbf{1}^\top + \gamma p^\top M & \gamma p^\top \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix} + \gamma p^0 = q^\top \begin{bmatrix} x_t \\ u_t \end{bmatrix} + q^0, \tag{9}$$

where $q = \begin{bmatrix} q^x \\ q^u \end{bmatrix} \coloneqq \begin{bmatrix} \mathbf{1} + \gamma M p \\ \gamma p \end{bmatrix}$, $q^0 \coloneqq \gamma p^0$ and

$$q^u = \gamma p = \gamma \left(\mathbf{1} + \gamma M p \right) = \gamma q^x, \tag{10}$$

$$q^0 = \gamma p^0 = \frac{\gamma \lambda}{1-\gamma} \min_i (\gamma p_i) = \frac{\gamma^2 \lambda}{1-\gamma} \min_i [q^x]_i. \tag{11}$$

It also holds that $J(x_t) = \min_{u_t \in \mathcal{U}(\lambda)} [c(x_t, u_t) + \gamma J(x_{t+1})] = \min_{u_t \in \mathcal{U}(\lambda)} Q(x_t, u_t)$. Plugging this last equation in the right hand side of (8) yields

$$Q(x_t, u_t) = c(x_t, u_t) + \gamma \min_{u_{t+1} \in \mathcal{U}(\lambda)} Q(x_{t+1}, u_{t+1}), \tag{12}$$

and the optimal policy is $u_t^* = \arg\min_{u_t \in \mathcal{U}(\lambda)} Q(x_t, u_t)$. One can think of (12) as the Bellman equation in the Q-factor [Sutton and Barto, 2018]. Inserting (9) in (12) yields

$$\begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} x_t \\ u_t \end{bmatrix} + q^0 = \mathbf{1}^\top x_t + \gamma \min_{u_{t+1} \in \mathcal{U}(\lambda)} \begin{bmatrix} q^x \\ q^u \end{bmatrix}^\top \begin{bmatrix} x_{t+1} \\ u_{t+1} \end{bmatrix} + \gamma q^0,$$

which is equivalent to

$$\left(q - \begin{bmatrix}\mathbf{1}\\ 0\end{bmatrix}\right)^\top \begin{bmatrix}x_t\\ u_t\end{bmatrix} = \gamma q^\top \begin{bmatrix}x_{t+1}\\ 0\end{bmatrix} + \beta \tag{13}$$

where $\beta \coloneqq (\gamma-1)q^0 + \gamma\lambda \min_i [q^u]_i$. Using (10)-(11) we notice that $\beta = (\gamma - 1)\frac{\gamma^2\lambda}{1-\gamma}\min_i [q^x]_i + \gamma^2\lambda \min_i [q^x]_i = 0$. We plug (10) in (13) and set $\beta = 0$ to get the model-free update

$$(q^x - \mathbf{1})^\top x_t = \gamma (q^x)^\top (x_{t+1} - u_t). \tag{14}$$

Despite the absence of the knowledge of processing rates, (14) makes it possible to get information about q^x by collecting triplets (x_t, u_t, x_{t+1}). In fact, collecting t data points gives

$$(q^x - \mathbf{1})^\top \begin{bmatrix} x_0 & \cdots & x_{t-1}\end{bmatrix} = \gamma (q^x)^\top \begin{bmatrix} x_1 - u_0 & \cdots & x_t - u_{t-1}\end{bmatrix} \tag{15}$$

Multiply (15) by $Z_t^\top$ from the right where $Z_t \coloneqq \begin{bmatrix} x_0 & \cdots & x_{t-1}\end{bmatrix}$ and defining the data correlation matrices

$$\Sigma_t = \sum_{k=0}^{t-1} x_k x_k^\top + \Sigma_0 \text{ and } \bar{\Sigma}_t = \sum_{k=0}^{t-1} (x_{k+1} - u_k) x_k^\top \tag{16}$$

gives the linear data-driven equation in the q^x parameter

$$(q^x - \mathbf{1})^\top \Sigma_t = \gamma (q^x)^\top \bar{\Sigma}_t, \tag{17}$$

where we have added a regularization term Σ_0 to render Σ_t invertible even for small value of t. (17) makes it possible to estimate the optimal q^x parameter directly from the data via

$$\begin{cases} q_t^x \coloneqq (I - \gamma \Sigma_t^{-1} \bar{\Sigma}_t^\top)^{-1}\mathbf{1}, \\ q_t^u \coloneqq \gamma q_t^x, \\ q_t^0 \coloneqq \frac{\gamma^2\lambda}{1-\gamma} \min_{i_t} [q^x]_{i_t}. \end{cases} \tag{18}$$

where the estimate is denoted by q_t^x. Lastly, if the true dynamics M is known we replace $\Sigma_t^{-1}\bar{\Sigma}_t^\top$ in (18) by M, yielding its model-based counterpart. (18) forms the basis of the proposed policy construction as shall be seen in the sequel.

2.5 Problem Formulation

In this letter, we assume that the dispatcher does not know the processing rates and has to learn them while scheduling dispatching policies in an online fashion. After dispatching a job at time t, it receives a noisy observation of the number of awaiting jobs across all units at the end of time slot $[t, t+1]$, denoted by x_{t+1} and given by

$$x_{t+1} = Mx_t + u_t + w_t. \tag{19}$$

Inspired by (18), we propose and analyze the performance of the following dispatching policies in controlling system (19)

$$\begin{cases} \Sigma_t = \Sigma_{t-1} + x_{t-1}x_{t-1}^\top, \quad \Sigma_0 \succ 0 \\ \bar{\Sigma}_t = \bar{\Sigma}_{t-1} + (x_t - u_{t-1})x_{t-1}^\top, \quad \bar{\Sigma}_0 = 0 \\ q_t^x = (I - \gamma \Sigma_t^{-1}\bar{\Sigma}_t^\top)^{-1}\mathbf{1}, \\ u_t = \lambda e_{i_t} + \epsilon_t \quad \text{where} \quad i_t = \arg\min_j [q_t^x]_j. \end{cases} \tag{20}$$

A control input u_{t-1} is applied to the system at state x_{t-1} leading to a new state x_t, and a triplet (x_{t-1}, u_{t-1}, x_t) is collected and used to update the correlation matrices $(\Sigma_t, \bar{\Sigma}_t)$ of the policy (20). After that, an estimate of the q^x parameter, denoted by q_t^x, is obtained, based on which the first component of the next policy u_t is extracted. The second component, ϵ_t, provides explorations towards learning the best processing rate. ϵ_t is also the probing action in dual control [Wittenmark, 1995].

REMARK 2.1
The introduction of $\Sigma_0 \succ 0$ ensures the invertibility of Σ_t for all t. However, it introduces an initial bias in the estimate q_t^x, which is diminishing as more data is collected. Later analysis, such as Theorems 1 and 2, therefore has error bounds which are only valid after some initial time t_0.

REMARK 2.2
Controllers of the type (20) are known as adaptive certainty equivalence controllers in the adaptive control literature [Åström and Wittenmark, 1995], since u_t could be found via the model estimate $\hat{M}_t := \Sigma_t^{-1}\bar{\Sigma}_t^\top$ in (18), disregarding the uncertainty incurred in the estimation. Another popular instant of adaptive certainty equivalence controllers is the self-tuning regulator (STR) [Åström and Wittenmark, 1973].

The assumption of a linear processing model is robustified by the presence of the additive disturbances w_t [Zhou and Doyle, 1998]. These could account for on/off periods for the units activity, extra traffic that is not under control and unmodelled dynamics corresponding to scenarios/behaviors not captured by the current linear model. Motivated by this, we refrain from putting any statistical assumptions on w_t.

3. Main Results

3.1 Positive Least-Squares Estimation

We start by defining $\tilde{\Sigma}_t := \sum_{k=0}^{t-1} w_k x_k^\top$. This implies that $\bar{\Sigma}_t = M\Sigma_t + \tilde{\Sigma}_t$ and therefore $\Sigma_t^{-1}\bar{\Sigma}_t^\top = \hat{M}_t = M + \tilde{M}_t$, where $\tilde{M}_t := \Sigma_t^{-1}\tilde{\Sigma}_t^\top$ is the model misspecification. It should be indicated that $\tilde{M}_t$ Need not be positive or diagonal.

Lemma 1 in Appendix B.1 indicates that the ground truth q^x parameter is positive. It is therefore desirable that its least-squares estimate, q_t^x, imitates that. We

show that as long as $\tilde{M}_t$ is small enough, it is indeed the case. The definition of $\tilde{M}_t$ gives $q_t^x = (I - \gamma M - \gamma \tilde{M}_t)^{-1}\mathbf{1}$. If we define $\mathcal{M} \coloneqq I - \gamma M - \gamma \tilde{M}_t$, a sufficient condition for the positivity of q_t^x is that $\mathcal{M}^{-1}$ is strictly row diagonally dominant with positive diagonal entries, since then $\mathcal{M}^{-1}\mathbf{1} > 0$. A sufficient condition ensuring that $\mathcal{M}$ is strictly row diagonally dominant with positive diagonal entries is presented next.

LEMMA 1
Let $\tilde{M}_t$ be such that $\|\tilde{M}_t\|_{1,\infty} < \frac{1}{\gamma} - \max_i M_{ii}$. Then, the matrix $\mathcal{M}$ is strictly row and column diagonally dominant with positive diagonal entries.

Proof. See appendix-B.2. ∎

REMARK 3.1
The strict row diagonal dominance of $\mathcal{M}$ ensures it is nonsingular [Johnson et al., 2023].

The diagonal entries of $\mathcal{M}^{-1}$ are positive since strictly diagonally dominant matrices possess the inverse diagonal property (IDP), i.e. matrices for which their diagonal entries sign matches their inverses' [Furtado et al., 2024]. However and according to the recent result in [Johnson et al., 2024], the inverse of a strictly column diagonally dominant matrix is a matrix that is strictly diagonally dominant of its row entries, a weaker form of diagonal dominance. Hence, to ensure that $\mathcal{M}^{-1}$ is strictly diagonally dominant in the desired sense, we impose a stronger assumption on $\tilde{M}_t$.

LEMMA 2
Let $\tilde{M}_t$ satisfies $\|\tilde{M}_t\|_{1,\infty} < \frac{1}{n-1}(\frac{1}{\gamma} - \max_i M_{ii})$. Then, both $\mathcal{M}$ and $\mathcal{M}^{-1}$ are strictly row diagonally dominant with positive diagonal entries.

Proof. See appendix-B.3. ∎

3.2 Scalability Properties by Virtue of Diagonal Dominance

In the event that n is very large, solving the system of linear equations for q_t^x could be done efficiently given the strict diagonal dominance property the matrix $\mathcal{M}$ enjoys. Further nice computational properties associated with this are:

(i) A very popular method for solving a system of linear equations is Gaussian elimination, achieved by finding an LU factorization of the matrix $\mathcal{M}$. Gaussian elimination without pivoting can be unstable. No pivoting is necessary for a strictly column diagonally dominant matrix [Golub and Van Loan, 2013].

(ii) For very large n, solving the system of linear equations via factorization can be expensive. It is more efficient to use iterative methods such as Jacobi or Gauss-Seidel methods. Jacobi and Gauss–Seidel methods are guaranteed to converge if the matrix is strictly diagonally dominant [Bagnara, 1995].

Next, we analyze the performance of the policy in (20).

3.3 Performance Analysis

We first provide error bounds in the (q^x, q^u, q^0) parameters estimation. This error bound is extracted assuming the diagonal dominance property of $\mathcal{M}$ established in section 3.1.

THEOREM 1
Consider $\rho \in \mathbb{R}_+$, $\gamma \in (0,1)$, a positive definite matrix Σ_t and a matrix $\bar{\Sigma}_t$. Suppose that M is a diagonal positive schur matrix and that for all $t \geq t_0$

$$\|\Sigma_t^{-1}\bar{\Sigma}_t^\top - M\|_{1,\infty} \leq \rho \quad \textit{for} \quad \rho < \tfrac{1}{\gamma} - \max_i M_{ii}. \tag{21}$$

Let (q^x, q^u, q^0) and (q_t^x, q_t^u, q_t^0) be given by

$$\begin{cases} q^x = (I-\gamma M)^{-1}\mathbf{1}, \\ q^u = \gamma q^x, \\ q^0 = \frac{\gamma^2\lambda}{1-\gamma}\min_i [q^x]_i, \end{cases} \qquad \begin{cases} q_t^x = (I-\gamma\Sigma_t^{-1}\bar{\Sigma}_t^\top)^{-1}\mathbf{1}, \\ q_t^u = \gamma q_t^x, \\ q_t^0 = \frac{\gamma^2\lambda}{1-\gamma}\min_i [q_t^x]_i. \end{cases}$$

Then, it holds that

$$\left\| \begin{bmatrix} q_t \\ q_t^0 \end{bmatrix} - \begin{bmatrix} q \\ q^0 \end{bmatrix} \right\| \leq \frac{\gamma\rho n(1-\gamma\min_i M_{ii})}{(1-\gamma\max_i M_{ii})(1-\gamma\max_i M_{ii}-\gamma\rho)} \begin{bmatrix} q \\ q^0 \end{bmatrix}.$$

Proof. See the appendix-B.4. ∎

REMARK 3.2
The inequality (21) can benefit from all data collected up until t_0. In the case the disturbance sequence w_t is chosen as i.i.d. Gaussian noise, a large value of t_0 typically will make it possible to satisfy (21) with a smaller value for ρ. Hence, more data gives better bounds. However, if faced with adversarial disturbances we may never learn the true dynamics since the adversary w could fool us forever.

REMARK 3.3
Assumption (21) will be valid as long as two conditions are met: 1) The condition number of Σ_t must not be too large. This is a condition on explorations. 2) The disturbance sequence w_t must be sufficiently small compared to x_t. If w_t corresponds to unmodelled dynamics, it means that such dynamics are subject to a gain bound from state to disturbance.

REMARK 3.4
The formula (21) could also be written as $\|(\sum_{k=0}^{t-1} x_k x_k^\top)^{-1}\sum_{k=0}^{t-1} x_k w_k^\top\|_{1,\infty} \leq \rho$. This bound, viewed as a gain bound from x_t to w_t, is a standard way of quantifying tolerance to unmodelled dynamics and time-variations in the plant in line with the robust control literature [Zhou and Doyle, 1998]. Besides its robustness guarantees, the policy in (20) continuously updates its belief on what the true q^x parameter is, which also makes it adapt to slow time-variations.

The main result of the letter is stated next.

THEOREM 2
Let (p, p_0) be as in (6)-(7) while $\gamma \in (0,1)$, $\rho \in \mathbb{R}_+$ and M being a diagonal positive Schur matrix. Let

$$\begin{cases} x_{t+1} = Mx_t + u_t + w_t, \\ q_t^x = (I - \gamma \Sigma_t^{-1} \bar{\Sigma}_t^\top)^{-1} \mathbf{1}, \\ u_t = \lambda e_{i_t} + \epsilon_t \quad \textit{where} \quad i_t = \arg\min_j [q_t^x]_j \end{cases}$$

and Σ_t and $\bar{\Sigma}_t$ are defined as in (16) while (21) is assumed to hold for all $t \geq t_0$. Then, it holds that

$$\sum_{t=t_0}^{\infty} \gamma^{t-t_0} \mathbf{1}^\top x_t - J(x_{t_0}) \leq \frac{1}{1-\gamma \max_i M_{ii}} \sum_{t=t_0}^{\infty} \gamma^{t-t_0+1} \mathbf{1}^\top |\epsilon_t + w_t| + \frac{2\gamma^2 \rho n \lambda / (1-\gamma)}{(1-\gamma \max_i M_{ii})(1-\gamma \max_i M_{ii} - \gamma\rho) - \gamma\rho n (1-\gamma \min_i M_{ii})}.$$

where $J(x_{t_0}) = p^\top x_{t_0} + p^0$ is the optimal cost.

proof. See the appendix-B.5. ∎

REMARK 3.5
Theorem 2 quantifies the suboptimality incurred in the cost function due to applying the data-driven controller instead of the optimal controller. The first term in the right hand side captures the cost incurred due to disturbances w_t and explorations ϵ_t. The second term captures the effect of model misspecification and vanishes if $\rho \to 0$.

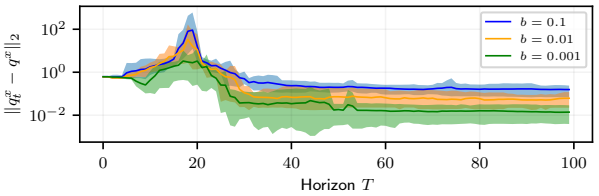
3.4 The Choice of Explorations ϵ_t

Though the choice of explorations is not the primary focus of the paper, standard strategies for Multi-Armed Bandit (MAB) problems, such as ϵ-greedy, ϵ-decreasing, adaptive ϵ-greedy, etc are also applicable in our setup. In graph settings, one could also adopt a graph-intuited approach, for instance forwarding most of the available job to the estimated best unit and using the remaining jobs to explore the rest of the units, for example uniformly as is done in the numerical experiments.

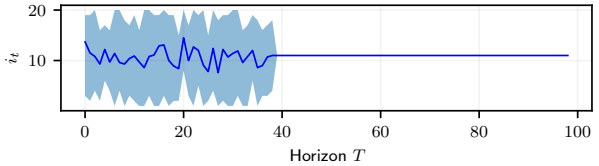
4. Numerical Examples

4.1 Performance on a Random Processing Network

We consider a processing network consisting of $n = 20$ units with processing rates and initial condition generated at random. We select $\lambda = 5$, $\gamma = 0.7$, $\Sigma_0 = 10^{-6} I$ and deploy the policy in (20) with continual ε_t-uniform explorations, i.e. that is if at



(a) The error in the q^x parameter estimation.



(b) The estimated best unit index in terms of processing rate at $b = 0.01$.

Figure 2. The online performance of the policy in (20) for different disturbance levels for 100 noise different realizations.

time t unit j is estimated to have the best processing rate, then the deployed policy would be

$$u_t = (\lambda - \varepsilon_t)e_j + (\mathbf{1} - e_j)\tfrac{\varepsilon_t}{n-1} = \lambda e_j + \underbrace{\left(\tfrac{\mathbf{1}-e_j}{n-1} - e_j\right)\varepsilon_t}_{\epsilon_t :=},$$

where $0 < \varepsilon_t \ll \lambda$ is randomly selected. The disturbances are drawn according to $w_t \sim U(0, b)$ for all t. The error in the q^x parameter estimation and the estimated best unit as a function of algorithm horizon T for 100 different disturbance realizations for different values of b are shown in Figure 2. Note the initial exploration phase before the controller identifies the true q^x parameter and that unit 11 possesses the best processing rate. Figure 3, on the other hand, plots a test of the condition in (21) while selecting $\rho = 0.9 \leq \frac{1}{\gamma} - \max_i M_{ii} = 1.35$. Note that the condition holds true for $t \geq t_0 = 42$.

4.2 Adaptation in the Face of Unmodelled Dynamics

We consider a two unit processing network

$$x_{t+1} = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.4 \end{bmatrix} x_t + u_t + w_t,$$

where $x_0 = 0$ and the second unit experiences service shut down, i.e. $\eta_2 = 1$ during the time interval $t \in [t_1, t_2]$ with $t_2 - t_1 = 20$ and $t_1 \geq 5n$ is selected at random. To

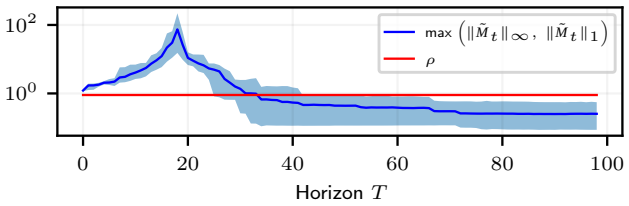


Figure 3. The condition in (21) for $b = 0.01$ and $\rho = 0.9$.

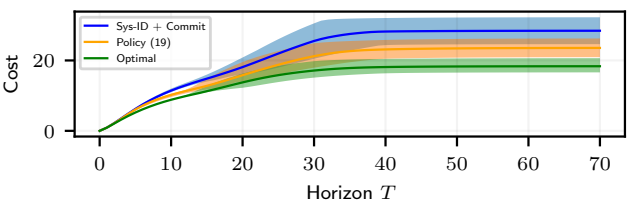


Figure 4. performance comparison between (20), the optimal policy and STC in the presence of unmodelled dynamics.

model this, we let $w_t = \begin{bmatrix} 0 & 0.6[x_t]_2 \end{bmatrix}^\top$ for $t_1 \leq t \leq t_2$ and 0 elsewhere. We deploy the policy in (20) with $\Sigma_0 = 10^{-6} I$, $\lambda = 5$, $\gamma = 0.9$ and ε_t-uniform explorations. We compare the cost of the policy in (20) to the cost of the optimal policy which switches to the first unit during unit 2 inactivity and to a sys-ID then commit (STC) approach with identification horizon of $5n = 10$. The comparison is shown in figure 4 for 100 different realizations of t_1. Note that our approach outperforms STC.

5. Conclusion

The letter presented an adaptive data-driven control scheme tailored towards learning the best processing rate in a multi-unit processing network. We quantified the cost suboptimality that the data-driven policy incurs as a function of the disturbance level and explorations.

A. Preliminaries

A.1 Constrained Optimal Control

We define, a general discrete time, infinite horizon, constrained optimal control problem

$$\begin{aligned} \text{Minimize} \quad & \sum_{t=0}^{\infty} \gamma^t g(x_t, u_t) \text{ over } \{u_t\}_{t=0}^{\infty} \\ \text{subject to} \quad & x_{t+1} = f(x_t, u_t); x_t \in \mathcal{X} \\ & u_t \in \mathcal{U}(x_t, \lambda), \quad \forall t \end{aligned} \tag{22}$$

where $f : \mathcal{X} \times \mathcal{U} \to \mathcal{X}$, x represents the n-dimensional state variables, u the m-dimensional control variable and λ some positive constant. If one could find a "cost function" $J : \mathcal{X} \to [0, \infty)$ that solves the Bellman equation given by $J(x) = \min_{u \in \mathcal{U}(x,\lambda)} \{g(x, u) + \gamma J(f(x, u))\}$ for all $x \in \mathcal{X}$, and for a minimizing argument u^*, then it holds that $\min_{u \in \mathcal{U}(x,\lambda)} \sum_{t=0}^{\infty} \gamma^t g(x_t, u_t) = \sum_{t=0}^{\infty} \gamma^t g(x_t, u_t^*) = J(x_0)$.

B. Proofs

B.1 Bounds on (q^x, q^0) Parameters

Bounds on the ground truth (q^x, q^0) parameters are given.

LEMMA 1
The parameters q^x and q^0 satisfy

(i) $\frac{1}{1-\gamma \min_i M_{ii}} \mathbf{1} \leq q^x \leq \frac{1}{1-\gamma \max_i M_{ii}} \mathbf{1}$;

(ii) $q^0 = \frac{\gamma^2 \lambda}{1-\gamma} \frac{1}{1-\gamma \min_i M_{ii}}$.

Proof. We have $q^x = \mathbf{1} + \gamma M p$, $q^0 = \frac{\gamma^2 \lambda}{1-\gamma} \min_i p_i$ and $p_i = \frac{\eta_i}{\eta_i - \gamma}$, $i = 1, \ldots, n$. It follows that

$$\begin{aligned} q^x = p \leq \frac{1}{1-\gamma \max_i M_{ii}} \mathbf{1} \quad \text{and} \quad q^x \geq \frac{1}{1-\gamma \min_i M_{ii}} \mathbf{1}, \\ q^0 = \frac{\gamma^2 \lambda}{1-\gamma} \frac{\max_i \eta_i}{\max_i \eta_i - \gamma} = \frac{\gamma^2 \lambda}{1-\gamma} \frac{1}{1-\gamma \min_i M_{ii}}. \end{aligned}$$

■

B.2 Proof of Lemma 1

According to [Evmorfopoulos, 2012], Lemma 2, the matrix $\mathcal{M}$ is strictly row (column) diagonally dominant with positive diagonal elements if $\|M + \tilde{M}_t\|_\infty < \frac{1}{\gamma}$ ($\|M + \tilde{M}_t\|_1 < \frac{1}{\gamma}$), which is ensured if $\|\tilde{M}_t\|_\infty < \frac{1}{\gamma} - \max_i M_{ii}$ ($\|\tilde{M}_t\|_1 < \frac{1}{\gamma} - \max_i M_{ii}$) as conditioned in the Lemma. ■

B.3 Proof of Lemma 2

Given that $\|\tilde{M}_t\|_{1,\infty} < \frac{1}{n-1}(\frac{1}{\gamma} - \max_i M_{ii})$, we have $\gamma(n-1)\sum_j |[\tilde{M}_t]_{ji}| < 1 - \gamma M_{ii}$ for $i = 1, \ldots, n$. Then,

$$\begin{aligned}\mathcal{M}_{ii} = 1 - \gamma M_{ii} - \gamma[\tilde{M}_t]_{ii} &\geq 1 - \gamma M_{ii} - \gamma(n-1)|[\tilde{M}_t]_{ii}| \\ &> (n-1)\sum\nolimits_{j\neq i}|\gamma[\tilde{M}_t]_{ji}| = (n-1)\sum\nolimits_{j\neq i}|\mathcal{M}_{ji}|. \quad (23)\end{aligned}$$

This implies that $\mathcal{M}$ is strictly column diagonally dominant. $\mathcal{M}^{-1}$ having positive diagonal elements follows from $\mathcal{M}$ enjoying IDP [Furtado et al., 2024]. Let $\mathcal{M}^{-1}_{ij}$ for $1 \leq i, j \leq n$ denote the entries of $\mathcal{M}^{-1}$. Under (23), it was shown in [Johnson et al., 2024] that $\mathcal{M}^{-1}_{ii} > (n-1)|\mathcal{M}^{-1}_{ij}|$ for all $1 \leq i, j \leq n$. Then, it holds for all i that

$$\sum\nolimits_{j\neq i}|\mathcal{M}^{-1}_{ij}| \leq \sum\nolimits_{j\neq i}\max_j|\mathcal{M}^{-1}_{ij}| \leq (n-1)\max_j|\mathcal{M}^{-1}_{ij}| < \mathcal{M}^{-1}_{ii},$$

indicating that $\mathcal{M}^{-1}$ is strictly row diagonally dominant with positive diagonal entries. ∎

B.4 Proof of Theorem 1

We have that q^x and q^x_t satisfy the following equations

$$q^x_t = \gamma(M + \tilde{M}_t)q^x_t + \mathbf{1} \quad \text{and} \quad q^x = \gamma M q^x + \mathbf{1}. \quad (24)$$

Taking the difference between the equations in (24) gives $q^x_t - q^x = \gamma M(q^x_t - q^x) + \gamma\tilde{M}_t q^x_t$. Adding and subtracting the term $\gamma\tilde{M}_t q^x$ to the last equation and solving for $q^x_t - q^x$ gives the least squares estimation error

$$q^x_t - q^x = \gamma\mathcal{M}^{-1}\tilde{M}_t q^x.$$

To extract an upper bound on the error, we start by extracting a norm bound on the inverse of the strictly row diagonally dominant matrix $\mathcal{M}$. From [Varga, 1976], it holds that $\|\mathcal{M}^{-1}\|_\infty \leq (\min_i|\mathcal{M}_{ii}| - \sum_{j\neq i}^n|\mathcal{M}_{ij}|)^{-1} =: (\min_i c_i)^{-1}$. Then,

$$\begin{aligned}c_i = 1 - \gamma M_{ii} - \gamma[\tilde{M}_t]_{ii} - \sum\nolimits_{j\neq i}|\gamma[\tilde{M}_t]_{ij}| &\geq 1 - \gamma M_{ii} \\ - \gamma|[\tilde{M}_t]_{ii}| - \sum\nolimits_{j\neq i}|\gamma[\tilde{M}_t]_{ij}| &= 1 - \gamma M_{ii} - \gamma\sum\nolimits_j|[\tilde{M}_t]_{ij}|.\end{aligned}$$

Therefore,

$$\begin{aligned}\min_i c_i \geq 1 - \gamma\max_i M_{ii} - \gamma\max_i\sum\nolimits_j|[\tilde{M}_t]_{ij}| &= 1 - \gamma\max_i M_{ii} - \gamma\|\tilde{M}_t\|_\infty \\ &= 1 - \gamma\max_i M_{ii} - \gamma\rho,\end{aligned}$$

and in consequence $\|\mathcal{M}^{-1}\|_\infty \leq (1-\gamma \max_i M_{ii} - \gamma\rho)^{-1}$. Since the ∞-norm is sub-multiplicative, it holds

$$\|\mathcal{M}^{-1}\tilde{M}_t\|_\infty \leq \|\mathcal{M}^{-1}\|_\infty \|\tilde{M}_t\|_\infty \leq \rho(1-\gamma \max_i M_{ii} - \gamma\rho)^{-1} =: \alpha,$$

from which we get that $|\mathcal{M}^{-1}\tilde{M}_t| \leq \alpha \mathbf{1}\mathbf{1}^\top$, which implies that $|q_t^x - q^x| \leq \gamma\alpha \mathbf{1}\mathbf{1}^\top q^x$. Next, we find a constant $\theta > 0$ such that $\gamma\alpha\mathbf{1}\mathbf{1}^\top q^x \leq \theta q^x$. According to Lemma 1 we get

$$\gamma\alpha\mathbf{1}\mathbf{1}^\top q^x \leq \gamma\alpha\mathbf{1}\mathbf{1}^\top(1-\gamma \max_i M_{ii})^{-1}\mathbf{1} = \gamma\alpha n(1-\gamma \max_i M_{ii})^{-1}\mathbf{1},$$

and therefore it suffices that $\gamma\alpha n(1-\gamma \max_i M_{ii})^{-1}\mathbf{1} \leq \theta q^x$. Since $\theta q^x \geq \theta(1-\gamma \min_i M_{ii})^{-1}\mathbf{1}$, it is then sufficient to ensure

$$\theta(1-\gamma \min_i M_{ii})^{-1}\mathbf{1} \geq \gamma\alpha n(1-\gamma \max_i M_{ii})^{-1}\mathbf{1},$$

and therefore picking $\theta = \frac{\gamma\alpha n(1-\gamma \min_i M_{ii})}{1-\gamma \max_i M_{ii}}$ implies that $\theta q^x \geq \gamma\alpha\mathbf{1}\mathbf{1}^\top q^x$ and eventually that $|q_t^x - q^x| \leq \theta q^x$. Since $q_t^u = \gamma q_t^x$ and $q^0 = \frac{\gamma^2\lambda}{1-\gamma}\min_i [q^x]_i$, it holds that $|q_t^u - q^u| \leq \theta q^u$ and $|q_t^0 - q^0| \leq \theta q^0$. Consequently, we get that

$$\left\| \begin{bmatrix} q_t \\ q_t^0 \end{bmatrix} - \begin{bmatrix} q \\ q^0 \end{bmatrix} \right\| \leq \frac{\gamma\rho n(1-\gamma \min_i M_{ii})}{(1-\gamma \max_i M_{ii})(1-\gamma \max_i M_{ii} - \gamma\rho)} \begin{bmatrix} q \\ q^0 \end{bmatrix}.$$

■

B.5 Proof of Theorem 2

Let $i = \arg\min_j [q^x]_j$ and drive the system by (20) we get

$$p^\top x_{t+1} + p^0 = p^\top(Mx_t + \lambda e_{i_t} + \epsilon_t + w_t) + p^0 \leq p^\top(Mx_t + \lambda e_{i_t}) + p^0 + p^\top|\epsilon_t + w_t|.$$

Multiply both sides by γ and use $\gamma p^\top M = (p-\mathbf{1})^\top$ to get

$$\begin{aligned}\gamma(p^\top x_{t+1} + p^0) &\leq (p-\mathbf{1})^\top x_t + \lambda\gamma p^\top e_{i_t} + \gamma p^0 + \gamma p^\top|\epsilon_t + w_t| \\ &= p^\top x_t + \gamma p^0 + \lambda\gamma p^\top e_{i_t} - \mathbf{1}^\top x_t + \gamma p^\top|\epsilon_t + w_t|.\end{aligned}$$

Also,

$$q^0 = \frac{\gamma^2\lambda}{1-\gamma}\min_j [q^x]_j = \frac{\gamma^2}{1-\gamma}(q^x)^\top\lambda e_i = \frac{\gamma^2}{1-\gamma}\lambda p^\top e_i.$$

Theorem 1 with $\theta \coloneqq \frac{\gamma\rho n(1-\gamma \min_i M_{ii})}{(1-\gamma \max_i M_{ii})(1-\gamma \max_i M_{ii} - \gamma\rho)}$ gives for $t \geq t_0$

$$\begin{aligned}\frac{\gamma^2}{1-\gamma}\lambda p^\top e_{i_t} \leq (1-\theta)^{-1}\frac{\gamma^2}{1-\gamma}\lambda(q_t^x)^\top e_{i_t} &= (1-\theta)^{-1}q_t^0 \\ &\leq (1+\theta)(1-\theta)^{-1}q^0 = (1+\theta)(1-\theta)^{-1}\gamma p^0,\end{aligned}$$

from which

$$\lambda\gamma p^\top e_{i_t} \leq (1+\theta)(1-\theta)^{-1}(1-\gamma)p^0,$$

and since $(1-\gamma)p^0 = \gamma\lambda \min_j p_j$, we get that

$$\begin{aligned}\lambda\gamma p^\top e_{i_t} \leq (1+\theta)(1-\theta)^{-1}\gamma\lambda \min_j p_j = \gamma\lambda \min_j p_j + \tfrac{2\theta}{1-\theta}\gamma\lambda \min_j p_j \\ = (1-\gamma)p^0 + \tfrac{2\theta}{1-\theta}\gamma\lambda \min_j p_j,\end{aligned}$$

and also that

$$\mathbf{1}^\top x_t \leq p^\top x_t + p^0 - \gamma(p^\top x_{t+1} + p^0) + \tfrac{2\theta}{1-\theta}\gamma\lambda \min_j p_j + \gamma p^\top |\epsilon_t + w_t|.$$

Since $\min_j p_j = \frac{1}{1-\gamma \min_j M_{jj}}$ and $p \leq \frac{1}{1-\gamma \max_i M_{ii}}\mathbf{1}$, we then get in consequence

$$\mathbf{1}^\top x_t \leq p^\top x_t + p^0 - \gamma(p^\top x_{t+1} + p^0) + \tfrac{2\theta\lambda\gamma}{(1-\theta)(1-\gamma \min_i M_{ii})} + \tfrac{\gamma}{1-\gamma \max_i M_{ii}}\mathbf{1}^\top |\epsilon_t + w_t|.$$

Multiply both sides by γ^{t-t_0}, summing over $t_0 \leq t \leq \infty$ and letting $J(x_{t_0}) \coloneqq p^\top x_{t_0} + p^0$ yields the desired result

$$\begin{aligned}\sum_{t=t_0}^{\infty} \gamma^{t-t_0}\mathbf{1}^\top x_t - J(x_{t_0}) \leq \tfrac{2\theta\lambda\gamma}{(1-\gamma)(1-\theta)(1-\gamma \min_i M_{ii})} \\ + \tfrac{1}{1-\gamma \max_i M_{ii}} \sum_{t=t_0}^{\infty} \gamma^{t-t_0+1}\mathbf{1}^\top |\epsilon_t + w_t|.\end{aligned}$$

■

References

Adler, S., M. Moharrami, and V. Subramanian (2022). "Learning a discrete set of optimal allocation rules in queueing systems with unknown service rates". *arXiv preprint arXiv:2202.02419*.

Åström, K. J. and B. Wittenmark (1973). "On self tuning regulators". *Automatica* **9**:2, pp. 185–199.

Åström, K. and B. Wittenmark (1995). *Adaptive Control (2 ed.)* English. Addison-Wesley. ISBN: 0-201-55866-1. DOI: 10.2307/1269433.

Bagnara, R. (1995). "A unified proof for the convergence of jacobi and gauss–seidel methods". *SIAM Rev.* **37**:1, pp. 93–97.

Bellman, R. (1966). "Dynamic programming". *Science* **153**:3731, pp. 34–37.

Bencherki, F. and A. Rantzer (2024). "Data-driven adaptive dispatching policies for processing networks". *IEEE Control Systems Letters* **8**, pp. 2841–2846. DOI: 10.1109/LCSYS.2024.3516637.

Bertsimas, D. and C. W. Kim (2023). "Optimal control of multiclass fluid queueing networks: a machine learning approach". *arXiv preprint arXiv:2307.12405*.

Bertsimas, D., E. Nasrabadi, and I. C. Paschalidis (2014). "Robust fluid processing networks". *IEEE Trans. Autom. Control* **60**:3, pp. 715–728.

Bradtke, S. J., B. E. Ydstie, and A. G. Barto (1994). "Adaptive linear quadratic control using policy iteration". In: *Proc. Am. Control Conf.* Vol. 3. IEEE, pp. 3475–3479.

Chen, M., R. Dubrawski, and S. P. Meyn (2004). "Management of demand-driven production systems". *IEEE Trans. Autom. Control* **49**:5, pp. 686–698.

Choudhury, T., G. Joshi, W. Wang, and S. Shakkottai (2021). "Job dispatching policies for queueing systems with unknown service rates". In: *MobiHoc '21*, pp. 181–190.

Cochran, J. K. and K. T. Roche (2009). "A multi-class queuing network analysis methodology for improving hospital emergency department performance". *Comput. Oper. Res.* **36**:5, pp. 1497–1512.

Cohen, A., V. Subramanian, and Y. Zhang (2024). "Learning-based optimal admission control in a single-server queuing system". *Stoch. Syst.*

Evmorfopoulos, N. (2012). "Some results on diagonally dominant matrices with positive diagonal elements". *Dept. Comput. Commun. Eng., Univ. Thessaly, Volos, Greece, Rep. TR-09-09-001*.

Furtado, S., C. R. Johnson, C. Marijuán, and M. Pisonero (2024). "Square matrices with the inverse diagonal property". *Kuwait J. Sci.* **51**:1, p. 100125.

Golub, G. H. and C. F. Van Loan (2013). *Matrix computations*. JHU press.

Johnson, C. R., C. Marijuán, and M. Pisonero (2023). "Diagonal dominance and invertibility of matrices". *Spec. Matrices* **11**:1, p. 20220181.

Johnson, C., C. Marijuán, M. Pisonero, and I. Spitkovsky (2024). "Diagonal entries of inverses of diagonally dominant matrices". *Linear Algebra Appl.* **692**, pp. 84–90.

Krishnasamy, S., R. Sen, R. Johari, and S. Shakkottai (2021). "Learning unknown service rates in queues: a multiarmed bandit approach". *Oper. Res.* **69**:1, pp. 315–330.

Li, Y. and A. Rantzer (2024). "Exact dynamic programming for positive systems with linear optimal cost". *IEEE Trans. Autom. Control*.

Liu, B., Q. Xie, and E. Modiano (2022). "Rl-qn: a reinforcement learning framework for optimal control of queueing systems". *ACM Trans. Model. Perform. Eval. Comput. Syst.* **7**:1, pp. 1–35.

Meyn, S. (2008). *Control techniques for complex networks*. Cambridge University Press.

Ohlin, D., E. Tegling, and A. Rantzer (2024). "Optimal control of linear cost networks". *Eur. J. Control*, p. 101068.

Qu, S., J. Wang, and J. Jasperneite (2018). "Dynamic scheduling in large-scale stochastic processing networks for demand-driven manufacturing using distributed reinforcement learning". In: *23rd IEEE Int. Conf. Emerg. Technol. Fact. Autom. ETFA*. Vol. 1. IEEE, pp. 433–440.

Stahlbuhk, T., B. Shrader, and E. Modiano (2021). "Learning algorithms for minimizing queue length regret". *IEEE Trans. Inf. Theory* **67**:3, pp. 1759–1781.

Sutton, R. S. and A. G. Barto (2018). *Reinforcement learning: An introduction*. MIT press.

Tassiulas, L. and A. Ephremides (1990). "Stability properties of constrained queueing systems and scheduling policies for maximum throughput in multihop radio networks". In: *29th Proc. IEEE Conf. Decis. Control*. IEEE, pp. 2130–2132.

Varga, R. S. (1976). "On diagonal dominance arguments for bounding $\|A^{-1}\|_\infty$". *Linear Algebra Appl.* **14**:3, pp. 211–217.

Walton, N. and K. Xu (2021). "Learning and information in stochastic networks and queues". In: *Tutorials in Operations Research: Emerging Optimization Methods and Modeling Techniques with Applications*. INFORMS, pp. 161–198.

Wittenmark, B. (1995). "Adaptive dual control methods: an overview". *Adaptive Systems in Control and Signal Processing 1995*, pp. 67–72.

Yang, Z., R. Srikant, and L. Ying (2023). "Learning while scheduling in multi-server systems with unknown statistics: maxweight with discounted ucb". In: *Int. Conf. Artif. Intell. Stat*. PMLR, pp. 4275–4312.

Zhou, K. and J. C. Doyle (1998). *Essentials of robust control*. Vol. 104. Prentice hall Upper Saddle River, NJ.

Printed by Media-Tryck, Lund 2025 NORDIC SWAN ECOLABEL 3041 0903



Department of Automatic Control
P.O. Box 118, 221 00 Lund, Sweden
www.control.lth.se

ISRN LUTFD2/TFRT--3284--SE
ISSN 0280–5316